Universidad del País Vasco Euskal Herriko Unibertsitatea
BILBOKO INGENIARITZA ESKOLA
ESCUELA DE INGENIERÍA DE BILBAO

MÁSTER UNIVERSITARIO EN INGENIERÍA INDUSTRIAL

TRABAJO FIN DE MÁSTER

EXPLORING THE FEASIBILITY OF SMALL MODULAR REACTORS (SMRS) FOR SUSTAINABLE POWER SUPPLY TO DATA CENTERS





Estudiante: Aldekoa Camba, Iker

Director/Directora: Herrero Villalibre, Saioa

Curso: 2024-2025

Fecha: Bilbao, 13 de julio de 2025

TRABAJO FIN DE MÁSTER
REALIZADO EN MOVILIDAD

ILLINOIS TECH

Lewis College of Science and Letters

ILLINOIS TECH | Lewis College of Science and Letters

ILLINOIS INSTITUTE OF TECHNOLOGY

LEWIS COLLEGE OF SCIENCE AND LETTERS

Supervisado y dirigido en destino por:

Jeff Terry

ABSTRACT

This thesis explores the feasibility of powering data centers with Small Modular Reactors (SMRs), focusing on technical suitability, environmental performance, and economic viability. The study applies a two-stage decision-making process: first, filtering out unqualified SMR designs using exclusion criteria, and then applying the Analytic Hierarchy Process (AHP) to rank the most promising alternatives. Key evaluation criteria include safety, energy availability, licensing maturity, fuel cycle characteristics, and deployment flexibility. The Xe-100, KP-FHR, and Aurora Powerhouse reactors emerged as the top candidates. The economic analysis shows that SMRs currently face high capital costs and are not yet competitive with renewables on a pure cost basis. However, their value proposition improves significantly in scenarios where resilience, dispatchability, and long-term decarbonization are prioritized—especially when public support mechanisms or multi-unit deployments are available. From an environmental perspective, SMRs offer ultra-low life-cycle carbon emissions, modest water use, compact land footprints, and manageable radioactive waste. These characteristics make them a compelling option for decarbonizing mission-critical digital infrastructure, particularly where renewable deployment is limited by intermittency, land constraints, or reliability concerns.

Keywords: Small Modular Reactors (SMRs), data centers, AHP, energy systems, sustainability, nuclear energy.

RESUMEN

Este trabajo analiza la viabilidad de alimentar centros de datos mediante Reactores Modulares Pequeños (SMRs), abordando su idoneidad técnica, impacto ambiental y viabilidad económica. Para ello, se emplea un enfoque en dos etapas: en primer lugar, se descartan los diseños no aptos mediante criterios de exclusión; y, en segundo lugar, se aplica el Proceso de Jerarquía Analítica (AHP) para clasificar los SMRs más prometedores. Entre los criterios evaluados destacan la seguridad, disponibilidad energética, madurez regulatoria, características del ciclo de combustible y flexibilidad de despliegue. Los diseños Xe-100, KP-FHR y Aurora Powerhouse resultan ser los más adecuados. A nivel económico, los SMRs presentan actualmente costes de capital elevados, lo que los sitúa en desventaja frente a las renovables en escenarios sin ayudas. No obstante, su propuesta de valor mejora significativamente cuando se priorizan la resiliencia, la despachabilidad y la descarbonización a largo plazo, especialmente si existen mecanismos de apoyo público o despliegues modulares a gran escala. Desde el punto de vista ambiental, los SMRs destacan por sus bajas emisiones de carbono en todo su ciclo de vida, bajo consumo de agua, reducido uso de suelo y una gestión de residuos radiactivos viable. Estas características los convierten en una opción atractiva para descarbonizar centros de datos, sobre todo en contextos donde las renovables encuentran limitaciones técnicas o logísticas.

Palabras clave: Reactores Modulares Pequeños (SMR), centros de datos, AHP, sistemas energéticos, sostenibilidad, energía nuclear.

LABURPENA

Lan honek data zentro handiak erreaktore modular txikien bidez elikatzeko bideragarritasuna aztertzen du, hiru ikuspegitik: egokitasun teknikoa, ingurumen-eragina eta bideragarritasun ekonomikoa. Horretarako, bi fasetako metodologia aplikatu da: lehenik, baldintza teknikoak betetzen ez dituzten diseinuak baztertzen dira; ondoren, Analytic Hierarchy Process (AHP) erabiliz, egokienak sailkatzen dira. Ebaluazio-irizpide nagusien artean daude segurtasuna, energia eskuragarritasuna, lizentzia-egoera, erregaiaren zikloa eta hedatzeko malgutasuna. Xe-100, KP-FHR eta Aurora Powerhouse diseinuak nabarmendu dira egokien bezala. Ekonomiari dagokionez, SMR-ak oraindik ez dira lehiakorrak energia berriztagarriekiko kostuen aldetik, bereziki laguntza publiko edo eskala handiko hedapenik gabe. Hala ere, haien abantailak handitu egiten dira erresilientzia, egonkortasuna eta epe luzerako deskarbonizazioa baloratzen diren kasuetan. Ingurumenaren ikuspegitik, SMR-ak oso emisio txikiak dituzte bizitza-zikloan, ur kontsumo moderatua, lur erabilera murriztua eta hondakin erradioaktiboen kudeaketa bideragarria. Horregatik, aukera interesgarria dira datu zentroetan deskarbonizazioa lortzeko, bereziki energia berriztagarriak aplikatzeko zailtasunak dauden inguruneetan.

Hitz gakoak: Erreaktore modular txikiak (SMR), datu-zentroak, AHP, energia-sistemak, iraunkortasuna, energia nuklearra.

TABLE OF CONTENTS

1 INTRODUCTION 11
2 BACKGROUND 13
3 OBJECTIVES AND SCOPE OF THE WORK 14
4 BENEFITS OF THE PROJECT AND CONTRIBUTION TO THE SUSTAINABLE DEVELOPMENT GOALS (SDGs) 15
5 LITERATURE REVIEW 17
5.1 Data Centers 17
5.1.1 Tier Classification of Data Centers 17
5.1.2 Size and Power Demand of Data Centers 18
5.1.3 Future Modularity of Data Centers 18
5.1.4 Energy Demand Growth and Real-World Data 19
5.1.5 Data Center Power Consumption and Distribution 20
5.2 Small Modular Reactors (SMRs) 21
5.2.1 Advantages of SMRs 21
5.2.2 Disadvantages of SMRs 22
5.2.3 Types of SMRs 22
5.2.4 Secondary applications 23
6 SELECTION AND RANKING OF SMRs 26
6.1 Methodology 26
6.1.1 Stage 1 27
6.1.2 Stage 2 40
6.2 Results 85
6.3 Sensitivity Analysis 86
6.3.1 Sensitivity to Criteria Weighting 87
6.3.2 Sensitivity to Subcriteria Weighting 88
6.3.3 Conclusions 92
7 ECONOMIC FEASIBILITY AND COST ANALYSIS 93
7.1 Introduction 93
7.1.1 Objective of the analysis 93
7.1.2 Key Metrics 93
7.1.3 Economic Advantages and Disadvantages of SMRs 94
7.2 Capital Expenditures (CAPEX) 95
7.2.1 Definition and Relevance 95
7.2.2 Overnight Capital Cost (OVC) 95

7.2.3 Interest During Construction (IDC) 96
7.2.4 CAPEX Summary Table 97
7.2.5 Data Center-Specific Considerations 98
7.3 Operational Expenditures (OPEX) 99
7.3.1 Definition and Scope 99
7.3.2 Total OPEX Estimates 99
7.3.3 Breakdown of OPEX by reactor type 99
7.3.4 Considerations for Data Centers 100
7.4 Cost Adjustment Factors 101
7.4.1 FOAK vs BOAK Coast Premiums 101
7.4.2 Learning Rates 102
7.4.3 Multi-unit deployment advantages 103
7.5 Levelized Cost of Electricity (LCOE) 104
7.5.1 LCOE as a Comparative Metric 104
7.5.2 Input Parameters and Assumptions 105
7.5.3 LCOE Calculation for Different Deployment Scenarios 106
7.6 Total Cost of Ownership (TCO) 111
7.6.1 TCO Definition and Scope 111
7.6.2 Methodology and Formula 111
7.6.3 Case A: SMR Standalone Configuration 112
7.6.4 Case B: SMR + Data Center Co-location 114
7.7 Comparative Analysis with Alternative Energy Sources 116
7.7.1 Selected Technologies: Coal, Gas, Solar PV, Wind, Grid Supply 116
7.7.2 CAPEX and OPEX Benchmark Ranges 116
7.7.3 LCOE Comparison: SMR vs Alternative Sources 121
7.7.4 Economic Implications of Resilience and Reliability 126
7.7.5 Emissions-Related Economic Impacts 128
7.8 Profitability Assessment 129
7.8.1 Revenue Model and EBITDA Estimation 129
7.8.2 Net Present Value (NPV) 129
7.8.3 Internal Rate of Return (IRR) 130
7.8.4 Interpretation and Discussion 130
7.8.5 Conclusion 131
7.9 Risk of Budget Overruns in Nuclear Projects 132
7.10 Conclusions 133

8 ENVIRONMENTAL IMPACT ANALYSIS 136
8.1 Introduction 136
8.2 Methodology: Life-Cycle Assessment (LCA) 136
8.2.1 Objective and Scope 136
8.2.2 Functional Unit 137
8.2.3 System Boundaries 137
8.2.4 Data Sources 139
8.3 Results: Carbon Footprint Comparison 139
8.3.1 SMR Emissions by Life-Cycle Phase 139
8.3.2 Comparison with Other Electricity Sources 140
8.4 Other Environmental Factors 141
8.4.1 Water Use 141
8.4.2 Nuclear Waste Management 142
8.4.3 Land Use 143
8.4.4 Site Selection and Environmental Constraints 144
8.5 Conclusions 145
9 PROJECT PLANNING 147
9.1 Description of Project Tasks 147
9.2 Gantt Chart 148
10 Project Budget Description 149
10.1 Labor Costs 149
10.2 Hardware Cost 150
10.3 Software Costs 150
10.4 Additional Costs 150
10.5 Total Cost 151
11 CONCLUSIONS 152
12 REFERENCES 154
APPENDIX 1: Global and Local Weights in the AHP Model 162

LIST OF TABLES

Table 1. Selected SMR designs 28
Table 2. Intensity values for pairwise comparison 45
Table 3. Pairwise comparison matrix of Safety Characteristics 46
Table 4. Pairwise comparison matrix of Reactor System Characteristics 47
Table 5. Pairwise comparison matrix of Fuel Cycle Characteristics 48
6. Pairwise comparison matrix of Waste Management 49
Table 7. Pairwise comparison matrix of Energy Availability 50
Table 8. Pairwise comparison matrix of Land Footprint 52
Table 9. Pairwise comparison matrix of Licensing Status 54
Table 10. Pairwise comparison matrix of Selection Criteria 58
Table 11. Pairwise comparison matrix of Fuel Safety 61
Table 12. Pairwise comparison matrix of Coolant Toxicity 62
Table 13. Pairwise comparison matrix of Source Term 63
Table 14. Pairwise comparison matrix of Configuration 64
Table 15. Pairwise comparison matrix of Component Qualification 65
Table 16. Pairwise comparison matrix of Seismic Resilience 66
Table 17. Pairwise comparison matrix of Refueling Time 67
Table 18. Pairwise comparison matrix of Enrichment 68
Table 19. Pairwise comparison matrix of Discharge Burnup 69
Table 20. Pairwise comparison matrix of Waste Streams 71
Table 21. Pairwise comparison matrix of Storage and Disposal Plan 72
Table 22. Pairwise comparison matrix of Unit Capacity Factor 73
Table 23. Pairwise comparison matrix of Module Independence 74
Table 24. Pairwise comparison matrix of EPZ Size 75
Table 25. Pairwise comparison matrix of Plant Size 76
Table 26. Pairwise comparison matrix of Scalability for Future Expansion 77
Table 27. Pairwise comparison matrix of Coolant Temperature 78
Table 28. Pairwise comparison matrix of Design Status 79
Table 29. Pairwise comparison matrix of Licensing Initiated 80
Table 30. Pairwise comparison matrix of Operational Experience 81
Table 31. Pairwise comparison matrix of Non-Standard Design Features 82
Table 32. Scoring of Reactors Part 1 83
Table 33. Scoring of Reactors Part 2 84
Table 34. Ranking of Reactors 85
Table 35. Adjusted Criteria Weighting 87
Table 36. New Reactor Ranking after Adjusted Criteria Weighting 87
Table 37. Adjusted Safety Characteristics Weight 88
Table 38. New Reactor Ranking after Adjusted Safety Characteristics Weight 89
Table 39. Adjusted Fuel Cycle Characteristics Weight 89
Table 40. New Reactor Ranking after Fuel Cycle Characteristics Weight 89
Table 41. Adjusted Energy Availability Weight 90
Table 42. New Reactor Ranking after Adjusted Energy Availability Weight 90
Table 43. Adjusted Licensing Status Weight 91
Table 44. New Reactor Ranking after Adjusted Licensing Status Weight 91
Table 45. Breakdown of OVC for PWR, SFR, HTGR and MSR 95

Table 46. IDC values for different Interest Rates and OVCs 97
Table 47. CAPEX values for different Interest Rates and OVCs 97
Table 48. Adjusted CAPEX for Data Centers 98
Table 49. Breakdown of OPEX for PWR, SFR, HTG and MSR 99
Table 50. Adjusted OPEX for Data Centers 100
Table 51. FOAK CAPEX values 102
Table 52. Learning Rates Scenarios 103
Table 53. CAPEX Exponents (MUE) and OPEX Multipliers (MOM) 104
Table 54. Adjusted OVC and OPEX for different Deployment Scenarios 104
Table 55. BOAK Single Unit LCOE values 107
Table 56. BOAK with 4 units LCOE values 108
Table 57. FOAK Single Unit LCOE values 109
Table 58. FOAK with 4 units LCOE values 110
Table 59. SMR Standalone TCO values for different Scenarios 113
Table 60. Adjusted CAPEX and OPEX for Data Centers for different Scenarios 114
Table 61. SMR + Data Center TCO values for different Scenarios 115
Table 62. CAPEX and OPEX values for different Energy Sources 117
Table 63. CAPEX and OPEX values for different SMR Scenarios with a 5% Discount Rate 117
Table 64. CAPEX and OPEX values for different SMR Scenarios with a 10% Discount Rate 118
Table 65. CAPEX and OPEX values for different SMR Scenarios with a 15% Discount Rate 119
Table 66. CAPEX and OPEX values for different SMR Scenarios with a 20% Discount Rate 120
Table 67. LCOE values for different Energy Sources 122
Table 68. LCOE values for different SMR Scenarios with a 5% Discount Rate 122
Table 69. LCOE values for different SMR Scenarios with a 10% Discount Rate 123
Table 70. LCOE values for different SMR Scenarios with a 15% Discount Rate 124
Table 71. LCOE values for different SMR Scenarios with a 20% Discount Rate 125
Table 72. TCO comparison for SMR vs Solar PV + Batteries 127
Table 73. Present Values (PV) for different Discount Rates 130
Table 74. NPV values for different Discount Rates 130
Table 75. IRR values for different CAPEX values 130
Table 76. Contingency Multiplier for different Project Types 132
Table 77. Greenhouse Gas (GHG) Emissions for every SMR Life-Cycle Phase 140
Table 78. Emission Factor and Annual Emissions for different Energy Sources 140
Table 79. Water Use by Energy Source 142
Table 80. Labor Costs during Phase 1 of the Project 149
Table 81. Labor Costs during Phase 2 of the Project 150
Table 82. Additional Costs during the Project 151
Table 83. Total Costs for the Project 151
Table 84. Global Weights 163
Table 85. Local Weights 163

LIST OF FIGURES

Figure 1. Data Center power supplied by SMRs 12
Figure 2. Sustainable Development Goals (SDGs) related to the Project 16
Figure 3. Modularity of Data Centers 19
Figure 4. Growth in Electricity Consumptions by Data Centers from 2020 to 2030 19
Figure 5. Approximate Power Consumption Breakdown in Data Centers 20
Figure 6. Scheme of District Heating 24
Figure 7. Water Desalination 24
Figure 8. Liquid Cooling in Data Centers 25
Figure 9. SMR Selection Methodology 26
Figure 10. i-SMR 29
Figure 11. NuScale Power Module 30
Figure 12. PWR-20 31
Figure 13. HTMR100 32
Figure 14. Xe-100 33
Figure 15. ARC-100 34
Figure 16. FLEX Reactor 35
Figure 17. IMSR400 36
Figure 18. KP-FHR 37
Figure 19. Stellarium 38
Figure 20. Aurora Powerhouse 39
Figure 21. AP-300 40
Figure 22. Selection Criteria and Subcriteria 41
Figure 23. Local Weights for Safety Characteristics' Subcriterions 46
Figure 24. Local Weight for Reactor System Characteristics' Subcriterions 48
Figure 25. Local Weights for Fuel Cycle Characteristics' Subcriterions 48
Figure 26. Local Weights for Waste Management's Subcriterions 50
Figure 27. Local Weights for Energy Availability's Subcriterions 51
Figure 28. Local Weights for Land Footprint's Subcriterions 52
Figure 29. Local Weights for Licensing Status' Subcriterions 54
Figure 30. Weight Distribution of Main Criteria 58
Figure 31. Global Weights of Selection Criteria and Subcriteria 59
Figure 32. Intensity of each of the Ratings Scale in Fuel Safety 61
Figure 33. Intensity of each of the Ratings Scale in Coolant Toxicity 62
Figure 34. Intensity of each of the Ratings Scale in Source Term 63
Figure 35. Intensity of each of the Ratings Scale in Configuration 64
Figure 36. Intensity of each of the Ratings Scale in Component Qualification 65
Figure 37. Intensity of each of the Ratings Scale in Seismic Resilience 66
Figure 38. Intensity of each of the Ratings Scale in Refueling Time 67
Figure 39. Intensity of each of the Ratings Scale in Enrichment 68
Figure 40. Intensity of each of the Ratings Scale in Discharge Burnup 70
Figure 41. Intensity of each of the Ratings Scale in Waste Streams 71
Figure 42. Intensity of each of the Ratings Scale in Storage and Disposal Plan 72
Figure 43. Intensity of each of the Ratings Scale in Unit Capacity Factor 73
Figure 44. Intensity of each of the Ratings Scale in Module Independence 74
Figure 45. Intensity of each of the Ratings Scale in EPZ Size 75

Figure 46. Intensity of each of the Ratings Scale in Plant Size 76
Figure 47. Intensity of each of the Ratings Scale in Scalability for Future Expansion 77
Figure 48. Intensity of each of the Ratings Scale in Coolant Temperature 78
Figure 49. Intensity of each of the Ratings Scale in Design Status 79
Figure 50. Intensity of each of the Ratings Scale in Licensing Status 80
Figure 51. Intensity of each of the Ratings Scale in Operational Experience 81
Figure 52. Intensity of each of the Ratings Scale in Non-Standard Design Features 82
Figure 53. Scoring of Reactors 85
Figure 54. CAPEX by Energy Source at 5% Discount Rate 117
Figure 55. OPEX by Energy Source at 5% Discount Rate 118
Figure 56. CAPEX by Energy Source at 10% Discount Rate 119
Figure 57. CAPEX by Energy Source at 15% Discount Rate 120
Figure 58. CAPEX by Energy Source at 20% Discount Rate 121
Figure 59. LCOE by Energy Source at 5% Discount Rate 122
Figure 60. LCOE by Energy Source at 10% Discount Rate 123
Figure 61. LCOE by Energy Source at 15% Discount Rate 124
Figure 62. LCOE by Energy Source at 20% Discount Rate 125
Figure 63. Cost of Components SMR vs Solar PV + batteries 127
Figure 64. Project's Gantt Diagram 148

LIST OF ACRONYMS

Acronym	*Meaning*
AHP	Analytic Hierarchy Process
AI	Artificial Intelligence
ARIS	Advanced Reactor Information System
BOAK	Between-of-a-Kind
CAPEX	Capital Expenditure
EBITDA	Earnings Before Interest, Taxes, Depreciation, and Amortization
EPZ	Emergency Planning Zone
ETS	Emissions Trading Scheme
FOAK	First-of-a-Kind
GHG	Greenhouse Gas
HALEU	High-Assay Low-Enriched Uranium
HLW	High-Level Waste
HTGR	High-Temperature Gas-Cooled Reactor
IAEA	International Atomic Energy Agency
IDC	Interest During Construction
IEA	International Energy Agency
ILW	Intermediate-Level Waste
INL	Idaho National Laboratories
IPCC	Intergovernmental Panel on Climate Change
IRR	Internal Rate of Return
IT	Information Technology
LCA	Life-Cycle Assessment
LCOE	Levelized Cost of Electricity
LEU	Low-Enriched Uranium
LLW	Low-Level Waste
LMFR	Liquid Metal Fast Reactor
LOCA	Loss Of Coolant Accident
LWR	Light Water Reactor
MOM	Multi-unit OPEX Multiplier
MSR	Molten Salt Reactor
MSRE	Molten Salt Reactor Experiment
MUE	Multi-Unit Exponent
NPV	Net Present Value
NRC	Nuclear Regulatory Commission
NREL	National Renewable Energy Laboratory
OPEX	Operational Expenditure
OVC	Overnight Cost
PDU	Power Distribution Unit
PWR	Pressurized Water Reactor
PV	Photovoltaic
SDG	Sustainable Development Goal
SFR	Sodium-cooled Fast Reactor
SMART	Selection Methodology for Advanced Reactor Technologies
SMR	Small Modular Reactor
SSE	Safe Shutdown Earthquake
TCO	Total Cost of Ownership
TRISO	Tristructural-Isotropic Fuel
TRL	Technology Readiness Level
UCF	Unit Capacity Factor

1 INTRODUCTION

In recent decades, data centers have become critical infrastructures supporting advanced technologies such as artificial intelligence (AI), big data processing, and cloud computing. This rapid evolution has led to a substantial increase in energy demand, posing economic and environmental challenges, particularly as reliance on fossil fuels persists. To ensure the sustainability of digital infrastructure, it is essential to explore innovative and low-carbon energy solutions capable of delivering reliable and continuous power.

Small Modular Reactors (SMRs) have emerged as a promising alternative. These next-generation nuclear technologies offer modularity, high power density, and enhanced safety features, while also enabling flexible siting and potential cost reductions through standardization and factory fabrication. SMRs are uniquely positioned to deliver uninterrupted, dispatchable, zero-carbon electricity and are especially suited for mission-critical facilities such as data centers, which operate with high reliability requirements and limited tolerance for power disruptions.

This project evaluates the feasibility of integrating SMRs as a primary power source for large-scale data centers, adopting a multidisciplinary approach that includes technical selection, economic modeling, and environmental impact assessment.

The study first outlines its main objectives and scope, emphasizing the need to evaluate the technical, economic, and environmental feasibility of SMRs in data center applications. It also identifies how this work contributes to key Sustainable Development Goals (SDGs), particularly SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation, and Infrastructure), and SDG 11 (Sustainable Cities and Communities), by proposing a clean and resilient power alternative for critical digital infrastructure.

It then continues with a background overview of the energy challenges in the digital sector and a review of SMR technologies. The selection process is structured in two stages: first, an exclusion-based filter reduces the initial list of over 80 SMR designs to those compatible with data center requirements; second, the most promising candidates are ranked using the Analytic Hierarchy Process (AHP), incorporating multiple criteria such as safety, energy availability, licensing maturity, fuel cycle characteristics, and deployment flexibility. A sensitivity analysis of the AHP model is also conducted to test the robustness of the rankings under varying stakeholder priorities and subcriterion weights.

The technical assessment is followed by an in-depth economic analysis exploring the capital and operational costs of SMRs, their levelized cost of electricity (LCOE), total cost of ownership (TCO), and profitability metrics such as EBITDA, NPV, and IRR. Multiple deployment scenarios are evaluated, including First-of-a-Kind and multi-unit configurations, with particular attention to resilience-related benefits relevant to data centers.

The environmental impact analysis is based on a life-cycle assessment (LCA) approach and includes a comparison of SMRs with other electricity sources in terms of carbon footprint, water consumption, radioactive waste generation, land use, and siting considerations. The analysis reveals the potential of SMRs to significantly reduce emissions and resource use while maintaining a compact footprint compatible with urban and brownfield deployment.

The report also includes a section outlining the project planning and task distribution, supported by a Gantt chart to visualize the timeline of key activities. In addition, a project budget estimation is provided to assess the resource requirements for further development and research, and an appendix is included with detailed AHP input data—such as global and local weights—to ensure transparency and reproducibility.

In summary, this study aims to provide a comprehensive evaluation of SMRs as a strategic energy solution for sustainable data centers. By integrating technical, economic, and environmental perspectives, it contributes to the growing body of research on decarbonization pathways for the digital sector and supports informed decision-making for future infrastructure planning.



Figure 1. Data Center power supplied by SMRs

2 BACKGROUND

Data centers have rapidly evolved from simple digital storage units into the backbone of modern technological infrastructure, supporting critical services such as artificial intelligence (AI), machine learning, cloud computing, and real-time analytics. This transformation has led to a surge in computational intensity, dramatically increasing electricity consumption. Current estimates indicate that data centers account for approximately 3% of global electricity demand [1], and this share is expected to rise significantly as AI and high-performance computing expand. The intensifying energy requirements pose not only operational and economic challenges but also raise serious environmental concerns, particularly when electricity is sourced from fossil-fuel-based grids.

Meeting the growing power needs of data centers in a sustainable and resilient manner has therefore become a pressing priority. In this context, SMRs have emerged as a promising technological alternative. These advanced nuclear systems combine compact size, modular construction, and inherent safety features to provide clean, uninterrupted, and dispatchable energy. Unlike large-scale nuclear power plants, SMRs can be sited close to demand centers, such as data facilities, enabling secure power supply with minimal transmission losses. Their lack of CO_2 emissions during operation and high energy density make them particularly suited for addressing the twin imperatives of decarbonization and energy reliability.

This interest is no longer theoretical. Major tech companies have already begun to engage with SMR developers to secure carbon-free, high-reliability energy for their operations. Google, for instance, has partnered with Kairos Power to support the development of fluoride salt-cooled high-temperature reactors [2]. Amazon Web Services (AWS) is collaborating with X-Energy to explore the deployment of helium-cooled high-temperature gas reactors [3]. Similarly, Oklo has entered agreements to supply its Aurora fast reactor technology to multiple data center operators [4]. Beyond the tech sector, utilities like EDF and Enel are actively investigating SMR integration into national grids [5], reinforcing the momentum behind this emerging class of nuclear technologies.

Despite these promising developments, significant uncertainties remain. The diversity of SMR designs—ranging from water-cooled and gas-cooled systems to molten salt and fast-spectrum reactors—presents both opportunities and technical challenges. Deployment pathways must also navigate regulatory complexity, public perception, and economic viability. In light of these factors, this study seeks to establish a structured framework for evaluating which SMR technologies are most suitable for data center applications.

By combining technical screening, economic analysis, and environmental impact assessment, this project aims to inform decision-making around the future role of SMRs in powering the digital economy. It responds to a growing need for sustainable infrastructure solutions at the intersection of nuclear energy and information technology.

3 OBJECTIVES AND SCOPE OF THE WORK

The purpose of this work is to evaluate the feasibility of using SMRs as a sustainable energy solution for powering modern data centers. The scope of the study includes a detailed analysis of the growing energy demands of data centers, the potential role of SMRs in meeting these needs, and the technical, economic, and environmental considerations involved. This study will explore various SMR technologies, focusing on their modularity, scalability, and safety features, which make them a promising option for energy-intensive facilities like data centers.

The primary objectives of this work are as follows:

1. **Assessment of Data Center Energy Demands:** To analyze the current and future energy consumption trends of data centers, focusing on factors such as size, tier classification, and power requirements (ranging from 1-5 MW to hyperscale data centers over 20 MW) [6]. This includes understanding the technological advancements and scalability of data centers and how they drive energy demand.

2. **Evaluation of SMR Technologies:** To explore the various types of SMRs, highlighting their design features, operational principles, and potential advantages over traditional energy sources. The study will also examine how the modular nature of SMRs allows for flexibility in meeting the specific energy needs of data centers.

3. **Technical Feasibility Study:** To identify the most suitable SMR designs for data centers through a multi-criteria evaluation using the Analytic Hierarchy Process (AHP), considering safety features, energy availability, licensing progress, fuel cycle characteristics, and deployment flexibility.

4. **Economic Viability and Cost Analysis:** To evaluate the economic implications of deploying SMRs for powering data centers. This will include a cost-benefit analysis of SMR installation, operation, and maintenance compared to conventional energy sources, as well as the potential savings from reduced energy costs and reliance on fossil fuels.

5. **Environmental Impact and Sustainability:** To analyze the potential environmental benefits of using SMRs, focusing on their low-carbon emissions and the role they can play in reducing the overall carbon footprint of data centers. The study will also consider the life cycle of SMR systems and their long-term sustainability as a power source for the data center industry.

The findings of this study will provide a comprehensive understanding of the potential of SMRs as a reliable and sustainable energy source for the rapidly growing data center industry. The results will offer valuable insights for decision-makers in the data center sector and the energy industry, as well as contribute to the broader conversation on sustainable energy solutions for high-demand industries.

4 BENEFITS OF THE PROJECT AND CONTRIBUTION TO THE SUSTAINABLE DEVELOPMENT GOALS (SDGs)

The primary benefit of this project is its potential to contribute to the ongoing efforts to sustainably address the growing energy demands of the global data center industry. As data centers expand to support increasing digital services, the need for reliable, scalable, and low-carbon energy solutions has become increasingly critical. By exploring the viability of SMRs as a power source for these facilities, this study aims to provide valuable insights into a novel and sustainable approach to meeting these demands.

Key benefits of this work include:

1. **Enhanced Understanding of Data Center Energy Needs**: The study offers a comprehensive analysis of current and future energy demands of data centers, highlighting the relationship between technological advancements, operational scale, and electricity consumption.

2. **Identification of Sustainable Energy Solutions**: By evaluating SMRs as a potential alternative, this work explores the benefits of modular nuclear energy as a low-carbon substitute for traditional energy sources. The findings could help define a more sustainable energy model for data centers, significantly reducing the industry's dependence on fossil fuels and supporting global efforts to mitigate climate change.

3. **Informed Decision-Making for Data Center Operators**: The economic feasibility analysis provides operators with critical information regarding the financial implications of adopting SMRs. The cost-benefit framework helps decision-makers evaluate potential advantages over conventional power sources, such as long-term energy savings, improved energy security, and reduced operational risks.

4. **Contribution to Energy Transition and Technological Innovation**: The integration of SMRs into data center operations could serve as a benchmark for broader nuclear energy applications across other high-demand sectors. This work contributes to the growing body of knowledge around the role of advanced nuclear technology in the transition to low-carbon energy systems, fostering innovation and encouraging future research.

5. **Environmental and Sustainability Impact**: The study improves understanding of how SMRs can reduce the carbon footprint of digital infrastructure, supporting global sustainability goals. By demonstrating the potential of SMRs to deliver clean, reliable power, it reinforces the relevance of nuclear energy in achieving net-zero emissions targets for energy-intensive industries.

In addition to these direct benefits, the project aligns with key global sustainability objectives outlined in the United Nations Sustainable Development Goals (SDGs) [7]. Specifically, it supports SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation

and Infrastructure), and SDG 11 (Sustainable Cities and Communities), which reflect broader environmental, technological, and societal priorities.

This work contributes to **SDG 7** by promoting SMRs as a clean, dispatchable, and scalable electricity source capable of powering critical infrastructure like data centers. As digital services expand, energy demand must be met with low-carbon, reliable supply options. SMRs offer a viable alternative to fossil fuels and intermittent renewables by delivering consistent power with negligible greenhouse gas emissions—helping to secure a cleaner and more resilient energy future.

It also aligns with **SDG 9** by fostering innovation in both the energy and digital sectors. The integration of SMRs into high-demand applications like data centers encourages modernization of industrial infrastructure and supports sustainable economic development. The multidisciplinary approach of this study—combining technical, economic, and environmental analysis—reinforces the role of innovation in energy planning and policy.

Lastly, the work supports **SDG 11** by identifying nuclear technologies that can be deployed near urban or peri-urban environments without requiring large land areas or extensive transmission networks. SMRs, thanks to their modularity and passive safety features, are well-suited to powering sustainable and smart cities. Their use could enhance the resilience of essential digital services while minimizing environmental impacts, contributing to more sustainable and technologically advanced urban communities.

In summary, this work not only provides valuable guidance for data center energy planning but also contributes meaningfully to global efforts toward decarbonization, technological innovation, and urban sustainability. By aligning technical research with international sustainability goals, the project underscores the role that advanced nuclear technologies can play in building a cleaner, smarter, and more resilient energy future.



Figure 2. Sustainable Development Goals (SDGs) related to the Project

5 LITERATURE REVIEW

This chapter reviews the current literature on data centers and SMRs. It examines key aspects of data centers, including tier classification, size and power demand, modularity, the rapid growth of energy consumption, and power distribution systems. Additionally, it explores SMRs, outlining what they are, their advantages and drawbacks, different types, and their potential secondary applications. This review provides the background needed to understand how emerging nuclear technologies could help meet the rising energy demands of modern data infrastructure.

5.1 Data Centers

Data centers are essential infrastructures that store, manage, and process vast amounts of data for various digital applications. They are critical to the functioning of the internet and the global economy, supporting services such as cloud computing, big data analytics, artificial intelligence, and more [8]. These facilities house large amounts of computing power, storage systems, and networking equipment, all working together to provide reliable, fast, and secure access to digital information. The demand for data centers has surged in recent years due to the exponential growth of data, driven by increased digitalization across industries, the rise of Internet of Things (IoT) devices, and the rapid expansion of cloud services [9].

5.1.1 Tier Classification of Data Centers

Data centers are classified into different tiers based on their level of redundancy and reliability. The Tier Classification System, developed by the Uptime Institute [10], consists of four primary levels:

- **Tier 1**: These data centers have a basic design with no redundancy. They are dependent on a single power and cooling source, meaning any failure can result in significant downtime. These centers are typically used for less critical applications where uptime is not crucial.
- **Tier 2**: Tier 2 centers have some redundancy, including a backup power source, but still rely on a single cooling path. While more reliable than Tier 1, they are still vulnerable to single points of failure, making them suitable for less demanding workloads.
- **Tier 3**: These data centers are designed with multiple independent power and cooling paths, ensuring continuous operation during maintenance or component failure. They provide higher reliability and are typically used for mission-critical operations that require higher uptime.
- **Tier 4**: The highest level of redundancy, Tier 4 data centers offer fully independent power and cooling systems, designed to guarantee no downtime. These centers are used for the most critical applications, where uptime is non-negotiable, offering 99.999% availability.

5.1.2 Size and Power Demand of Data Centers

Data centers vary significantly in size and the amount of power they require. The power consumption of a data center is typically measured in megawatts (MW), and the required power is dependent on the scale and the applications they support. The different categories based on power needs include [6]:

- ***1-5 MW:*** These smaller data centers typically serve regional needs or less intensive computing tasks. They may support local businesses, smaller cloud services, or specific workloads like small-scale AI or web hosting. They are often more flexible and easier to deploy in various locations.
- ***5-20 MW:*** Medium-sized data centers often support larger enterprises or cloud providers offering services to multiple customers. These data centers can handle more intensive workloads, such as big data analytics, large-scale SaaS offerings, or enterprise IT infrastructures. They represent a significant investment in both space and power infrastructure but are not as massive as hyperscale data centers.
- ***> 20 MW (Hyperscale):*** Hyperscale data centers are the largest and most power intensive. These centers, typically built by major technology companies like Google, Amazon, and Microsoft, can exceed 100 MW in power capacity. They are designed to handle massive amounts of data, support large-scale cloud computing, and provide high-density computing resources. Hyperscale facilities are a vital component of the backbone of the digital economy, supporting billions of users and millions of devices worldwide.

5.1.3 Future Modularity of Data Centers

The future of data centers lies in their ability to scale quickly and efficiently [11]. Modularity is becoming an increasingly important design principle, enabling data centers to be deployed and expanded with flexibility. Modular data centers use standardized, prefabricated units, such as modular servers, power systems, and cooling units, that can be added or removed as needed. This allows for more efficient scaling in line with growing demand without large capital expenditures. In addition, modularity supports the decentralization of data storage, a key element of edge computing, where smaller data centers are deployed closer to end-users to reduce latency and improve service reliability.

As technology evolves, modular data centers will become more adaptable, enabling rapid deployment in different locations, and allowing for easy upgrades in terms of computing power and energy efficiency. The rise of SMRs could further complement the modularity of data centers by providing a reliable, scalable, and low-carbon energy source, making it possible to power large-scale facilities while reducing the environmental impact of energy consumption.

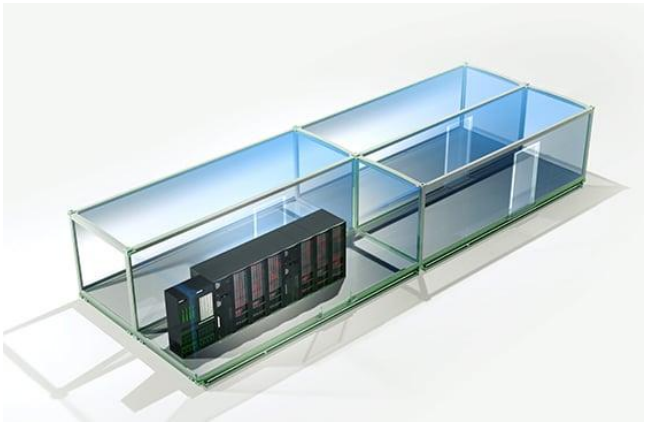


Figure 3. Modularity of Data Centers

5.1.4 Energy Demand Growth and Real-World Data

The energy demand of data centers has seen exponential growth over the past decade. Current estimates indicate that data centers account for around 3% of global electricity consumption, a figure that is expected to rise dramatically in the coming years [1]. According to the International Energy Agency (IEA), the demand for data center electricity will increase by up to 50% by 2030, driven by the expanding use of cloud services, artificial intelligence, and data storage [12].

For instance, Google reported in 2020 that their data centers consume around 7.4 TWh of electricity annually, roughly equivalent to the energy usage of a medium-sized country [13]. Similarly, Amazon Web Services (AWS) and Microsoft Azure are two of the largest cloud providers, both of which operate hyperscale data centers that require multiple gigawatts of power to maintain operations. As data traffic continues to grow, these companies have committed to increasing the efficiency of their data centers and transitioning to renewable energy sources. Despite these efforts, the challenge of balancing energy demand with sustainability remains a pressing issue for the industry.

The following chart shows the estimated growth in global electricity consumption by data centers from 2020 to 2030, with the United States and China accounting for the largest share [14].

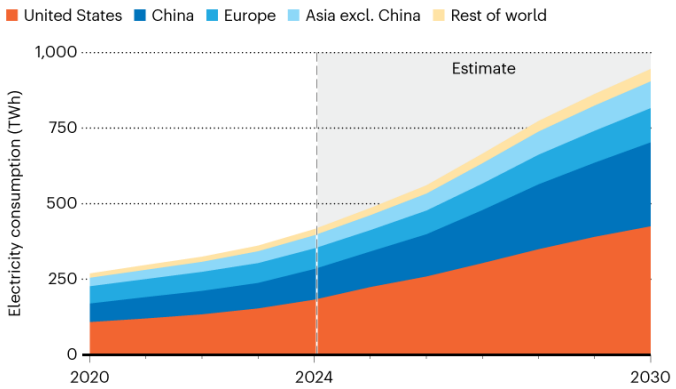


Figure 4. Growth in Electricity Consumptions by Data Centers from 2020 to 2030

5.1.5 Data Center Power Consumption and Distribution

Data centers rely on substantial electricity to keep operations running. A large share of power—often over 50%—is used by cooling systems that maintain safe temperatures and prevent hardware failures [15]. Servers also require constant power for CPUs, memory, and storage devices, especially when processing large datasets. In parallel, networking equipment like routers, switches, and firewalls consume energy to sustain uninterrupted connectivity. Additional loads come from backup power systems (UPS and generators), security systems (cameras, alarms, and access controls), and lighting needed for safe maintenance.

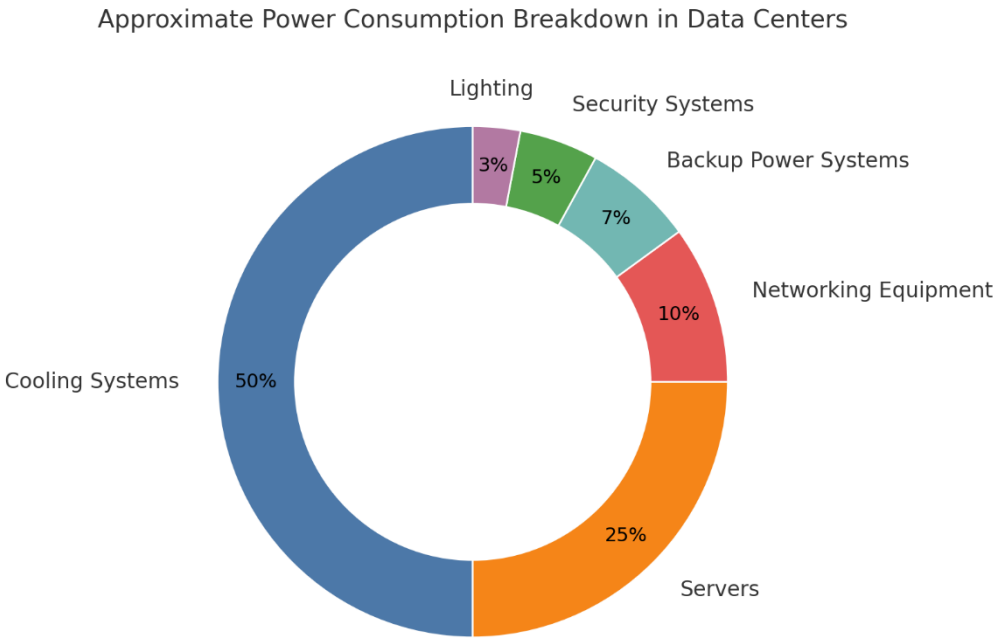


Figure 5. Approximate Power Consumption Breakdown in Data Centers

Delivering this power safely involves a coordinated distribution process. Electricity enters the facility through a utility feed at high voltage (typically 13.8–33 kV), often sourced redundantly from multiple grids. Switchgear manages and routes the incoming power to transformers, which step voltage down to usable levels. From there, uninterruptible power supplies (UPS) condition electricity, protecting equipment from surges and voltage fluctuations. Power distribution units (PDUs) convert and distribute the power to server racks, while remote power panels (RPPs) act as local hubs with protective breakers and fuses. Finally, rack-mounted PDUs deliver electricity directly to the IT equipment inside each server rack [15].

This layered system ensures stable, reliable, and redundant power distribution critical to continuous data center operations.

5.2 Small Modular Reactors (SMRs)

The use of SMRs as a potential energy solution for industries with high and growing power demands, such as data centers, has attracted significant attention in recent years. SMRs are a class of nuclear reactors that are designed to be smaller, safer, and more flexible than traditional large-scale nuclear reactors [16]. These reactors have a modular design that allows for scalability and can be deployed incrementally to meet the specific energy needs of different applications [17]. Their potential to supply clean, reliable, and cost-effective energy makes them an attractive option for industries seeking sustainable alternatives to fossil fuel-based power sources.

5.2.1 Advantages of SMRs

SMRs offer several key advantages over conventional nuclear reactors and other energy sources:

1. **Safety:** One of the most significant advantages of SMRs is their enhanced safety features. Many SMR designs utilize passive safety systems that do not require active intervention or external power sources to shut down safely in case of an emergency [18]. For instance, the use of natural convection for cooling and self-regulating control mechanisms significantly reduces the risk of overheating or radiation leaks. This safety design makes SMRs particularly attractive for deployment in areas with stringent safety regulations or where large reactors are not feasible.

2. **Modularity and Scalability:** SMRs can be built in modular units, allowing for flexible scaling based on demand [17]. This means that SMRs can be deployed incrementally, with each unit providing power as needed. This modularity also allows for easier financing and more cost-effective construction, as smaller units can be deployed and tested before expanding the system.

3. **Small Footprint and Flexibility in Location:** Unlike large nuclear reactors, which require vast areas for construction and are often situated far from consumption points, SMRs have a smaller physical footprint and can be deployed near energy consumers [19]. This flexibility in location makes SMRs ideal for powering decentralized systems, such as remote communities or energy-intensive industries like data centers.

4. **Reduced Capital Costs and Shorter Construction Timelines:** Due to their smaller size and modular construction, SMRs are typically less expensive to build and can be completed faster than traditional reactors [20]. This reduces the financial risks associated with nuclear power projects and makes them more accessible to a broader range of potential investors and operators.

5. **Low Carbon Emissions:** Like all nuclear reactors, SMRs produce no direct greenhouse gas emissions during operation [21]. This makes them a viable option for reducing the carbon footprint of high-energy industries, particularly as the world seeks to transition away from fossil fuels and meet climate goals.

5.2.2 Disadvantages of SMRs

While SMRs offer many advantages, they also face certain challenges and limitations when compared to traditional nuclear reactors and other energy sources:

1. **Economies of Scale:** While the modular design of SMRs allows for flexibility, they may not achieve the same economies of scale as larger reactors [22]. This could make the cost per unit of energy higher than conventional nuclear plants when producing electricity at large scales, potentially limiting their competitiveness in markets with low energy demand.
2. **Regulatory Challenges:** The introduction of new nuclear technologies, including SMRs, requires navigating complex regulatory frameworks that vary by country [23]. The licensing process for SMRs may be slower than for conventional reactors, and there may be public resistance to nuclear power, even with the improved safety features of SMRs.
3. **Fuel and Waste Management:** While SMRs can offer advantages in terms of waste management, their fuel needs and the long-term disposal of nuclear waste remain a significant challenge. Some SMR designs require specialized fuel, which could lead to additional supply chain complexities and concerns over long-term storage of nuclear waste [24].
4. **Public Perception:** Nuclear energy, regardless of its advantages, continues to face public skepticism due to concerns about safety, environmental impact, and the legacy of nuclear accidents [25]. SMRs, despite their enhanced safety, may not entirely overcome these challenges.

5.2.3 Types of SMRs

There are several different types of SMR designs, each with distinct operational characteristics and advantages. Some of the most notable SMR designs include:

1. **Water-Cooled SMRs:** These reactors are based on light water as the coolant and, in some designs, also as the neutron moderator. Water-cooled SMRs are an evolution of traditional large nuclear reactors, but with a smaller, modular design [26]. They retain the high reliability of light water reactors while offering the benefits of a smaller footprint, reduced construction costs, and improved safety features. However, they still face challenges with scale, cost competitiveness, and waste management.
2. **Molten Salt Reactors (MSR):** MSRs use molten salt as both coolant and, in some cases, the fuel itself. This allows MSRs to operate at higher temperatures and pressures than water-cooled SMRs, increasing their efficiency [27]. MSRs also have the advantage of being able to process nuclear fuel more efficiently and produce less long-lived nuclear waste. However, they face challenges related to materials compatibility with molten salt and the development of suitable fuel cycles.

3. **High-Temperature Gas Reactors (HTGR):** HTGRs use helium gas as a coolant and graphite as a neutron moderator. They can operate at much higher temperatures than water-cooled SMRs, which increases their efficiency and allows them to produce both electricity and process heat for industrial applications [28]. HTGRs are also inherently safe due to the passive cooling system. However, their development is still in the early stages, and cost competitiveness remains a key challenge.

4. **Liquid Metal Fast Reactors (LMFR):** LMFRs use liquid metals, such as sodium or lead, as coolant. These reactors can operate at very high temperatures, making them suitable for generating electricity or hydrogen production. LMFRs are fast neutron reactors, which can recycle nuclear fuel and reduce nuclear waste. However, the use of liquid metals presents engineering challenges, such as material corrosion and the handling of high-temperature, liquid-metal systems [29].

5.2.4 Secondary applications

Beyond electricity generation, SMRs offer a range of secondary applications that can significantly contribute to decarbonization, energy efficiency, and technological resilience. These applications include district heating, desalination, and industrial cooling—areas where SMRs can provide continuous, low-emission thermal energy in a compact and scalable form.

5.2.4.1 District Heating

District heating involves centralized production and distribution of thermal energy to meet the space heating and hot water needs of residential, commercial, and industrial buildings [30]. Traditional systems often rely on natural gas or coal-fired boilers, making them significant sources of carbon emissions.

SMRs present a low-carbon alternative by providing a constant and controllable heat source, which is ideal for district heating networks. Their scalability allows for deployment in urban centers or smaller communities, with the potential to:

- Reduce fossil fuel dependency and associated emissions.
- Improve energy security, especially in regions with limited access to imported fuels.
- Enable sector coupling by integrating with electricity, transportation (e.g., EV charging hubs), and building systems.
- Recover waste heat from SMR power generation through cogeneration (CHP), boosting overall system efficiency.

Notably, countries like Finland and smart are already exploring SMRs as long-term solutions for decarbonizing their district energy infrastructure [31].

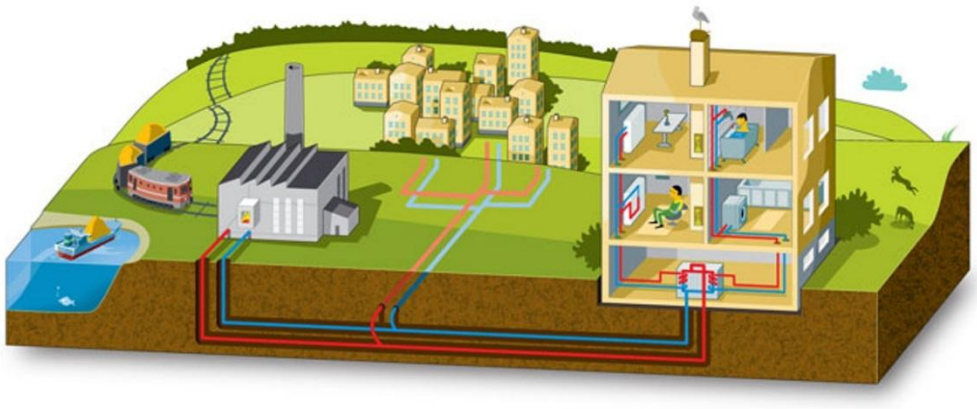


Figure 6. Scheme of District Heating

5.2.4.2 Desalination

Desalination is becoming increasingly important as global freshwater supplies face pressure from climate change, population growth, and industrial demand. However, most desalination technologies are energy-intensive, with high operating costs and carbon footprints when powered by fossil fuels.

SMRs can provide both thermal energy and electricity to drive desalination processes [32]:

- Multi-Effect Distillation and Multi-Stage Flash use low-grade thermal energy to evaporate seawater in multiple stages, significantly improving thermal efficiency.
- Reverse Osmosis is electrically driven and can benefit from the stable, non-intermittent electricity that SMRs provide.

In arid coastal nations or island economies, SMR-powered desalination could provide a self-sufficient and clean water supply, supporting agriculture, tourism, and industrial development.



Figure 7. Water Desalination

5.2.4.3 Cooling

SMRs can also play a critical role in industrial and large-scale cooling, particularly for data centers, which are increasingly energy-hungry due to the global growth in cloud computing and AI workloads.

As data centers strive to improve energy efficiency, liquid cooling technologies have emerged as a powerful alternative to traditional air cooling [33]. SMRs can support these systems by providing the thermal energy required to drive absorption, adsorption, or thermoelectric refrigeration cycles, enabling more sustainable and efficient heat management.

- **Absorption and Adsorption Cooling:** These systems use a heat source—such as low-grade thermal energy from an SMR—to drive refrigeration cycles without relying on compressors. In absorption cooling, refrigerants like water or ammonia are absorbed into a secondary fluid (e.g., lithium bromide) [34], while adsorption systems use solid materials like silica gel. These technologies are ideal for pairing with SMRs, as they make efficient use of continuous thermal output to support round-the-clock cooling.
- **Thermoelectric Cooling:** Using the Peltier effect [35], thermoelectric cooling modules can provide localized cooling for sensitive equipment in edge data centers. While less efficient for large-scale applications, they can benefit from a steady, distributed power supply from SMRs in remote or grid-constrained environments.
- **Hybrid Cooling Strategies:** By combining liquid cooling with thermally driven refrigeration, SMRs enable data centers to minimize reliance on grid electricity for cooling purposes, reduce water consumption, and achieve greater operational resilience.

This integration of SMRs with modern cooling technologies not only supports the sustainability goals of high-tech infrastructure but also enhances reliability, especially in regions with constrained electricity grids or extreme climates.

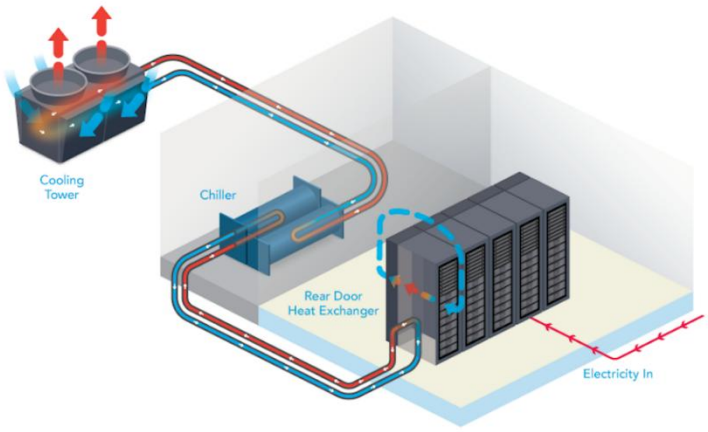


Figure 8. Liquid Cooling in Data Centers

6 SELECTION AND RANKING OF SMRs

6.1 Methodology

In this study, the selection process for identifying the most suitable Small Modular Reactors for data center applications will be conducted in two distinct stages.

Stage 1 involves an initial screening of the over 80 SMR designs available in the IAEA database [36]. The first step will be to review each reactor design based on its technical characteristics, including its energy output, cooling systems, fuel types, and modularity, in line with the specific requirements of data centers. The exclusion criteria will then be applied to eliminate reactors that do not meet the minimum specifications or those that are incompatible with the operational needs of data centers. These criteria include dependence on active safety systems, fuel enrichment and high toxic by-products, technology maturity, installation compatibility in the US and Europe, modular redundancy capability, and power level suitability. This stage will significantly reduce the number of reactors under consideration, narrowing it down to a few viable options.

Stage 2 will focus on selecting the three most suitable SMRs from the remaining pool. In this stage, a more detailed evaluation will take place, using the **Analytic Hierarchy Process (AHP)** to objectively compare and rank the SMR designs []. This analysis will be conducted with SMART (Selection Methodology for Advanced Reactor Technologies), a software developed by *IDOM Consulting, Engineering, Architecture* specifically for the selection of SMRs for different applications. The selection criteria will include factors such as the reactor's energy capacity (specifically matching the energy requirements of data centers), operational safety, technical characteristics and environmental impact. The different designs of SMRs will be carefully compared to understand how they align with the specific needs of data centers, such as reliability, scalability, and sustainability. Additionally, the modularity of the reactors will be considered, as data centers require flexible and scalable energy solutions that can be easily adjusted to meet future growth.

By the end of Stage 2, this process will result in a refined selection of the three most suitable SMRs for powering data centers, providing a comprehensive foundation for further research or implementation.

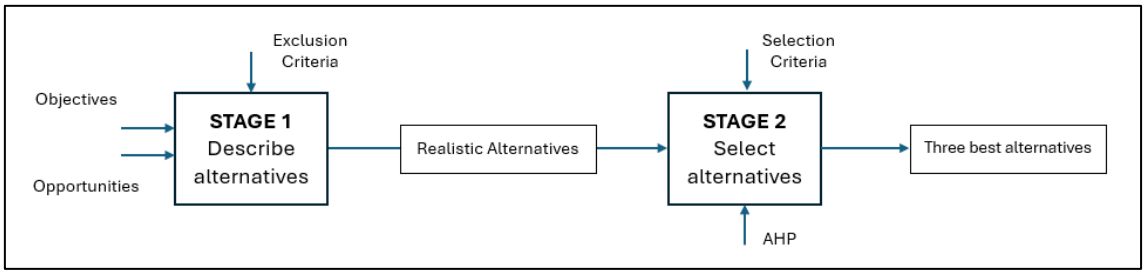


Figure 9. SMR Selection Methodology

6.1.1 Stage 1

As previously mentioned, in this first step, most of the SMR designs that do not meet the necessary requirements to supply data centers will be excluded from this analysis. The reactor data used for this screening were obtained from the IAEA Advanced Reactors Information System (ARIS) [36], ensuring consistency with internationally recognized reactor specifications and development status.

6.1.1.1 Exclusion Criteria

1. Dependence on active safety systems: A key feature of Generation IV reactors is their reliance on passive safety systems or passive shutdown mechanisms, which operate without human intervention and are designed to prevent any release of radioactivity into the environment via air or water. Reactor concepts that do not incorporate this feature or fail to clearly define their safety systems are excluded from this analysis.

2. Fuel Enrichment and Highly Toxic bi-products: Reactor designs must ensure that fuel enrichment remains below 20% of uranium-235 and that no significant highly radioactive bi-products are produced. The limit on uranium enrichment is established by international agencies, such as the IAEA, to enhance safety, security, and non-proliferation objectives [38].

3. Technology is not sufficiently mature: Out of the more than 80 SMR concepts reviewed, only a few have reached a significant Technology Readiness Level (TRL), with most still in the early stages of development and not yet ready for licensing. However, new concepts are continually being introduced, and some of these could advance to the demonstration stage (prototype or licensing) before many of those listed. This selection focuses on those with a high Technology Readiness Level suitable for commissioning.

4. Installation Compatibility in the U.S. and Europe**:** This criterion excludes Russian and Chinese SMR designs due to their low compatibility with installation regulations and standards in the U.S. and Europe [39]. In addition to technical and regulatory challenges, there are significant political considerations. Geopolitical tensions and sanctions can affect technological cooperation and knowledge transfer between these countries and Western nations. Using Russian or Chinese designs could complicate diplomatic and commercial relations, as well as face potential legal and security restrictions. Therefore, it is preferable to select SMRs that are already aligned with Western regulations and standards to ensure smoother integration and avoid political complications.

5. Modular Redundancy Capability**:** The selected SMR must support the installation of multiple modules. This ensures that if one module shuts down or requires a refueling outage, the other modules can continue to operate, providing uninterrupted power supply to the data center. Additionally, the SMR should allow for the selection of the number of modules to be installed, enabling customization to meet the specific power demand of

each data center. This criterion is essential for maintaining high availability and reliability, which are critical for data center operations.

6. Power Level Suitability**:** The selected reactors must operate within a suitable power range to meet the energy demands of data centers. SMRs that operate at very low power levels (less than 20 MWe) are excluded, as they would not be able to provide sufficient energy for large or expanding data centers. Additionally, the SMR should not exceed a maximum power output of 200 MWe to avoid potential operational and management complexities. Ensuring the SMR operates within these power limits is essential for maintaining the efficiency and reliability of data center operations.

6.1.1.2 Feasible Reactors Candidates

The following table presents the SMRs selected for detailed analysis after passing the exclusion criteria. Although the AP-300 does not meet the exclusion criteria, it is also included due to multiple research studies conducted at Illinois Institute of Technology. The table includes key information about each SMR, such as the designing organization, the country of origin, and its electrical power output in electric megawatts (MWe).

	Design organization	***Country***	***MWe***
WATER COOLED SMRs (LAND BASED)			
i-SMR	KHNP & KAERI	Republic of Korea	170
NuScale Power Module	NuScale Power Inc.	USA	77
PWR-20	Last Energy	USA	20
AP-300	Westinghouse	USA	330
HIGH TEMPERATURE GAS COOLED SMRs			
HTMR100	STL Nuclear	South Africa	35
Xe-100	X-Energy LLC	USA	82,5
LIQUID METAL COOLED FAST NEUTRON SPECTRUM SMRs			
ARC-100	ARC Nuclear Canada Inc.	Canada	100
Aurora Powerhouse Product Line	Oklo Inc.	USA	15 - 50
MOLTEN SALT SMRs			
FLEX Reactor	MoltexFLEX	UK	24
IMSR400	Terrestrial Energy Inc.	Canada / USA	195
KP-FHR	Kairos Power	USA	75
Stellarium	Stellaria	France	110

Table 1. Selected SMR designs

The next section provides a detailed description of each of the 12 reactors selected [40]:

- **i-SMR (KHNP and KAERI, Republic of Korea)**

The i-SMR is a Korean light-water modular reactor (LWR) with a power output of 170 MWe, developed by KHNP and KAERI. This versatile reactor is designed for various applications, including electricity generation, industrial heat supply, seawater desalination and hydrogen production. The standard design consists of four modules, and its total electricity output being comparable to that of a traditional fossil fuel plant.

The i-SMR design features a simplified configuration with an integrated Reactor Coolant System (RCS), eliminating the need for large pipes and minimizing the risk of large break loss-of-coolant accidents (LOCA). It operates without boron, improving safety and reducing maintenance, and relies on a fully passive safety system that requires no external power or operator intervention.

The reactor generates 520 MWth, with a 24-month refueling cycle and a maximum fuel rod burn-up of 62,000 MWd/MTU. Safety features include passive emergency core cooling, passive containment cooling, and auxiliary feedwater systems, as well as a multi-level defense approach to prevent accidents. The containment vessel ensures no radioactive releases, while the spent fuel pool cooling system maintains safe fuel temperatures. The design ensures a core damage frequency of less than 10^{-9} per module-year, with an expected availability factor over 95%, and simplifies load-following operations to improve efficiency.

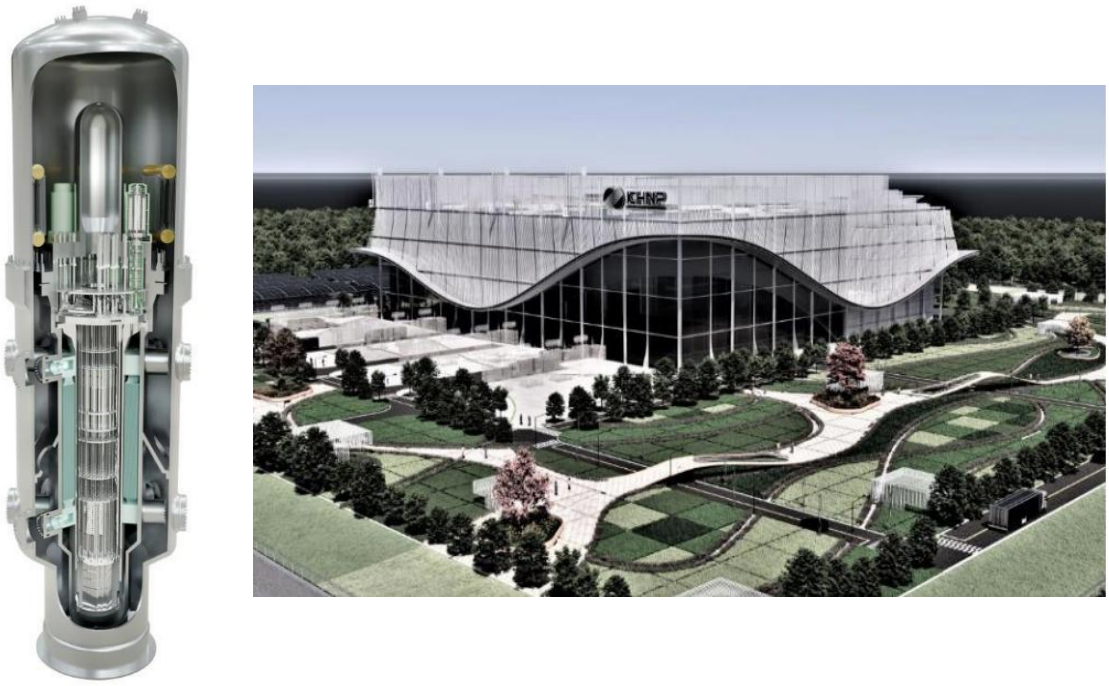


Figure 10. i-SMR

- **NuScale Power Module™ (NuScale Power Inc., United States of America)**

The NuScale Power Module™ (NPM) is a small, pressurized-water reactor (PWR) designed for scalability in SMR plants. NuScale modules are flexible, allowing configurations such as the 4-NPM (308 MW(e)), 6-NPM (462 MW(e)), and 12-NPM (924 MW(e)). Each NPM operates independently within a multi-module setup, controlled from a single control room. The U.S. Nuclear Regulatory Commission (NRC) approved the operation of up to 12 reactors with only 3 operators and no Shift Technical Advisor.

NuScale's design prioritizes passive safety, using natural circulation for core coolant flow and eliminating the need for coolant pumps. The nuclear steam supply system (NSSS) includes a reactor core, helical coil steam generators, and a pressurizer, all housed in a containment vessel submerged in a reactor pool. The reactor core uses 37 fuel assemblies with UO2 fuel and soluble boron for reactivity control. Reactivity is controlled through both soluble boron and 16 control rods, divided into control and shutdown groups.

The design emphasizes safety with a fully passive system that can cool the reactor indefinitely without the need for external power or operator action. The decay heat removal system (DHRS) uses natural circulation to remove heat, while the emergency core cooling system (ECCS) ensures core cooling in the event of a loss-of-coolant accident (LOCA).

NuScale's approach includes robust containment and spent fuel cooling safety, with a used fuel pool providing up to 10 years of storage. The plant operates with a nominal capacity factor of 95%, and refueling takes 10 days.

The design is supported by extensive testing, including a one-third scale prototype facility, and has achieved a Technology Readiness Level (TRL) of 8 for most subsystems. The NRC granted design approval in 2020, making NuScale the first SMR to receive such certification.

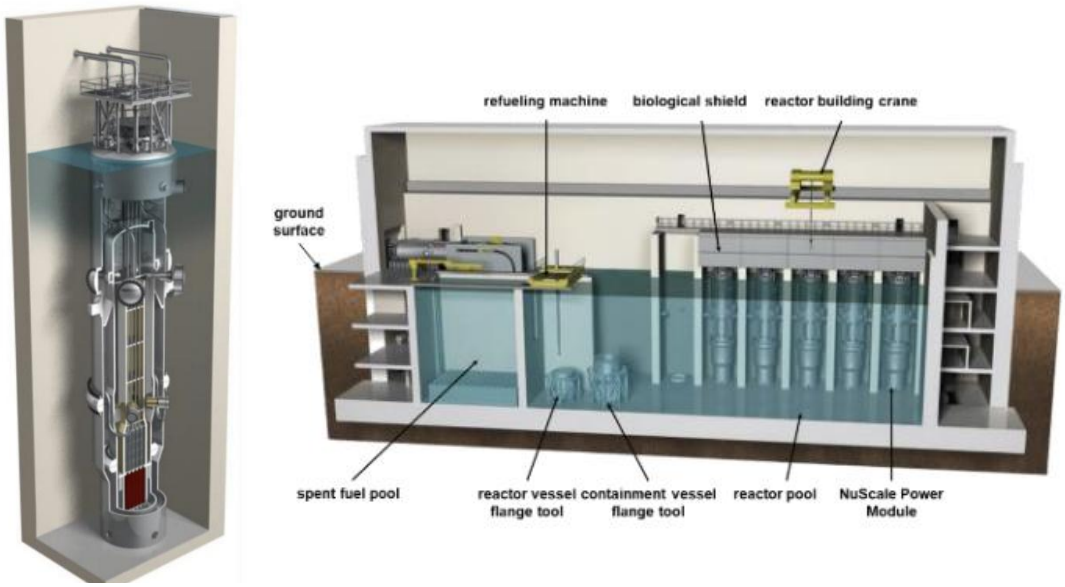


Figure 11. NuScale Power Module

- **PWR-20 (Last Energy, United States of America)**

The Last Energy's PWR-20 is a modular Pressurized Water Reactor (PWR) with a nominal output of 20 MWe, designed for distributed baseload power applications, particularly for industrial and combined heat and power (CHP) scenarios. It emphasizes modularity and standardization, utilizing off-the-shelf components and proven PWR technology to reduce costs, minimize risks, and streamline construction. The reactor's compact design and modular nature allow for flexible deployment in various settings, facilitating rapid installation and minimizing operational disruptions.

The reactor core is organized with a standard 17x17 fuel assembly grid, using low-enrichment uranium fuel to ensure stable and controlled nuclear reactions. The reactivity control system combines control rods and burnable poisons to maintain core stability throughout the fuel cycle. The Nuclear Steam Supply System operates with a single-loop configuration, optimizing thermal management and simplifying heat transfer. A forced-convection cooling system circulates coolant, ensuring efficient heat dissipation, while an air-cooled tertiary loop enhances cooling efficiency.

Safety is a key feature of the PWR-20, with a fully passive safety system that requires no operator intervention during emergencies. The reactor includes multiple safety layers to handle Design Basis Conditions (DBC) and Design Extension Conditions (DEC), ensuring core stability and safety even in extreme conditions. Its containment system consists of a 500-metric-ton iron cask, designed to prevent radioactive leaks, and spent fuel is passively cooled during the six-year refueling cycle.

The PWR-20's design also incorporates advanced automation systems, such as Programmable Logic Controllers (PLC) and Distributed Control Systems (DCS), to optimize plant operations and safety monitoring. The modular construction approach accelerates assembly and reduces space requirements, making the reactor adaptable to different site conditions. With licensing processes underway, the PWR-20 is expected to enter commercial operation of a reference plant in 2025, offering a reliable, cost-effective solution for distributed power generation.



Figure 12. PWR-20

- **HTMR100 (STL Nuclear (Pty) Ltd., South Africa)**

The HTMR100 (High Temperature Modular Reactor) is a 35 MWe pebble bed high temperature gas cooled rector designed by the south African STL Nuclear. This reactor is moderated by graphite and cooled by forced helium flow, and it produces high-quality steam for a turbine system with various applications such as electricity generation, cogeneration, and process heating. Its design is based on proven technology, and its size and fuel cycle simplify design and operation. The modular units ensure continuous and fast production, easy transport, and enhanced licensing process.

The HTMR100's main design features include a top-to-bottom helium flow through the core, collecting heated gas in a hot plenum and transferring it to a helical coil steam generator for power conversion. The reactor core comprises a steel pressure vessel, graphite reflector blocks, and neutron absorber rods, designed for 4 MPa pressure. Fuel elements are 60 mm diameter spheres with TRISO-coated particles, providing a long burn-up cycle. Fresh fuel is loaded via a charge lock, and reactivity is controlled by eighteen neutron absorber rods. The reactor pressure vessel is built to ASME standards, with internal structures ensuring stable core geometry and efficient heat transfer. Helium coolant circulates through the core, with electric blowers and a secondary Rankine cycle for power generation.

It is designed to operate safely without relying on automated systems or operators. It features passive safety systems like the Reactor Cavity Cooling System (RCCS) and thermosiphons for spent fuel cooling. The primary fission product barrier is the SiC layer of the TRISO-coated fuel particles. The design minimizes risks of water and air ingress, ensuring safety under various conditions. The plant aims for high availability of steam and electricity, with excess used for desalination. The Automation System (ATS) includes safety and non-safety systems for protection, control, monitoring, and human-system interfaces.

The reactor building is partially submerged for protection and designed to withstand earthquakes. Underground tunnels connect buildings, keeping spent fuel below ground. The reactor can use various fuel types and cycles, with a waste management plan ensuring safe disposal of spent fuel elements and minimal environmental impact.

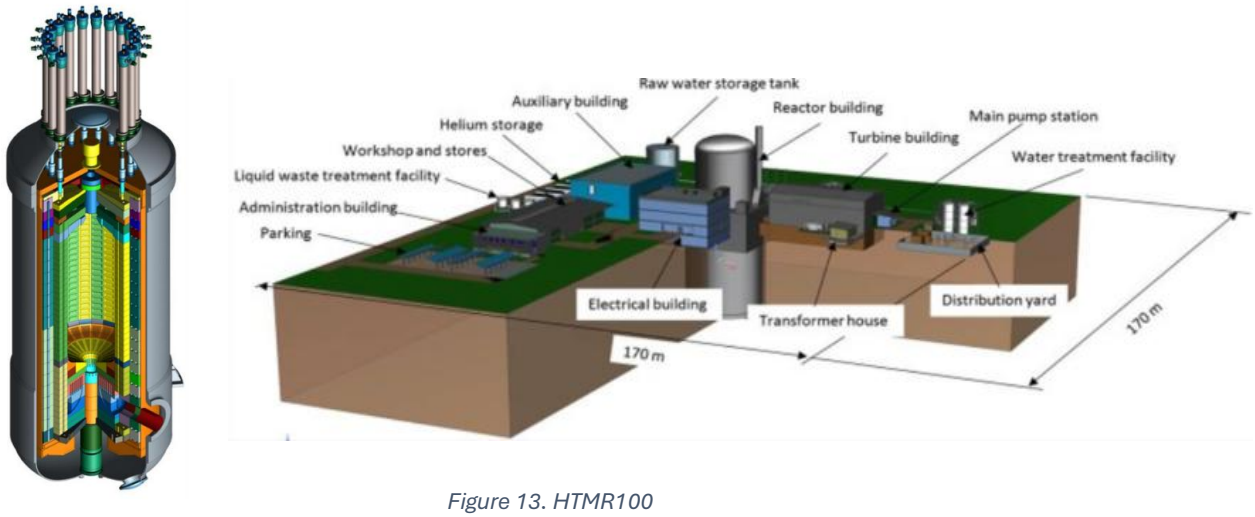


Figure 13. HTMR100

- **Xe-100 (X Energy, United States of America)**

The X Energy's Xe-100 is a pebble bed high temperature gas-cooled reactor with a power output of 82,5 MWe. One of the most distinguishing features is its continuous refueling system with low enriched spheres passing through the core six times to reach high burnups. It is designed for various applications, such as process heating, desalination, electricity, and co-generation.

The Xe-100 reactor core is a graphite pebble fuel element with 220,000 elements, each coated with 18,000 UCO TRISO particles. It is moderated by online refueling to maintain excess reactivity and ensure safe shutdown limits. The core's geometry, power density, heavy metal loading, and enrichment level have been optimized for decay heat removal during severe accident scenarios. The fuel element consists of TRistructural ISOtropic particles embedded in a graphite matrix pebble, enriched at 15.5 wt%. The fuel handling system (FHS) is a closed system with four subsystems, ensuring 100% accountability. The reactor uses a strong negative temperature coefficient for nuclear stability and has a 60-year life, allowing start-up during load-follow operation and Xeon decay effects.

The design has inherent safety features, including non-metallic fuel elements, helium, and graphite core structures. These elements allow for clean helium circuits and plant operations with low contamination. The reactor protection system (RPS) is credited with safety systems, including coated particle fuel elements, core support structures, RPV, and reactor building. Key safety features include a low power density reactor core with high thermal capacity, core meltdown proof, and no core damage frequency. Additionally, multiple independent fission product barriers, such as a filtered and vented reactor building, provide redundancy and diversity. The worst-case accident scenario is the Depressurized Loss of Forced Coolant (DLOFC), where no fuel damage will be experienced.

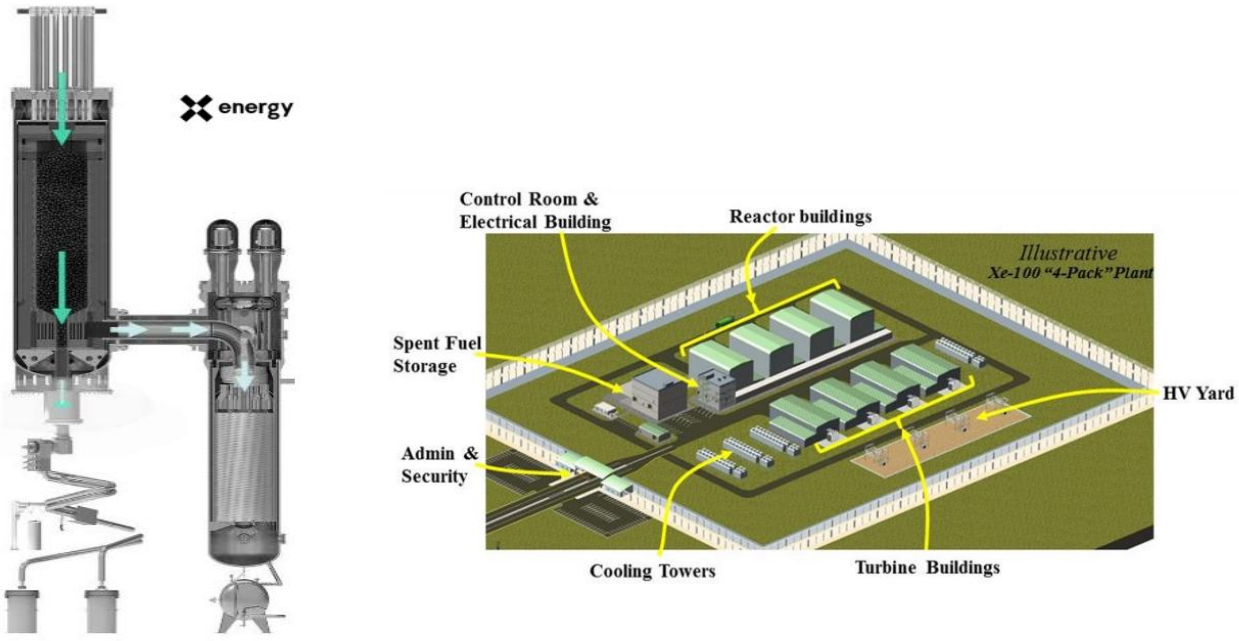


Figure 14. Xe-100

- **ARC-100 (ARC Nuclear Canada Inc., Canada)**

The ARC-100 is an advanced 100 MW(e) sodium-cooled fast reactor developed by ARC Nuclear Canada, based on the proven design of the EBR-II reactor from Argonne National Laboratory in Idaho. It addresses key challenges in nuclear energy by offering affordability, inherent safety, fuel recycling, and long refueling cycles of up to 20 years. Its metallic uranium fuel enhances safety by enabling passive shutdown mechanisms. The reactor targets grid-scale markets in developed nations as well as remote and off-grid locations, industrial heat, hydrogen production, and desalination.

The design prioritizes passive safety features, minimizing the need for operator intervention. It incorporates five layers of defense-in-depth, including a large thermal inertia sodium pool and redundant safety systems. The core consists of uranium-zirconium metallic fuel with an average enrichment of 13.1%, housed in a pool-type reactor vessel with a surrounding guard vessel to contain leaks. The power conversion system uses a sodium-to-steam heat exchanger, generating superheated steam for a 115 MWe air-cooled turbine. Heat removal is managed by passive systems like the Reactor Vessel Auxiliary Cooling System (RVACS) and Direct Reactor Auxiliary Cooling System (DRACS), ensuring decay heat removal without external power.

The ARC-100's ability to recycle its own fuel reduces nuclear waste concerns, and its simplified, low-maintenance design enhances economic viability. With a 60-year design lifetime, it offers a reliable, flexible, and proliferation-resistant clean energy solution, well-suited for a wide range of applications.

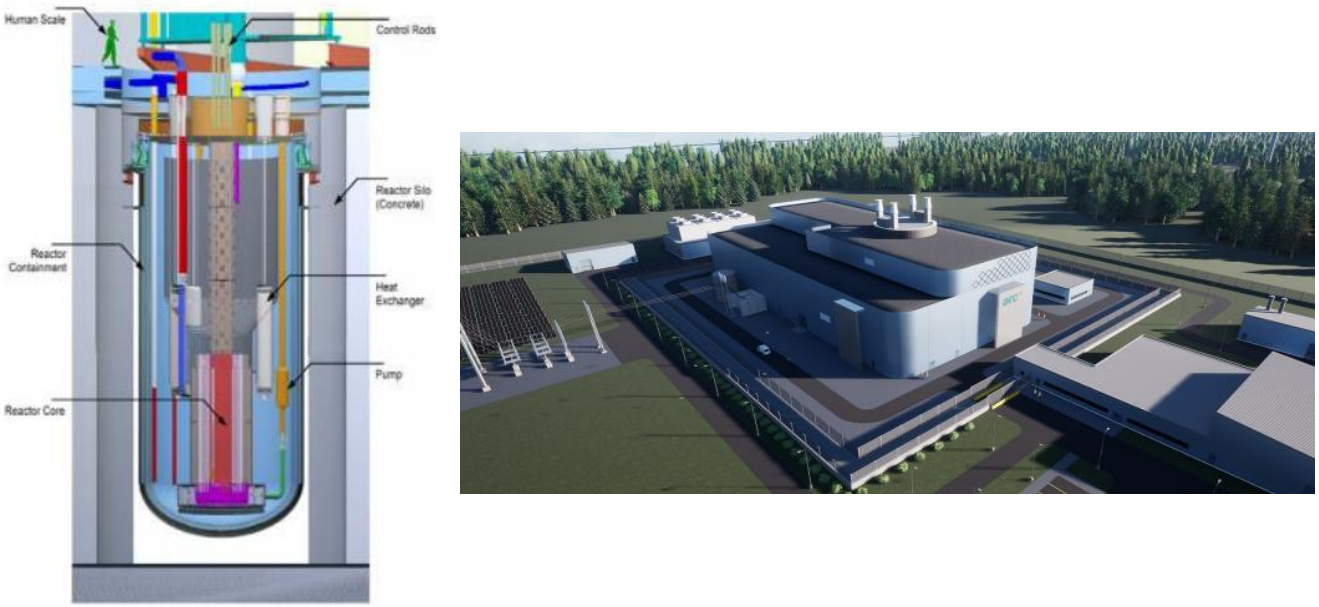


Figure 15. ARC-100

- **FLEX Reactor (MoltexFLEX, United Kingdom)**

The FLEX reactor, developed by MoltexFLEX Ltd., is a molten salt reactor (MSR) designed for cost-effective, clean energy production. It features a patented two-fluid design and is graphite-moderated. Unlike conventional nuclear reactors, the FLEX reactor operates at near-atmospheric pressure, significantly reducing containment costs and improving inherent safety through passive systems.

Designed for modularity, the reactor enables factory production of key components, simplifying transportation and installation while reducing costs and construction time. It builds on existing nuclear technology, minimizing the need for extensive research and accelerating deployment.

Each FLEX reactor produces 60 MWth, with a high-temperature output (795°C) suitable for electricity generation, industrial heat applications, and hydrogen production. The system also integrates with MoltexFLEX's GridReserve® thermal storage, enhancing flexibility and efficiency.

The reactor core uses separate fuel and coolant salts, with fuel contained in vented tubes, reducing contamination risks. Reactivity control is achieved through burnable absorbers and an innovative temperature-driven neutron absorption system. Safety is enhanced by passive heat removal and a design that inherently decreases reactivity in case of core damage.

Fuel consists of low-enriched uranium (LEU) in a molten salt mixture, optimizing performance and safety. The FLEX reactor prioritizes cost reduction, rapid deployment, and global scalability, making it a promising solution for reducing CO_2 emissions.

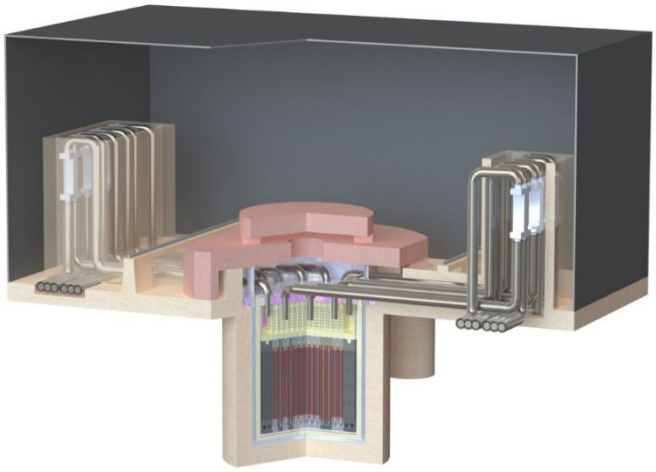




Figure 16. FLEX Reactor

- **IMSR400 (Terrestrial Energy Inc., Canada / United States of America)**

The Integral Molten Salt Reactor (IMSR) by Terrestrial Energy is a Gen IV advanced reactor designed for commercial use in the early 2030s. This thermal-spectrum, graphite-moderated reactor operates at near-atmospheric pressure with a thermal capacity of 884 MW, providing 390 MWe of electrical power or 822 MWth of thermal power at 585°C, with a 56-year plant lifespan. The IMSR uses Low-Enriched Uranium (SALEU) fuel with less than 5% U-235, and molten fluoride salt serves as both fuel and coolant.

The IMSR is designed to provide power for electric grids and industrial applications. Its design is based on the Molten Salt Reactor Experiment (MSRE) conducted at Oak Ridge National Laboratory in the 1950s-70s. The system includes a nuclear steam supply system with a multi-loop primary circuit, secondary and tertiary coolant circuits, and heat exchangers for power generation and process heat.

The reactor features inherent safety attributes, such as a negative reactivity temperature coefficient to prevent accidents, and passive safety mechanisms that do not rely on traditional control rods. Its containment system ensures no release of fission products due to its operation at atmospheric pressure and its use of molten salt for heat dissipation.

With a reactor vessel operating at low pressure and advanced primary pumps based on MSRE designs, the IMSR offers simplified safety and control systems. The plant's design eliminates the need for off-site emergency responses, meeting international safety standards and achieving a 95% capacity factor with minimal outage days. It utilizes passive cooling for spent fuel and integrates redundant systems for high reliability.

Terrestrial Energy continues to test and validate key materials and systems, including fuel salt properties, graphite interactions, and alloy corrosion, ensuring the reactor's readiness for future deployment.



Figure 17. IMSR400

- **KP-FHR (Kairos Power, United States of America)**

Kairos Power's fluoride salt-cooled high-temperature reactor (KP-FHR) is a 75 MWe novel technology that combines TRISO fuel in pebble form with low-pressure fluoride salt coolant. This reactor design features an efficient steam cycle to convert heat from fission into electricity. The KP-FHR is designed to be highly available, low-maintenance, and cost-competitive with natural gas, offering dispatchable power that enhances grid resiliency and security. It aims to complement variable renewable energy sources, contributing to a clean energy system. The company has also received its construction permit for Hermes, a 35 MWth reactor that will demonstrate its capability to produce low-cost nuclear heat.

The reactor's design philosophy combines TRISO particle fuel with molten fluoride salt coolant (Flibe), creating a high-temperature, low-pressure reactor with passive safety systems. This approach reduces reliance on high-cost nuclear components, utilizing conventional technologies to lower capital costs. The reactor features a spherical pebble core with a graphite reflector and uses a reactivity control system that includes control and shutdown elements, which are designed to insert safely in the event of a loss of power. The core operates at low pressure, and the vessel is made of thin-walled stainless steel, ensuring safety and operational efficiency. The fuel used is ceramic, which can withstand extremely high temperatures without damage.

In terms of safety, it incorporates robust features such as a passive decay heat removal system and an emergency core cooling system that is driven by fundamental physics rather than engineered systems. The reactor's design eliminates the need for high-pressure containment structures, with the TRISO fuel providing a functional containment barrier for fission products. The reactor is designed for high safety margins, with the intrinsic safety characteristics of the fuel and coolant simplifying the overall design and reducing the need for extensive safety systems. The system includes a passive auxiliary cooling system that relies on thermal radiation and natural convection for heat removal during accident events, ensuring that cooling is maintained for at least 72 hours without external power.

The KP-FHR operates on a continuous refueling cycle, with fuel pebbles being extracted from the core, inspected, and either reinserted or sent for storage. Used TRISO pebble fuel, along with other waste, is packaged for dry interim storage and eventual disposal or recycling. The waste management approach ensures that all remaining waste qualifies for low-level disposal, aligning with national waste policies.

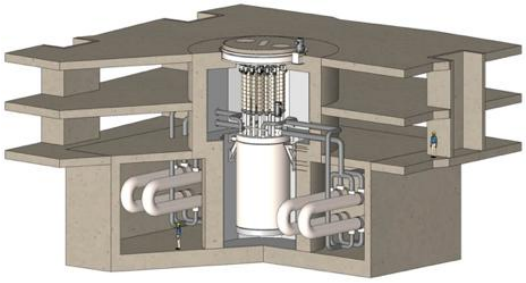


Figure 18. KP-FHR

- **Stellarium (Stellaria, France)**

The Stellarium is a 250 MWth / 110 MWe molten chloride salt fast breeder reactor developed by Stellaria, a spin-off of the French CEA and Schneider Electric. It is designed for energy-intensive industries like mines, refineries, and data centers, helping them decarbonize their energy supply. The reactor uses various fuels such as Uranium, Thorium, Plutonium, and Minor Actinides, contributing to closing the nuclear fuel cycle by continuously breeding and burning these materials.

The Stellarium reactor features a low-pressure design and operates at atmospheric pressure. It uses molten salt as the primary coolant, with heat exchanged through a secondary loop to avoid radiological hazards. The reactor can produce electricity or steam, depending on the application, and uses a supercritical steam turbine for high efficiency (~50%).

Its safety approach includes passive shutdown features, with molten salt's negative reactivity coefficient providing automatic reactivity control in the event of a temperature rise. The reactor is designed to have minimal radioactive release in case of severe accidents and can safely manage decay heat.

Stellarium is intended for industrial applications where it can handle fuel cycles with plutonium and minor actinides. It includes an efficient waste management plan where fission products and minor actinides are separated and reused or disposed of safely in geological repositories.

With robust safety features, flexible fuel cycle options, and efficient heat generation, Stellarium aims to provide a reliable energy solution for sectors requiring large-scale, low-carbon energy.



Figure 19. Stellarium

- **Aurora Powerhouse Product Line (Oklo Inc., United States of America)**

The Aurora reactor, developed by Oklo Inc., is a 15 or 50 MWe sodium fast compact and innovative fast fission reactor which uses metallic fuel. It is designed to provide emission-free, reliable, and cost-effective electricity and heat. Oklo's approach focuses on serving various sectors, including data centers, defense, industrial, manufacturing, real estate, and oil and gas, via power purchase agreements.

One of Aurora's key design features is its inherent safety characteristics, which include low decay heat removal via passive systems, built-in reactivity feedbacks to control power during temperature or overpower events, and multiple barriers to prevent fission product release. The reactor operates at atmospheric pressure and does not require water for safety-related cooling, which minimizes safety concerns. Additionally, the reactor is designed for rapid installation, typically within 18 months, and has a power output license of 40+ years.

The first proposed site for Aurora is at the Idaho National Laboratory (INL) in southeast Idaho, with the site use permit received in 2019 and provisional approval for the specific location in late 2021. The reactor's layout consists of a primary building with supporting structures and radiators located externally. These radiators exchange heat with the ambient air, which serves as the ultimate heat sink for the power cycle. The inherent safety design means that an evacuation zone beyond the site boundary is not required.

Aurora's design is currently in the detailed design phase, with completed fuel prototyping, heat transport testing, and ongoing development of advanced fuel recycling and fabrication technologies. Oklo is the first company to have had a custom combined license accepted by the U.S. Nuclear Regulatory Commission (NRC) for review. The company has been actively collaborating with the NRC since 2016 for both pre-application and review activities.

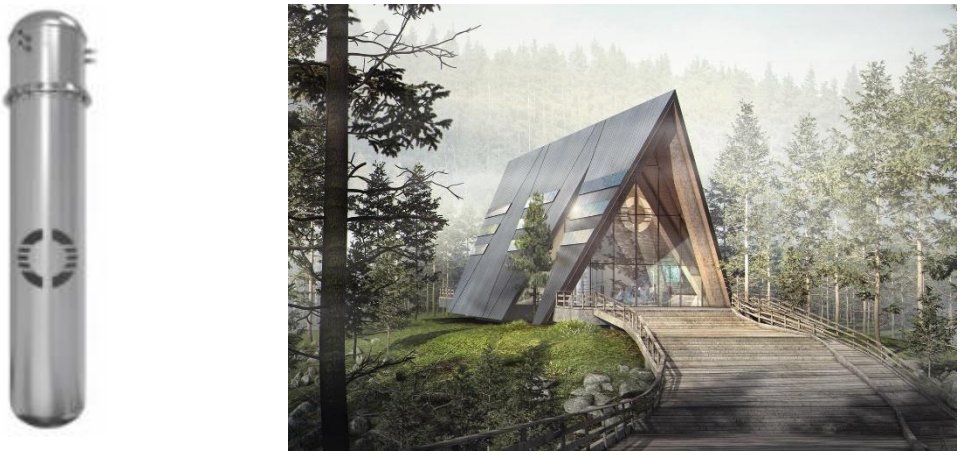


Figure 20. Aurora Powerhouse

- **AP-300 (Westinghouse Electric Company LLC, United States of America)**

The Westinghouse AP300 is a 330 MWe, Generation III+ PWR derived from the well-established AP1000 technology. It features passive safety systems powered by natural forces like gravity and natural circulation, eliminating reliance on active components such as pumps or diesel generators. This design allows the reactor to operate safely without AC power or external support systems for at least 72 hours during emergencies.

The AP300 design builds on the extensive experience gained from AP1000 projects, offering high certainty in construction, operation, and licensing. It uses well-tested components, including the Westinghouse 17x17 Robust Fuel Assembly, and supports long fuel cycles of up to four years. The MSHIM control strategy minimizes chemical control adjustments and enables flexible operation, including load-following capabilities.

Its compact layout reduces building size and material use, while strategically separating safety-related systems to ensure reliability. All safety-critical systems are housed within the Nuclear Island. The plant includes advanced instrumentation and control systems, using the Common Q™ safety platform and a reliable distributed control system architecture.

The AP300 has undergone extensive design validation through testing performed during the AP1000 development and licensing processes. It meets international regulatory standards and is on track to receive design certification by 2027, with the first construction expected around 2030. The fuel design incorporates proven features that enhance performance, safety, and economic efficiency across its operational life.



Figure 21. AP-300

6.1.2 Stage 2

In this second step, the 11 reactors that met the exclusion criteria will be assessed based on 10 selection criteria and 24 subcriteria, carefully defined to address the most relevant characteristics for powering data centers.

In the next figure, the selected criteria and subcriteria are displayed:

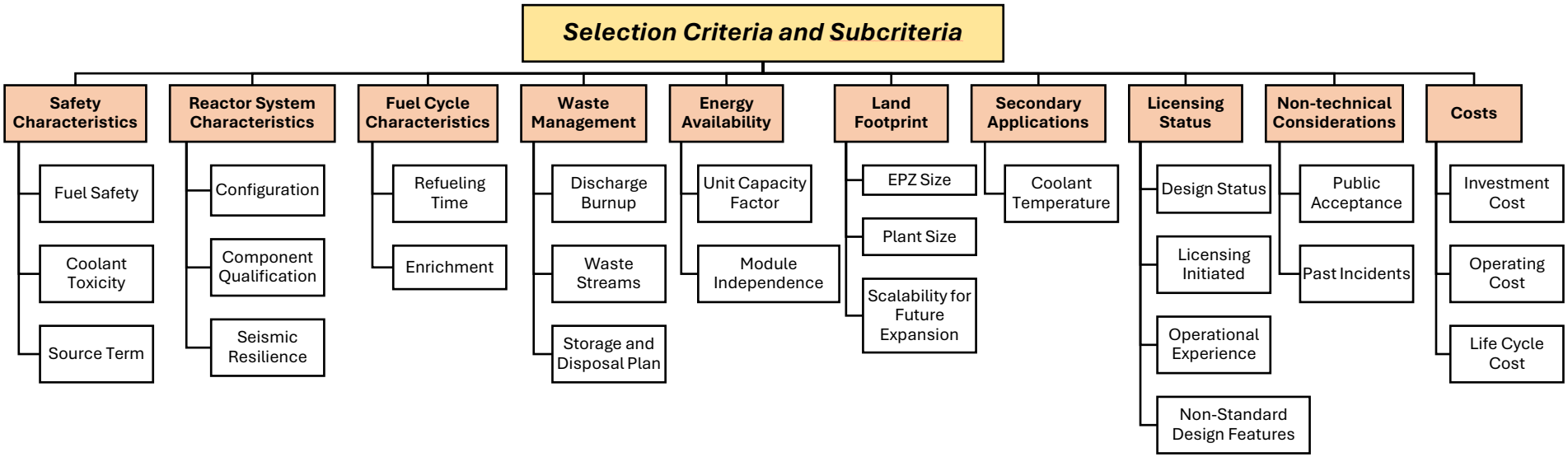


Figure 22. Selection Criteria and Subcriteria

6.1.2.1 Selection Criteria and Subcriteria

In the following section, the mentioned criteria and subcriteria will be briefly described:

1. **Safety Characteristics:** Safety is a fundamental priority in nuclear reactor evaluation, as the potential risks of accidents must be minimized to ensure the protection of human health and the environment. This criterion addresses the reactor's ability to operate reliably and safely under both normal and accidental conditions. It consists of three subcriteria:
    a. **Fuel Safety:** This subcriterion focuses on the robustness of the reactor's nuclear fuel under various conditions. It examines the ability of the fuel to retain its structural integrity during normal operation, transients, and accident scenarios.
    b. **Coolant Toxicity:** The coolant used in a reactor plays a critical role in transferring heat and maintaining operational safety, but its chemical properties can pose risks to personnel, the public, and the environment. This subcriterion evaluates the environmental and health hazards associated with potential leaks or spills of the coolant.
    c. **Source Term:** The source term represents the amount and composition of radioactive material that could potentially be released during an accident. This subcriterion evaluates the type and behavior of radionuclides generated in the reactor and their containment measures.
2. **Reactor System Characteristics:** This criterion evaluates the technical and design aspects of the reactor system, focusing on its engineering aspect. It is refined into three subcriteria also:
    a. **Configuration:** The configuration of the secondary system, whether it includes one or multiple loops (utilizing natural or forced convection) or follows an integral design with the secondary system housed within the reactor vessel, significantly impacts the overall complexity and performance of the system.
    b. **Component Qualification:** This subcriteria considers the qualification process of new materials and components, which could affect the timeline of the project.
    c. **Seismic Resilience:** It measures the reactor's ability to withstand earthquakes, ensuring it remains safe and operational by maintaining structural integrity and core cooling.
3. **Fuel Cycle Characteristics:** This criterion evaluates the various aspects of the nuclear fuel cycle, which includes the refueling time and its enrichment:
    a. **Refueling Time:** This refers to the duration required to replace spent nuclear fuel with fresh fuel in a reactor. Efficient refueling processes can minimize downtime and enhance the overall operation efficiency of the reactor.
    b. **Enrichment:** Enrichment is the process of increasing the proportion of fissile isotopes, such as uranium-235, in nuclear fuel. Higher enrichment

levels can improve fuel efficiency and reactor performance but require advanced technology and handling procedures.

4. **Waste Management:** This criterion assesses the environmental impact of the reactor, focusing on the management and disposal of radioactive waste. It consists of three different subcriteria:
    a. **Discharge Burnup:** Discharge burnup measures the amount of energy extracted from nuclear fuel before it is considered spent. Higher burnups rates indicate more efficient fuel use and less waste generation.
    b. **Waste Streams:** Waste streams refer to the types and quantities of radioactive waste produced during nuclear fuel cycle. Effective management of these waste streams is crucial for minimizing environmental impact.
    c. **Storage and Disposal Plan:** The disposal plan outlines the strategy for safely managing and disposing of radioactive waste, including interim storage, transportation, and final disposal in geological repositories.
5. **Energy Availability:** This criterion evaluates the reliability and efficiency of the reactor in producing energy, and it only consists of one subcriterion:
    a. **Unit Capacity Factor**: The unity capacity factor is the ratio of the actual energy produced by a reactor to the maximum possible energy it could produce over a specific period. High-capacity factors include reliable and efficient reactor operation.
    b. **Module Independence:** This subcriterion assesses whether each reactor module operates independently or is coupled with others. Independent modules are preferred for ensuring reliable, uninterrupted power supply.
6. **Land Footprint:** This criterion examines the physical space required for the reactor and its associated facilities. It is refined in three subcriteria:
    a. **EPZ Size**: The Emergency Planning Zone (EPZ) size refers to the area around a nuclear plant where emergency measures are planned in case of an accident. Smaller EPZ sizes can indicate advanced safety features, lower risk profiles and greater flexibility in sitting the plant.
    b. **Plant Size:** Plant size refers to the physical footprint of the nuclear plant, including all facilities and infrastructure. Smaller plant sizes can reduce land use and environmental impact.
    c. **Scalability for Future Expansion:** Scalability assesses the potential of increasing the plant's capacity or adding additional units to meet future energy demands, which might be a crucial factor for data centers.
7. **Secondary applications**: This criterion evaluates the potential for the reactor to support additional applications beyond electricity generation, such as district heating, desalination or hydrogen production. It consists of only one subcriterion:
    a. **Coolant Temperature:** Coolant outlet temperature affects the feasibility of secondary applications. Higher coolant temperatures can enable more efficient and diverse applications; however, they also pose greater engineering challenges and require more advanced materials.

8. **Licensing Status:** This criterion assesses the progress and readiness of the reactor design in terms of regulatory approval. It is refined in three subcriteria:
    a. **Design Status:** Design status refers to the current stage of the reactor's design, from conceptual to detailed design and even under construction. Advanced design status can reduce development risks and accelerate the licensing process.
    b. **Licensing Initiated:** Licensing initiated indicates whether the process of obtaining regulatory approval for the reactor has begun. Early initiation of licensing can shorten the time to market.
    c. **Operational Experience:** Operational experience refers to the amount of experience gained from operating the same type of reactors, providing valuable insights into safety and performance.
    d. **Non-Standard Design Features:** Non-Standard Design Features assesses whether a reactor includes unconventional design elements that may complicate the licensing process. Standardized designs are moderately more favorable, as they align more easily with existing regulatory frameworks and reduce approval risk.
9. **Non-technical Considerations:** This criterion examines factors that influence the acceptance and feasibility of the reactor beyond technical aspects. While this is an important factor, it will not be considered in the current selection process. The decision of the best SMR for data centers should ideally involve input from data center companies, electric utilities, SMR designers, regulatory bodies and the general public – whose perspectives are not yet available. Additionally, comparing past incidents across reactor types (PWR, HTGR, SFR and MSR) is not feasible, as PWRs have been the predominant reactor type in operation. However, this criterion is explained here for potential inclusion in future evaluations, and it consists of two subcriteria:
    a. **Public Acceptance:** Public acceptance measures the level of support or opposition from the public regarding the deployment of each reactor. Highly public acceptance is crucial for successful implementation.
    b. **Past Incidents:** Past incidents refer to historical events or accidents involving similar reactors. These incidents can influence public perception and regulatory scrutiny.
10. **Costs:** This last criterion evaluates the economic aspects of the reactor, including initial and ongoing expenses. Cost-related criteria will also be excluded from the initial selection due to lack of sufficiently mature data for reliable estimates. Additionally, cost figures provided by vendors vary significantly, as some report costs for First Of A Kind (FOAK) reactors, while other provide estimates for Nth Of A Kind (NOAK). Further inconsistences arise from differing cost metrics, such as the Levelized Cost of Energy (LCOE), which is commonly used for technology comparisons and policymaking. Given these variations, cost analysis will be addressed in a later stage, once more consistent and well-defined data is available, ensuring accurate assessments and avoiding double counting of cost components.

However, a broader cost analysis will be conducted in the next section, considering SMRs as a whole. It is refined in three different subcriterion:

a. **Investment Cost:** Investment cost is the initial capital required to construct the reactor and associated facilities. Lower investment costs can improve economic viability of nuclear projects.
b. **Operating Cost:** Operating cost refers to the ongoing expenses of running and maintaining the reactor. Efficient operation and cost management are key to long-term sustainability.
c. **Life Cycle Cost**: Life cycle cost is the total cost of the reactor over its entire lifespan, including construction, operation, maintenance and decommissioning. It provides a comprehensive assessment of the economic performance of the nuclear project.

6.1.2.2 Weighting of Selection Subcriteria

Since each subcriteria of all criteria have a different importance level regarding powering the data centers, there must be a weighting of these subcriteria to prioritize those that are essentials for this application. First, intensity values will be assigned on the pairwise comparison of the subcriteria within the same criteria, which are displayed in Table 2 [41]. Next, a pairwise comparison of the criteria is made to obtain the weight of each criterion.

Intensity of importance	***Definition***	***Explanation***
1	Equal importance	Two activities contribute equally to the objective
2	Weak or slight importance	Experience and judgment slightly favor one activity over another
3	Moderate importance	
4	Moderate plus importance	Experience and judgment strongly favor one activity over another
5	Strong importance	
6	Strong plus importance	An activity is favored very strongly over another; its dominance demonstrated in practice
7	Very strong importance	
8	Very, very strong importance	The evidence favoring one activity over another is of the possible order of affirmation
9	Extreme importance	
Reciprocals of above	If activity i has one of the above non-zero numbers assigned to it when compared with activity, then j has the reciprocal value when compared with I	

Table 2. Intensity values for pairwise comparison

6.1.2.2.1 Safety Characteristics

As mentioned before, safety is one of the most crucial aspects of nuclear reactors. This criterion is composed of 3 subcriteria: *Fuel Safety, Coolant Toxicity* and *Source Term*. The intensity of each of the subcriteria has been decided by pairwise comparison with the

SMART software (on the scale from 1 to 9 seen in Table 2). Thus, with three subcriteria, three pairwise comparisons were performed:

- *Fuel Safety* is **equally to moderately** more important than *Coolant Toxicity*: while *Coolant Toxicity* is also a concern, fuel-related incidents can have far more severe consequences, potentially leading to a radioactive release and core damage.
- *Fuel Safety* is **moderately to strongly** more important than *Source Term*: while *Source Term*, which refers to the types and amounts of radioactive or hazardous material released to the environment following an accident, is certainly of importance, *Fuel Safety* is a core aspect of SMR design, aiming to prevent such releases from occurring in first place.
- *Coolant Toxicity* is **equally to moderately** more important than *Source Term*: while *Source Term* is related to the types and amounts of radioactive or hazardous material released to the environment following an accident, *Coolant Toxicity* is a constant operational concern with immediate safety implications for the crew, facility, and environment, making it a higher priority in the selection subcriteria.

With these comparisons, the pairwise comparison matrix can be built, as shown in Table 3, and the local weights for each subcriterion are obtained, as shown in Figure 23. Among the Safety Characteristics criteria, *Fuel Safety* ranks as the most significant, contributing 57.14% of total weight. *Coolant Toxicity* follows, making up approximately one-third of the overall score.

	Fuel Safety	Coolant Toxicity	Source Term
Fuel Safety	1.00	2.00	4.00
Coolant Toxicity	0.50	1.00	2.00
Source Term	0.25	0.50	1.00

Table 3. Pairwise comparison matrix of Safety Characteristics

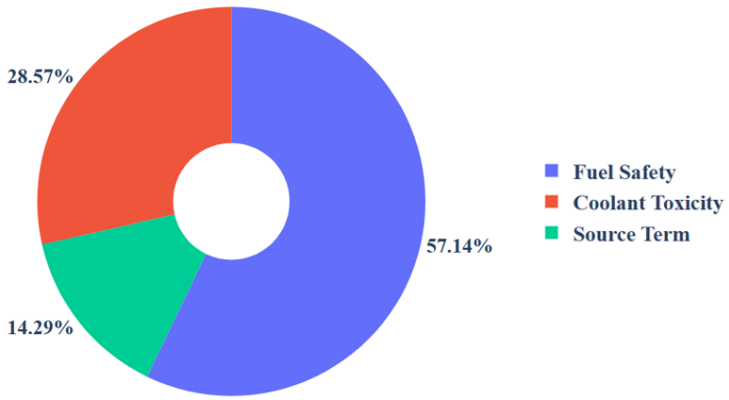


Figure 23. Local Weights for Safety Characteristics' Subcriterions

6.1.2.2.2 Reactor System Characteristics

Assessing and analyzing all aspects of the reactor system itself is essential for ensuring optimal performance. This criterion is composed of 3 subcriteria: *Configuration, Component Qualification* and *Seismic Resilience*. The intensity of each of the subcriteria has been obtained by pairwise comparison with the SMART software (on a scale from 1 to 9 seen in Table X). Therefore, with three subcriteria, three pairwise comparisons were also made:

- *Configuration* is **moderately** more important than *Component Qualification*: *Configuration* directly influences system complexity, performance, and integration with the data center. The choice of loops, convection method, or integral design affects efficiency and safety, making it more critical to overall feasibility. While *Component Qualification* impacts timelines, it does not fundamentally change reactor operation, giving *Configuration* a higher priority.
- *Configuration* is **strongly** more important than *Seismic Resilience*: *Configuration* fundamentally defines the reactor's complexity, efficiency and operational feasibility, directly impacting heat transfer, safety and integration with the data center. While *Seismic Resilience* is crucial for structural integrity in earthquake-prone areas, all designs have met a minimum safety standard, making it a less decisive factor. Since *Configuration* affects reactor performance across all deployments, it holds significantly greater importance.
- *Component Qualification* is **moderately** more important than *Seismic Resilience*: Qualification of new materials and components directly impacts the project timeline, regulatory approval, and long-term reliability of the reactor. While *Seismic Resilience* is important for ensuring structural integrity in earthquake-prone areas, all designs have met a minimum safety standard, making it a less critical differentiator. Since *Component Qualification* influences the feasibility and deployment speed of the reactor, it holds moderate priority over Seismic Resilience.

With these comparisons, the pairwise comparison matrix can be built, as shown in Table 4, and the local weights for each subcriteria are obtained, as shown in Figure 24. The most important subcriterion, in the *Reactor System Characteristics* criterion, is *Configuration* accounting for 58.16% of the score, followed by *Component Qualification*, that accounts for 30.9% of the score. The consistency ratio for this pairwise matrix is 0.00376.

	Configuration	**Component Qualification**	**Seismic Resilience**
Configuration	1.00	3.00	5.00
Component Qualification	0.33	1.00	3.00
Seismic Resilience	0.20	0.33	1.00

Table 4. Pairwise comparison matrix of Reactor System Characteristics

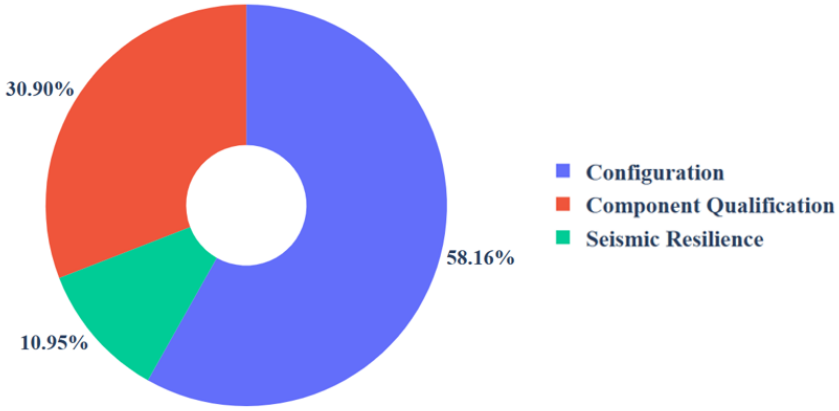


Figure 24. Local Weight for Reactor System Characteristics' Subcriterions

6.1.2.2.3 Fuel Cycle Characteristics

Fuel Cycle Characteristics is a description and evaluation of all characteristics related to the fuel cycle itself. This criterion is composed of 2 subcriteria: *Refueling Time* and *Enrichment*. With two subcriteria, only one pairwise comparison is performed:

- *Refueling Time* is **equally to moderately** more important than *Enrichment*: Being refueling a critical operation and considering that all the reactor designs follow the maximum achievable level of *Enrichment* for civil applications, *Enrichment* is not as restrictive as *Refueling Time*.

With these comparisons, the pairwise comparison matrix can be built as shown in Table 5, and the local weights for each subcriteria are obtained, as shown in Figure 25. The most important subcriterion, in the *Fuel Cycle Characteristics* criterion, is *Refueling Time* accounting for 66.67% of the score, followed by *Enrichment*, that accounts for 33.33% of the score. The consistency ratio for this pairwise matrix is 0, as there are only 2 subcriteria.

	Refueling Time	**Enrichment**
Refueling Time	1.00	2.00
Enrichment	0.50	1.00

Table 5. Pairwise comparison matrix of Fuel Cycle Characteristics

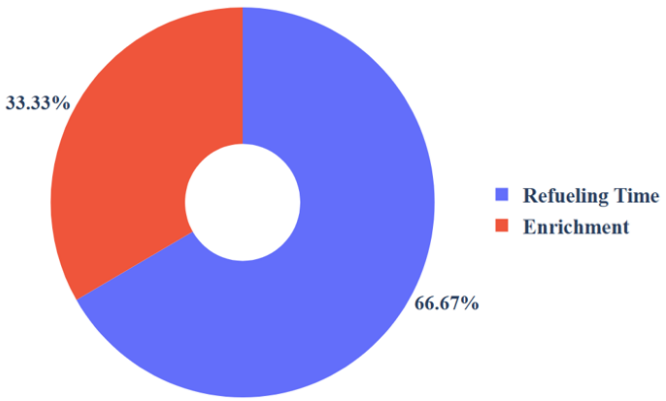


Figure 25. Local Weights for Fuel Cycle Characteristics' Subcriterions

6.1.2.2.4 Waste Management

Characteristics related to Waste Management. Waste is an important criterion because it can be a potential showstopper in certain political and economic scenarios. With three subcriteria, *Discharge Burnup, Waste Streams* and *Storage and Disposal Plan*, three pairwise comparisons are performed:

- *Discharge Burnup* is **moderately** more important than *Waste Streams*: The *Discharge Burnup* directly reduces the amount of spent fuel generated, leading to less waste overall. Higher burnup rates improve fuel efficiency and decrease the need for frequent refueling, simplifying waste management. While *Waste Streams* are important for handling radioactive waste, *Discharge Burnup* plays a more fundamental role in determining how much waste is produced, making it a higher priority.

- *Discharge Burnup* is **moderately to strongly** more important than *Storage and Disposal Plan*: The *Discharge Burnup* directly reduces the amount of spent fuel and radioactive waste that needs to be managed. Higher burnup rates lead to greater fuel efficiency, decreasing the volume of waste and easing long-term disposal challenges. While the *Storage and Disposal Plan* is essential for ensuring safe waste handling, its complexity and cost depend on the quantity of waste produced. Since *Discharge Burnup* minimizes waste at its source, it has a stronger impact on overall waste management.

- *Waste Streams* is **moderately** more important than *the Storage and Disposal Plan*: The type and volume of radioactive waste generated dictates the complexity of waste handling and disposal methods. Effective management of *Waste Streams* helps minimize environmental impact and determines the feasibility of different disposal strategies. While the *Disposal Plan* is necessary for final waste containment, its success relies on properly managing Waste Streams first, making them moderately more important.

With these comparisons, the pairwise comparison matrix can be built as shown in Table 6, and the local weights for each subcriteria are obtained, as shown in Figure X. The most important subcriterion, in the *Waste Management* criterion, is *Discharge Burnup* accounting for 53.96% of the score, followed by *Waste Streams*, that accounts for 29.7% of the score. The consistency ratio for this pairwise matrix is 0.00936.

	Discharge Burnup	**Waste Streams**	**Storage and Disposal Plan**
Discharge Burnup	1.00	3.00	4.00
Waste Streams	0.33	1.00	3.00
Storage and Disposal Plan	0.25	0.33	1.00

6. Pairwise comparison matrix of Waste Management

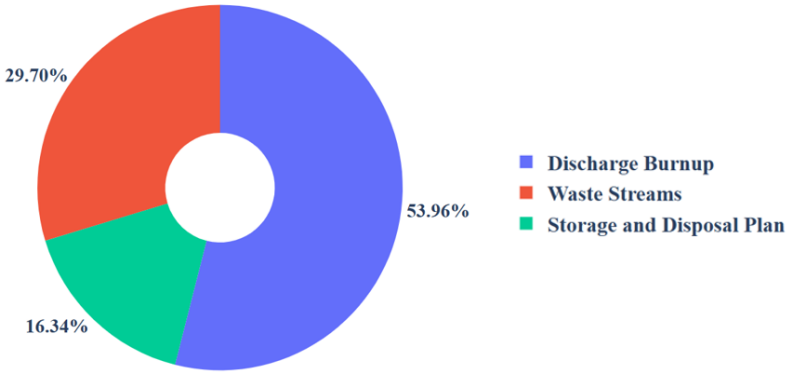


Figure 26. Local Weights for Waste Management's Subcriterions

6.1.2.2.5 Energy Availability

Energy availability is crucial in a data center to ensure continuous operation, prevent downtime, and maintain optimal performance for critical systems. In this case, with two subcriteria, only one pairwise comparison is needed:

- *Unit Capacity Factor* is **moderately** more important than *Module Independence*: The *Unit Capacity Factor* directly reflects a reactor's ability to deliver reliable, continuous power over time, which is crucial for meeting the constant energy demands of data centers. While *Module Independence* enhances system resilience by allowing isolated operation of modules, it is ultimately a supporting feature. Without a consistently high capacity factor, modular independence offers limited value, making overall power performance the more decisive factor.

With these comparisons, the pairwise comparison matrix can be built as shown in Table X, and the local weights for each subcriteria are obtained, as shown in Figure X. The most important subcriterion, in the *Energy Availability* criterion, is the *Unit Capacity Factor* accounting for 75% of the score, followed by *Module Independence*, that accounts for 25% of the score. The consistency ratio for this pairwise matrix is 0, as there are only 2 subcriteria.

	Unit Capacity Factor	**Module Independence**
Unit Capacity Factor	1.00	2.00
Module Independence	0.50	1.00

Table 7. Pairwise comparison matrix of Energy Availability

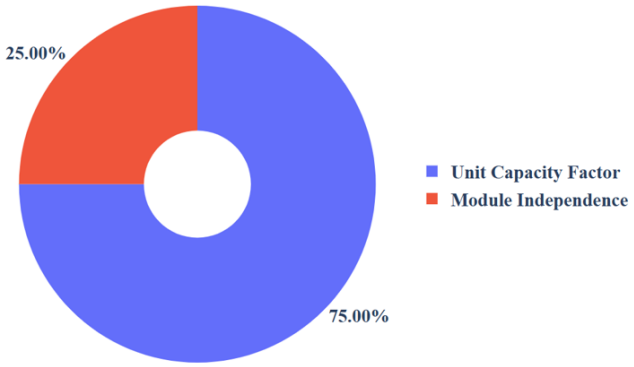


Figure 27. Local Weights for Energy Availability's Subcriterions

6.1.2.2.6 Land Footprint

The amount of land required for a data center is a critical factor, as it impacts site selection, construction costs, and overall feasibility. With three subcriteria, *EPZ Size, Plant Size* and *Scalability for Future Expansion*, three pairwise comparisons are performed:

- *EPZ Size* is **moderately** more important than *Plant Size*: The *EPZ Size* directly influences regulatory approval, public perception, and overall site flexibility. A smaller EPZ allows for more sitting options, reduces the complexity of emergency planning, and minimizes disruptions to surrounding areas. While *Plant Size* is important for optimizing land use, EPZ restrictions can impose significant limitations on where a facility can be built, making it a more critical consideration in site selection.

- *Scalability for Future Expansion* is **moderately** more important than *EPZ Size*: Scalability *for Future Expansion* holds greater importance than *EPZ Size* because data centers require long-term energy planning and the ability to adapt to increasing demand. While a smaller EPZ can facilitate sitting and regulatory approvals, the ability to scale up the plant's capacity or add additional units ensures long-term operational viability. A plant with limited scalability may face challenges in meeting future energy needs, making expansion potential a key factor in sustainability and strategic planning.

- *Scalability for Future Expansion* is **strongly** more important than *the Plant Size*: The ability to increase power generation directly impacts a data center's long-term sustainability and competitiveness. While minimizing *Plant Size* is beneficial for reducing land use and environmental impact, an insufficient scalable plant could severely limit future growth and require alternative energy sources. Ensuring that the plant can accommodate future expansion is essential for maintaining continuous and cost-effective power supply, making it a far more critical consideration than physical footprint of the facility.

With these comparisons, the pairwise comparison matrix can be built as shown in Table X, and the local weights for each subcriteria are obtained, as shown in Figure X. The most important subcriterion, in the *Land Footprint* criterion, is *Scalability for Future Expansion* accounting for 63.7% of the score, followed by *EPZ Size*, that accounts for 25.83% of the score. The consistency ratio for this pairwise matrix is 0.03919.

	EPZ Size	Plant Size	Scalability for Future Expansion
EPZ Size	1.00	3.00	0.33
Plant Size	0.33	1.00	0.20
Scalability for Future Expansion	3.00	5.00	1.00

Table 8. Pairwise comparison matrix of Land Footprint

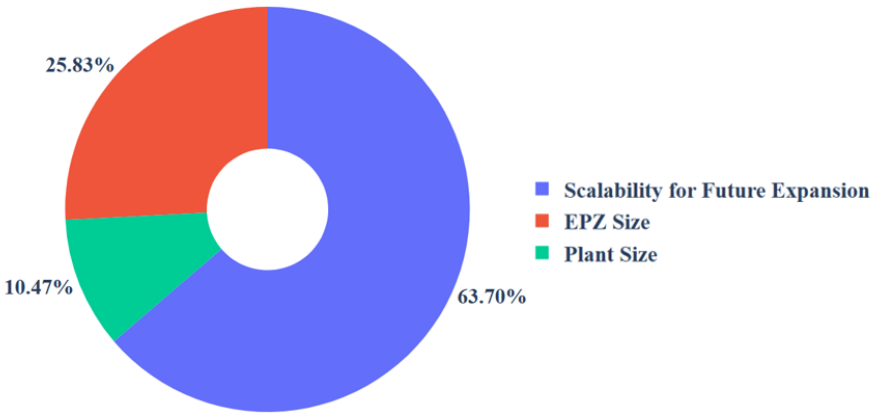


Figure 28. Local Weights for Land Footprint's Subcriterions

6.1.2.5.7 Secondary Applications

Secondary applications might play a key role in a data center by enhancing energy efficiency and optimizing resource utilization. In this case, since there is only one subcriterion, which is the *Coolant Temperature*, no pairwise comparison is needed.

6.1.2.5.8 Licensing Status

The goal of this criterion is to avoid selecting reactor technologies that may encounter approval challenges or have deployment timelines that extend too far into the future. This criterion is composed of four subcriteria: *Design Status, Licensing Initiated*, *Operational Experience* and *Non-Standard Design Features*, and therefore, six pairwise comparisons are performed:

- *Design Status* is **equally** important as *Licensing Initiated*: Both factors play a crucial role in role in determining how quickly and smoothly a reactor can move forward deployment. A well-advanced design reduces technical uncertainties and streamlines regulatory approval, while early initiation of the licensing process

ensures that regulatory hurdles are addressed in a timely manner. Since both elements contribute significantly to reducing development risks and accelerating commercialization, they hold equal importance in the decision-making process.

- *Operational Experience* is **equally to moderately** more important than *Design Status:* Proven performance and safety records provide confidence in the reactor's reliability and regulatory acceptance. While an advanced *Design Status* is crucial for expediting deployment, a lack of real-world operational data can introduce uncertainties in performance and safety evaluations. *Operational Experience* demonstrates that technology has been tested in real conditions, reducing risks and increasing the likelihood of smooth licensing and commercial success.

- *Design Status* is **equally to moderately** more important than *Non-Standard Design Features:* A reactor's level of design maturity plays a direct role in determining its readiness for deployment and licensing. Advanced design status reflects that major engineering challenges have already been addressed, reducing technical uncertainties. While non-standard features can complicate the approval process, a well-developed design—whether conventional or innovative—demonstrates a higher degree of feasibility. Therefore, although both factors affect licensing complexity, the overall progress of the design is slightly more critical in assessing a reactor's deployment potential.

- *Operational Experience* is **equally to moderately** more important than *Licensing Initiated*: Real-world data on reactor performance and safety can significantly influence regulatory approval and market acceptance. While starting the licensing process early can reduce delays, it does not necessarily guarantee approval if the reactor lacks a demonstrated track record. A technology with extensive *Operational Experience* is more likely to gain regulatory confidence, ensuring that a more predictable and efficient licensing process compared to one that is still unproven.

- *Licensing Initiated* is **equally to moderately** more important than *Non-Standard Design Features:* Beginning the licensing process signals a formal commitment to regulatory compliance and allows for early identification of potential issues. While reactors with non-standard features may face more challenges in gaining approval, simply having started the licensing process provides a clear pathway for addressing them. Therefore, although non-standard designs introduce complications, taking proactive steps toward licensing is slightly more important for ensuring regulatory progress.

- *Operational Experience* is **moderately** more important than *Non-Standard Design Features*: Real-world operational data provides strong evidence of safety, reliability, and regulatory acceptance. Even if a reactor has unconventional design features, demonstrated experience significantly reduces uncertainty in both performance and licensability. Regulators are more likely to trust a design that has proven itself in practice, regardless of its novelty. As a result, operational experience plays a more decisive role in supporting both licensing success and market adoption.

With these comparisons, the pairwise comparison matrix can be built as shown in Table X, and the local weights for each subcriteria are obtained, as shown in Figure X. The most important subcriterion, in the *Licensing Status* criterion, is *Operational Experience* accounting for X% of the score, followed by *Design Status* and *Licensing Initiated*, that each account for X% of the score.

	Design Status	Licensing Initiated	Operational Experience	Non-Standard Design Features
Design Status	1.00	1.00	0.50	2.00
Licensing Initiated	1.00	1.00	0.50	2.00
Operational Experience	2.00	2.00	1.00	3.00
Non-Standard Design Features	0.50	0.50	0.33	1.00

Table 9. Pairwise comparison matrix of Licensing Status

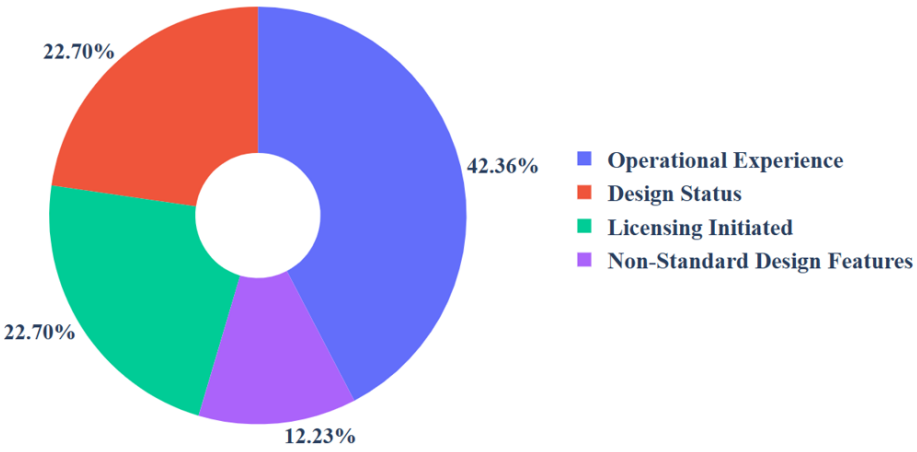


Figure 29. Local Weights for Licensing Status' Subcriterions

6.1.2.3 Weighting of Selection Criteria

The next step involves performing a pairwise comparison of each selection criterion. Given that there are 8 selection criteria, 28 pairwise comparisons must be performed:

- *Safety Characteristics* is **moderately to strongly** more important than *Reactor System Characteristics:* Safety is the highest priority in nuclear reactors, as it ensures the protection of human health and the environment. While the design and engineering of the reactor are essential, they must primarily support the goal of maintaining safe and reliable operations.
- *Safety Characteristics* is **moderately** more important than *Fuel Cycle Characteristics:* Although the fuel cycle influences reactor sustainability and efficiency, safety remains the fundamental concern. No matter how optimized the

fuel cycle is, operational risks must be minimized first to ensure the reactor's viability.

- *Safety Characteristics* is **moderately** more important than *Waste Management*: While proper waste management is crucial for long-term environmental safety, reactor safety directly impacts accident prevention and operational reliability, making it a slightly higher priority in overall decision-making.
- *Safety Characteristics* is **equally to moderately** more important than *Energy Availability:* The ability to provide continuous, reliable power is essential for data centers, but if a reactor cannot operate safely, its energy output becomes irrelevant. Safety ensures long-term viability, making it at least as important, if not slightly more so, than energy availability.
- *Safety Characteristics* is **moderately** more important than *Land Footprint:* A compact land footprint is beneficial for sitting flexibility, but safety takes precedence. Ensuring the reactor operates securely under all conditions is more critical than minimizing its physical footprint.
- *Safety Characteristics* is **moderately to strongly** more important than *Secondary Applications:* While secondary applications such as district heating or hydrogen production add value, they must never compromise reactor safety. Ensuring safe and controlled operation remains the foremost priority.
- *Safety Characteristics* is **moderately** more important than *Licensing Status:* Regulatory approval is essential for deployment, but a reactor must first incorporate robust safety features to meet licensing requirements. Without a strong safety foundation, approval is unlikely.
- *Fuel Cycle Characteristics* is **moderately** more important than *Reactor System Characteristics:* The fuel cycle directly affects operational efficiency, sustainability, and waste generation, making it slightly more significant than general reactor system design, which primarily supports operational functions.
- *Waste Management* is **moderately** more important than *Reactor System Characteristics:* Safe and effective radioactive waste disposal is essential to long-term environmental sustainability, placing it above purely technical design aspects in importance.
- *Energy Availability* is **moderately** more important than *Reactor System Characteristics:* The primary goal of the reactor is to provide reliable energy, especially for data centers, where downtime is unacceptable. Design considerations are secondary to ensuring a continuous energy supply.
- *Land Footprint* is **moderately** more important than *Reactor System Characteristics:* A compact reactor design improves deployment feasibility and site selection, making it slightly more important than general reactor system engineering.

- *Reactor System Characteristics* is **equally to moderately** more important than *Secondary Applications:* While secondary applications add value, they depend on a well-designed reactor system to function effectively. A strong engineering foundation must be in place before additional applications can be considered.
- *Licensing Status* is **moderately to strongly** more important than *Reactor System Characteristics:* A reactor's design must align with regulatory requirements to gain approval. Without compliance, even the most advanced design cannot be deployed.
- *Fuel Cycle Characteristics* is **equally to moderately** more important than *Waste Management:* An optimized fuel cycle helps reduce waste production, making it slightly more critical than waste disposal alone.
- *Energy Availability* is **equally to moderately** more important than *Fuel Cycle Characteristics:* A well-optimized fuel cycle supports reliable energy generation, but ensuring continuous power output remains the top priority.
- *Fuel Cycle Characteristics* is **equally to moderately** more important than *Land Footprint:* A highly efficient fuel cycle improves reactor performance and waste management, making it slightly more critical than simply minimizing land usage.
- *Fuel Cycle Characteristics* is **moderately to strongly** more important than *Secondary Applications:* The reactor's core function—fuel management—determines its efficiency, sustainability, and waste production, making it significantly more important than additional applications.
- *Licensing Status* is **equally to moderately** more important than *Fuel Cycle Characteristics:* A reactor must have a well-optimized fuel cycle, but without regulatory approval, it cannot move forward for deployment.
- *Energy Availability* is **moderately** more important than *Waste Management:* While safe waste disposal is essential, the primary function of an SMR for data centers is to provide a stable power supply, making energy availability a higher priority.
- *Waste Management* is **equally to moderately** more important than *Land Footprint:* Proper radioactive waste handling ensures long-term environmental safety, which is slightly more critical than reducing the reactor's physical footprint.
- *Waste Management* is **moderately** more important than *Secondary Applications:* Managing radioactive waste safely is a core regulatory and operational requirement, making it more critical than additional applications.
- *Licensing Status* is **moderately** more important than *Waste Management:* While waste management is a regulatory concern, obtaining licensing approval is the first step before any reactor can be built or operated.

- *Energy Availability* is **moderately** more important than *Land Footprint:* Ensuring that the reactor provides uninterrupted power is far more important than how much land it occupies. Energy supply stability remains a top priority.
- *Energy Availability* is **moderately** more important than *Secondary Applications:* The primary function of an SMR is to generate power reliably. While secondary applications add value, they do not outweigh the need for continuous energy production.
- *Energy Availability* is **equally to moderately** more important than *Licensing Status:* A reactor must be licensed to operate, but if it cannot provide reliable power, it is not a viable option for critical applications like data centers.
- *Land Footprint* is **moderately** more important than *Secondary Applications:* A The physical space requirements of a reactor impact sitting feasibility, making them slightly more important than its potential for supporting additional applications.
- *Licensing Status* is **equally to moderately** more important than *Land Footprint:* While a compact land footprint improves deployment flexibility, regulatory approval remains a fundamental requirement for operation.
- *Licensing Status* is **moderately to strongly** more important than *Secondary Applications:* Without regulatory approval, secondary applications become irrelevant, making licensing a significantly higher priority in the decision-making process.

	Safety Characteristics	Reactor System Characteristics	Fuel Cycle Characteristics	Waste Management	Energy Availability	Land Footprint	Secondary Applications	Licensing Status
Safety Characteristics	1.00	5.00	3.00	4.00	2.00	4.00	5.00	3.00
Reactor System Characteristics	0.20	1.00	0.25	0.33	0.25	0.33	2.00	0.25
Fuel Cycle Characteristics	0.33	4.00	1.00	3.00	0.50	3.00	4.00	0.50
Waste Management	0.25	3.00	0.33	1.00	0.33	2.00	3.00	0.33
Energy Availability	0.50	4.00	2.00	3.00	1.00	3.00	4.00	2.00
Land Footprint	0.25	3.00	0.33	0.50	0.33	1.00	3.00	0.33
Secondary Applications	0.20	0.50	0.25	0.33	0.33	0.33	1.00	0.25
Licensing Status	0.33	4.00	2.00	3.00	0.50	3.00	4.00	1.00

Table 10. Pairwise comparison matrix of Selection Criteria

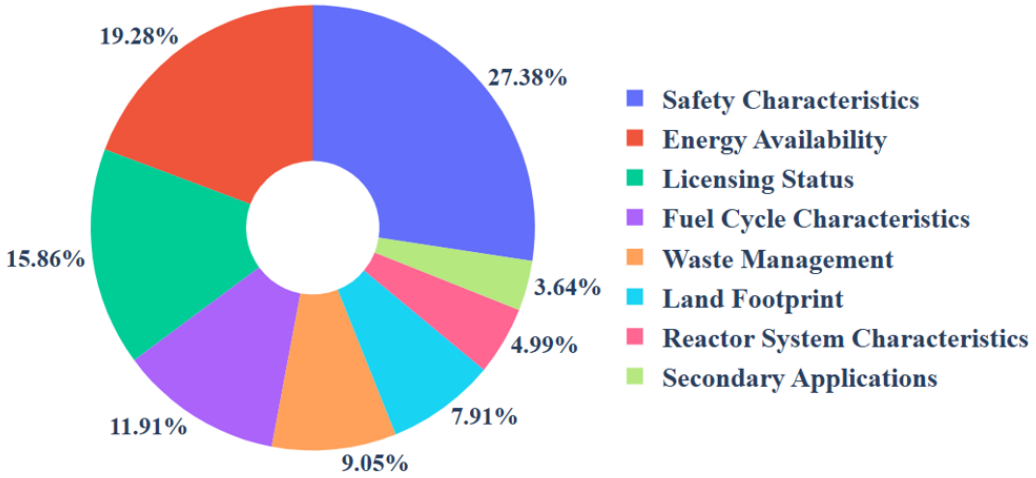


Figure 30. Weight Distribution of Main Criteria

6.1.2.4 Global Weights of Selection Criteria and Subcriteria

After completing the pairwise comparison process, the global weights for each subcriterion can be determined by multiplying the local weight of the subcriterion by the weight of the parent criterion. Figure 31 illustrates the global weights of the criteria (inner ring) and subcriteria (outer ring).

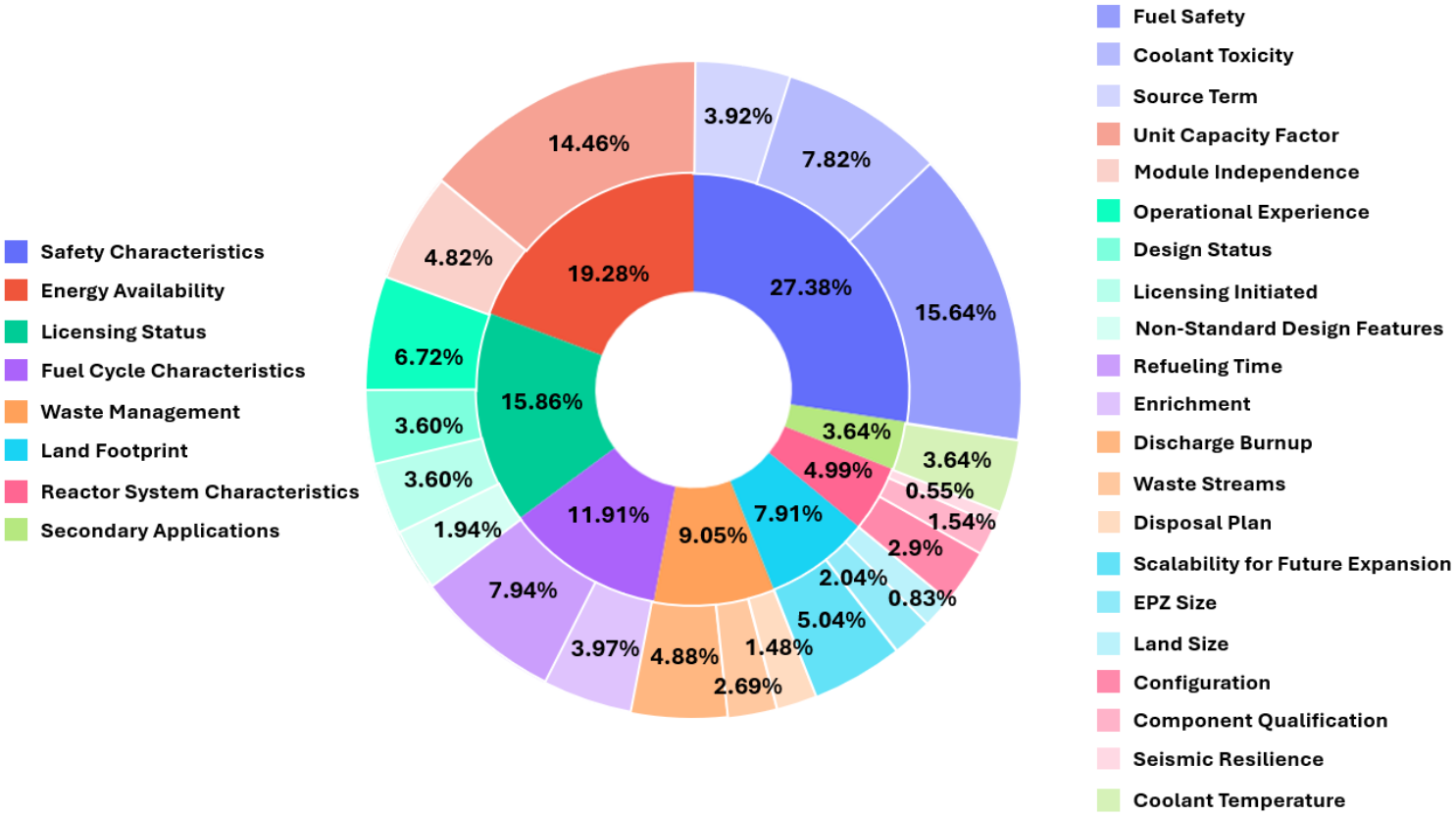


Figure 31. Global Weights of Selection Criteria and Subcriteria

The results highlight that safety, energy availability, and licensing status are the most critical factors in selecting an SMR for data center energy supply. Among them, *Safety Characteristics* hold the highest weight, emphasizing the necessity of a reactor that can operate reliably under both normal and accident scenarios. Within this category, *Fuel Safety* emerges as the most influential subcriterion, reflecting concerns about the integrity of nuclear fuel in various conditions. Data centers require uninterrupted, stable energy, making the *Unit Capacity Factor*—which measures a reactor's ability to deliver power consistently—the second most significant single subcriterion across all categories. This strong preference underlines that energy security and reliability are non-negotiable when considering nuclear power as a solution.

The regulatory readiness of an SMR is also a decisive factor, with licensing status receiving considerable weight. Reactors that are closer to approval or already operational have a clear advantage, as lengthy regulatory processes could delay deployment and undermine their feasibility. *Operational Experience* is another important aspect, as proven designs with a history of performance are preferred over experimental or conceptual models. Closely

linked to this is *Design Status*, which favors reactors that have moved beyond early-stage development, reducing the risk of unforeseen technological or financial obstacles.

Beyond regulatory and safety considerations, *Fuel Cycle Characteristics* play a major role in the assessment, particularly in terms of *Refueling Time* and *Enrichment*. Shorter refueling times mean less downtime, ensuring continuous power delivery to energy-intensive data centers. Lower fuel enrichment is generally preferred due to its cost advantages and the limited number of countries capable of handling highly enriched fuel. While higher enrichment can improve efficiency and reduce fuel consumption, it requires advanced infrastructure and regulatory approvals that may complicate deployment. Similarly, aspects of waste management, such as *Discharge Burnup* and *Waste Streams*, hold moderate weight, indicating an interest in long-term sustainability but not at the expense of core operational priorities.

Other factors, such as *Land Footprint* and *Secondary Applications*, are given relatively lower importance. While the physical size of the plant and its EPZ can influence siting decisions, these considerations are secondary compared to ensuring reliable and safe operation. Likewise, the ability of a reactor to support additional applications, such as hydrogen production or district heating, is viewed as a potential benefit rather than a decisive factor in selection. Similarly, *Reactor System Characteristics*, such as *Seismic Resilience* and *Component Qualification*, while relevant, do not carry as much weight as safety, licensing, and energy output, likely due to modern SMRs already integrating robust engineering measures.

Ultimately, the analysis underscores that data centers require an SMR that is not only safe and regulatory-ready but also capable of delivering uninterrupted power with minimal downtime. While aspects like land use, scalability, and secondary applications may influence final decisions, they remain secondary to fundamental operational requirements. These findings provide a clear direction for selecting the most suitable SMR for this application, ensuring a balance between safety, efficiency, and feasibility in nuclear energy deployment for data centers.

6.1.2.5 Rating Scales of Selection Subcriteria

Unlike a traditional AHP, where the next step would typically involve pairwise comparisons for each subcriterion across all reactor designs, a rating scale approach has been adopted instead. Each reactor is scored individually for each subcriterion. The main reasons for this modification are the uncertainty of the data used and the nature of the subcriteria: some are quantitative and relatively easy to compare, while others are qualitative and inherently more challenging to evaluate.

This approach allows for greater flexibility and consistency in assessing a diverse set of reactors, ensuring that both quantitative and qualitative factors are adequately addressed.

6.1.2.5.1 Safety Characteristics

6.1.2.5.1.1 Fuel Safety

The safety of nuclear fuel in accident scenarios is a crucial factor in SMR evaluation, as different fuel types vary in their ability to retain fission products and withstand extreme conditions. To assess this, a four-tier rating scale is used: *High, Medium, Low,* and *Very Low*, reflecting each fuel's resilience and potential risks.

TRISO fuel is rated *High* due to its exceptional ability to contain fission products within its multi-layered particle structure, even at extreme temperatures, making it the safest option [42]. Molten salt fuels receive a *Medium* rating because they naturally trap fission products within the liquid salt medium, although certain compositions introduce risks, such as potential reactions with air or water [43]. Metallic fuels are classified as *Low*, as they have a lower melting point and higher susceptibility to degradation under extreme conditions, which increases the risk of structural failure [44]. Finally, conventional uranium oxide pellets are rated *Very Low*, as they are prone to cracking, swelling, and mechanical interaction with cladding when exposed to high temperatures, potentially leading to fission product release [45].

The intensity of each of the ratings in the scale (Table 2) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	High	Medium	Low	Very Low
High	1.00	3.00	6.00	8.00
Medium	0.33	1.00	3.00	4.00
Low	0.17	0.33	1.00	3.00
Very Low	0.13	0.25	0.33	1.00

Table 11. Pairwise comparison matrix of Fuel Safety

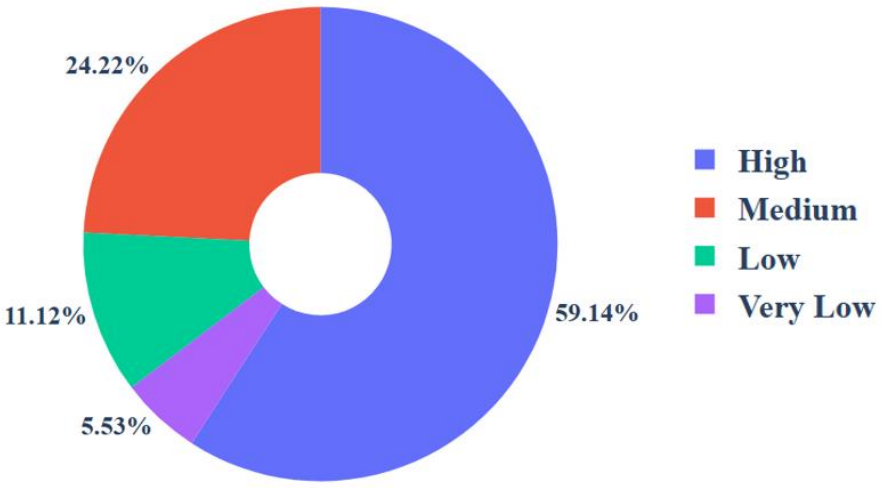


Figure 32. Intensity of each of the Ratings Scale in Fuel Safety

6.1.2.5.1.2 Coolant Toxicity

The toxicity of a reactor's coolant is a critical factor in ensuring worker safety and minimizing environmental risks in case of spills or leaks. Exposure to certain coolants can pose health hazards, making their toxicological properties an important consideration, particularly during Loss of Coolant Accidents (LOCA) or maintenance-related spills. To assess this, a four-tier rating scale is used: *High, Medium, Low,* and *Very Low*.

Molten salts are assigned the *High* toxicity rating due to their complex chemical composition, which can include reactive and corrosive elements such as fluorides or chlorides [46]. In case of leaks, they pose significant chemical hazards, with no specific antidote available, making treatment mainly symptomatic. Sodium is rated *Medium*, as it is highly reactive with water and air, potentially causing violent exothermic reactions that can lead to toxic sodium hydroxide formation and fire hazards [47]. However, exposure effects can be mitigated with appropriate safety measures.

Water, the most widely used coolant, is rated *Low* in toxicity since it is inherently non-hazardous under normal conditions. However, in high-radiation environments, it can become activated, leading to the formation of tritiated water, which, while less dangerous than chemical toxicity, still requires controlled handling [48]. Helium is rated *Very Low*, as it is an inert, non-toxic gas that does not react with other substances, making it the safest coolant option in terms of toxicity [49].

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	High	Medium	Low	Very Low
High	1.00	0.33	0.20	0.14
Medium	3.00	1.00	0.33	0.20
Low	5.00	3.00	1.00	0.33
Very Low	7.00	5.00	3.00	1.00

Table 12. Pairwise comparison matrix of Coolant Toxicity

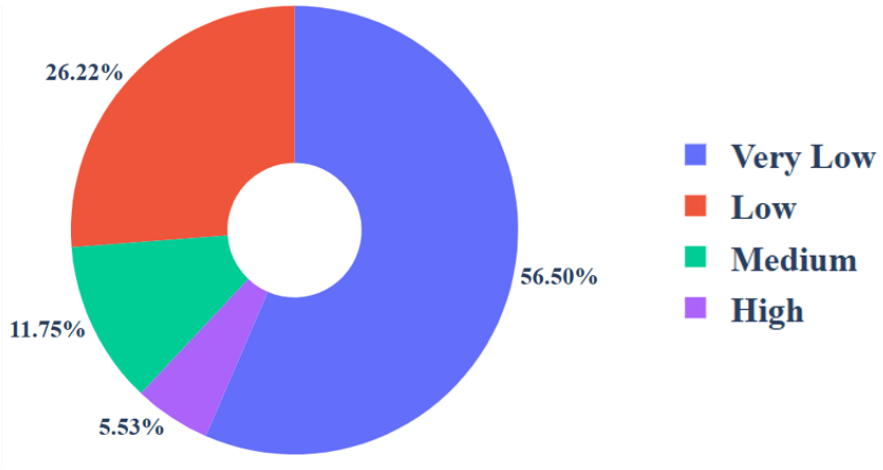


Figure 33. Intensity of each of the Ratings Scale in Coolant Toxicity

6.1.2.5.1.3 Source Term

The source term refers to the types and amounts of radioactive or hazardous substances that could be released into the environment following an accident, directly influencing the radiological consequences of a reactor design. Various factors, such as fuel type, coolant properties, and operating conditions, affect the potential release of fission products in the event of a system failure. Given the varying containment capabilities of different reactor technologies, a four-tier rating scale is used: *High, Medium, Low,* and *Very Low*.

MSRs with flowing fuel salt receive a *High* rating due to the presence of fission products within the liquid fuel itself. In the event of a leak, there is a direct pathway for radioactive release, making these designs more vulnerable to severe contamination risks [50]. MSRs with solid fuel (such as TRISO-based designs) and sodium-cooled reactors are rated *Medium*, as they provide better containment but still carry notable risks. While TRISO particles have superior retention of fission products, MSRs using them still involve a molten salt coolant, which can contribute to source term concerns. Sodium-cooled reactors also fall into this category due to sodium activation and potential for aerosol release if coolant leaks occur [47].

Water-cooled reactors receive a *Low* rating because, despite the potential formation of activated water, the source term is relatively contained within the reactor vessel and primary loop, and water itself does not transport fission products as easily as molten salts [48]. Helium-cooled reactors are rated *Very Low*, as helium is an inert gas that does not become radioactive or contribute to fission product transport [49], making it the safest option in terms of potential radiological release.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	High	Medium	Low	Very Low
High	1.00	0.33	0.17	0.14
Medium	3.00	1.00	0.33	0.20
Low	6.00	4.00	1.00	0.33
Very Low	8.00	6.00	3.00	1.00

Table 13. Pairwise comparison matrix of Source Term

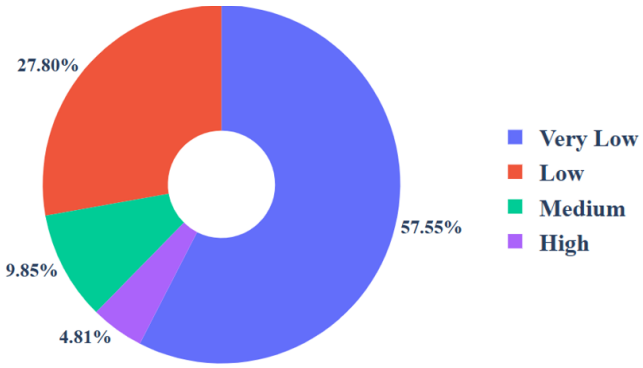


Figure 34. Intensity of each of the Ratings Scale in Source Term

6.1.2.5.2 Reactor System Characteristics

6.1.2.5.2.1 Configuration

This subcriterion evaluates whether the secondary system operates with one or multiple loops (using either natural or forced convection) or follows an integral design, where the secondary system is housed within the reactor vessel.

Integral reactor designs are generally preferred, as they reduce system complexity, minimize the amount of required equipment, and lower the risk of mechanical failure. In contrast, loop-based systems, whether using natural or forced convection, introduce additional components, increasing potential points of failure. While both loop configurations are rated similarly, forced convection is given a slightly higher intensity due to the performance limitations of natural convection [51].

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Integral	Loop with Forced Convection	Loop with Natural Convection
Integral	1.00	8.00	4.00
Loop with Forced Convection	0.13	1.00	0.20
Loop with Natural Convection	0.25	5.00	1.00

Table 14. Pairwise comparison matrix of Configuration

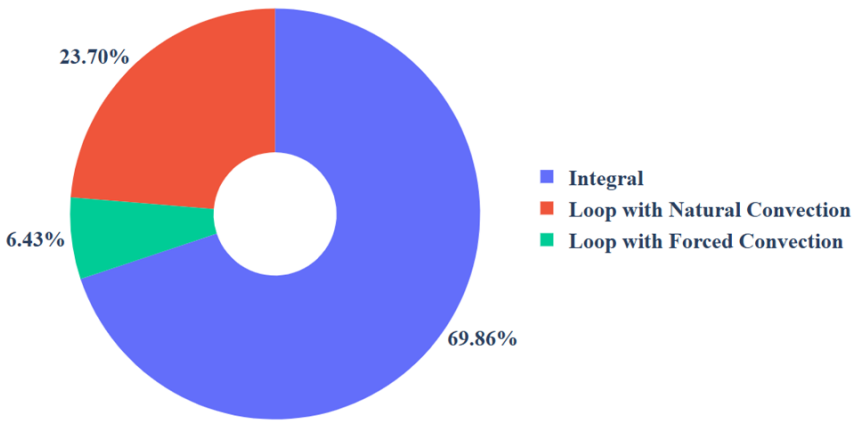


Figure 35. Intensity of each of the Ratings Scale in Configuration

6.1.2.5.2.2 Component Qualification

This subcriterion assesses the need for qualification of new components essential for the proper operation of the nuclear reactor, such as pumps, heat exchangers, and structural materials. The qualification process can introduce delays in deployment, as new materials and systems must undergo rigorous testing to meet regulatory and operational standards. Given the complexity and variability across reactor designs, a three-tier rating scale—*Low, Medium,* and *High*—is applied.

Low-rated designs are those that rely on existing, well-established technologies and materials, minimizing the need for additional qualifications. This includes pressurized water reactors (PWRs) and some compact reactors that use previously certified components, allowing for a smoother regulatory process. *Medium*-rated designs incorporate some new materials or systems that require qualification, though they also make use of already approved technologies where possible. These include reactors that introduce new heat transport mechanisms or innovative thermal storage solutions but still rely on qualified materials for critical components.

The *High* rating is assigned to reactors requiring significant material and component qualification, particularly those that introduce novel cooling systems, advanced heat exchangers, or unconventional structural materials. These designs face the most regulatory and engineering hurdles, potentially leading to longer development and deployment timelines [52].

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	High	Medium	Low
High	1.00	0.33	0.17
Medium	3.00	1.00	0.25
Low	6.00	4.00	1.00

Table 15. Pairwise comparison matrix of Component Qualification

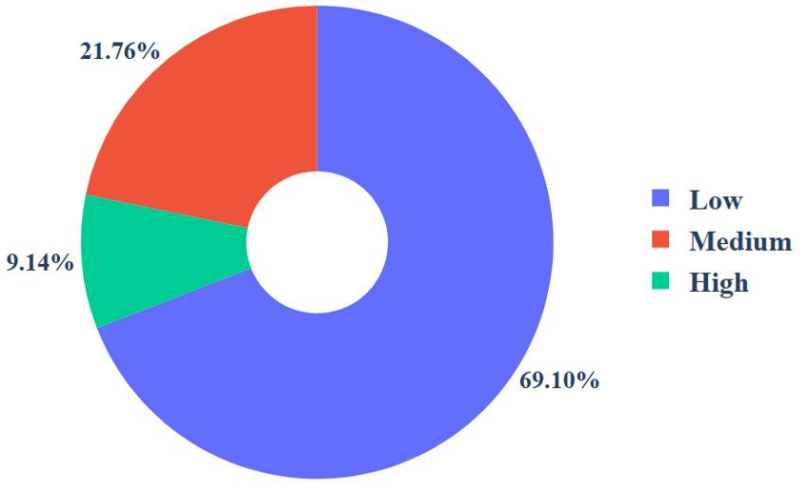


Figure 36. Intensity of each of the Ratings Scale in Component Qualification

6.1.2.5.2.3 Seismic Resilience

Seismic resilience evaluates a reactor's ability to withstand earthquakes while maintaining structural integrity and ensuring core cooling. This is a critical safety consideration, particularly for SMRs intended for deployment in seismically active regions. The classification is based on the Safe Shutdown Earthquake (SSE) value, which represents the maximum ground acceleration a reactor can endure while safely shutting down.

Two rating categories are used: *High* and *Standard*. Reactors classified as *High* have an SSE greater than 0.3 g, indicating enhanced structural reinforcement and seismic isolation technologies that allow them to operate in areas with higher seismic risks. These designs typically incorporate advanced shock-absorbing foundations or passive cooling mechanisms that enhance safety during extreme seismic events [53].

Reactors rated as *Standard* meet the 0.3 g SSE threshold, which aligns with the minimum seismic requirements for most commercial reactors. While still engineered to withstand significant earthquakes, these designs may have fewer structural adaptations for extreme seismic conditions, making them more suitable for locations with moderate or low seismic activity.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	High: > 0.3 g	**Standard: = 0.3g**
High: > 0.3 g	1.00	3.00
Standard: = 0.3g	0.33	1.00

Table 16. Pairwise comparison matrix of Seismic Resilience

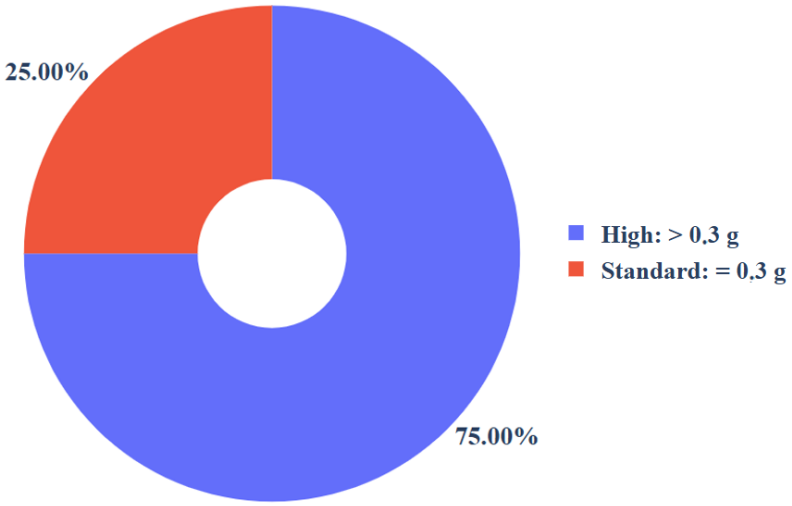


Figure 37. Intensity of each of the Ratings Scale in Seismic Resilience

6.1.2.5.3 Fuel Cycle Characteristics

6.1.2.5.3.1 Refueling Time

Refueling time is a key factor in SMR selection for data centers, as these facilities typically have a lifespan of 15–20 years [54]. The need for refueling during this period can impact operational continuity, maintenance logistics, and overall deployment feasibility. Based on this, reactors are classified into three categories, ranked from most to least favorable: *Online Refueling, No Refueling,* and *Periodical Refueling*.

Online Refueling is considered the most advantageous option, as these reactors can be refueled without shutting down, ensuring continuous power supply [55]—a critical factor for data centers, where even short outages can lead to significant financial and operational disruptions. This minimizes downtime and eliminates planned shutdowns related to fuel replacement.

No Refueling is the second-best option, referring to reactors designed to operate for at least 15 years without the need for refueling. Since this matches or exceeds the expected lifespan of a data center, it effectively removes fuel replacement as a logistical concern. These reactors are typically designed with long-life fuel cores, reducing maintenance complexity over their operational period.

Periodical Refueling is the least favorable category, as it requires refueling operations within a timeframe shorter than 15 years. This means that, for most data centers, at least one scheduled shutdown for refueling would be necessary, introducing potential interruptions and logistical challenges. While still viable, these designs require more planning and operational coordination to ensure minimal disruption.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Online Refueling	No Refueling	Periodical Refueling
Online Refueling	1.00	3.00	9.00
No Refueling	0.33	1.00	7.00
Periodical Refueling	0.11	0.14	1.00

Table 17. Pairwise comparison matrix of Refueling Time

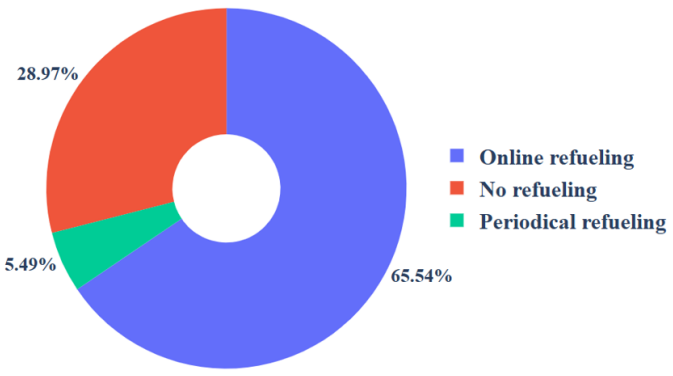


Figure 38. Intensity of each of the Ratings Scale in Refueling Time

6.1.2.5.3.2 Enrichment

Uranium enrichment is the process of increasing the percentage of U-235 in nuclear fuel, requiring the uranium to be converted into a gaseous state to take advantage of the different chemical and physical properties of its isotopes. The level of enrichment directly affects fuel performance, cost, and international regulatory concerns [56].

From a political perspective, uranium enrichment is a highly sensitive issue, as access to enrichment technology is closely monitored under international safeguards and IAEA inspections due to its potential link to nuclear weapons proliferation. Additionally, higher enrichment levels lead to increased costs and negative public perception, making fuel with lower enrichment levels more desirable from an economic and regulatory standpoint.

For these reasons, natural uranium and low-enriched uranium (LEU) receive the highest ratings, as they are widely available, cost-effective, and pose fewer geopolitical concerns. On the other hand, High-Assay Low-Enriched Uranium (HALEU), while still below the 20% enrichment limit for civilian use, receives the lowest rating due to higher costs, stricter regulatory oversight, and increased proliferation concerns.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	High (10-20%)	Medium (5-10%)	Low (0.7-5%)	Natural (0.7%)
High (10-20%)	1.00	0.50	0.25	0.17
Medium (5-10%)	2.00	1.00	0.50	0.25
Low (0.7-5%)	4.00	2.00	1.00	0.50
Natural (0.7%)	6.00	4.00	2.00	1.00

Table 18. Pairwise comparison matrix of Enrichment

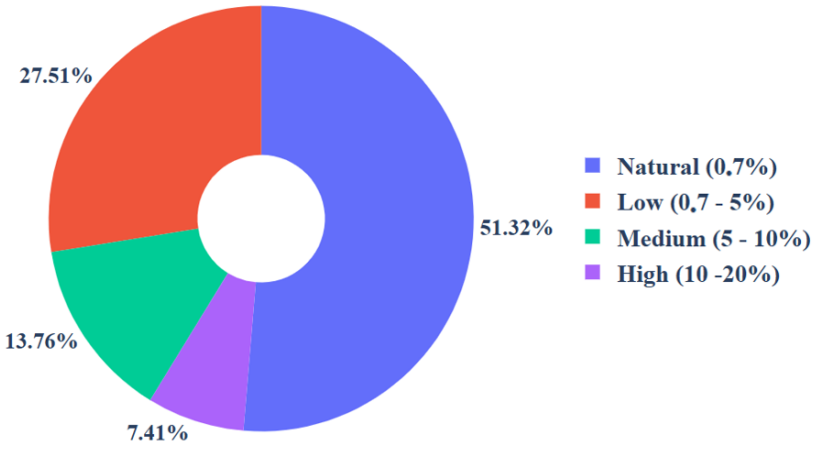


Figure 39. Intensity of each of the Ratings Scale in Enrichment

6.1.2.5.4 Waste Management

6.1.2.5.4.1 Discharge Burnup

Discharge burnup measures how much energy is extracted from nuclear fuel before it is considered spent, directly impacting fuel efficiency, waste generation, and operational sustainability. Higher burnup levels indicate better fuel utilization, reducing refueling frequency and the volume of radioactive waste produced [57]. However, achieving high burnup requires advanced fuel materials capable of withstanding prolonged radiation exposure and high temperatures.

A five-tier rating scale is used, ranked from best to worst: *Very High*, *High*, *Medium*, *Low*, and *Very Low*. *Very High* burnup (>100 MWd/kg) is the most desirable, as it maximizes fuel efficiency and minimizes waste. *High* burnup (80–100 MWd/kg) also performs well but with slightly lower efficiency. *Medium* burnup (60–80 MWd/kg) offers a balanced approach, maintaining reasonable fuel usage with manageable technological requirements. *Low* burnup (40–60 MWd/kg) results in increased fuel consumption and waste, while *Very Low* burnup (<40 MWd/kg) is the least favorable, leading to frequent refueling, higher operational costs, and increased radioactive waste production.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Very High (>100 MWd/kg)	**High (80-100 MWd/kg)**	**Medium (60-80 MWd/kg)**	**Low (40-60 MWd/kg)**	**Very Low (<40 MWd/kg)**
Very High (>100 MWd/kg)	1.00	2.00	3.00	4.00	5.00
High (80-100 MWd/kg)	0.50	1.00	2.00	3.00	4.00
Medium (60-80 MWd/kg)	0.33	0.50	1.00	2.00	3.00
Low (40-60 MWd/kg)	0.25	0.33	0.50	1.00	2.00
Very Low (<40 MWd/kg)	0.20	0.25	0.33	0.50	1.00

Table 19. Pairwise comparison matrix of Discharge Burnup

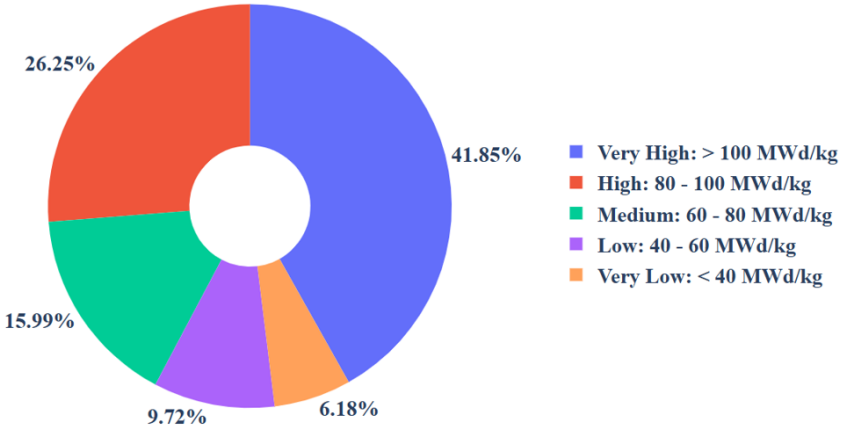


Figure 40. Intensity of each of the Ratings Scale in Discharge Burnup

6.1.2.5.4.2 Waste Streams

Waste streams refer to the types and volumes of radioactive materials generated throughout a reactor's lifecycle, including spent fuel, coolant, moderator, fission products, and irradiated materials. Effective waste management is essential for minimizing environmental impact, reducing storage requirements, and ensuring long-term sustainability [58]. Based on waste volume and complexity of treatment, a three-tier rating scale is used: *High, Medium,* and *Low*, where *Low* is the most favorable and *High* is the least desirable.

High waste streams can be associated with High-Temperature Gas-Cooled Reactors (HTGRs) and Pressurized Water Reactors (PWRs). HTGRs produce significant volumes of irradiated graphite moderator, which is challenging to process and dispose of, often requiring permanent storage. PWRs, as well as other water-cooled reactors, generate large amounts of spent nuclear fuel that require long-term storage or reprocessing. While PUREX reprocessing allows some fuel recovery, it does not eliminate the need for extensive waste handling and containment.

Medium waste streams are found in Molten Salt Reactors (MSRs). While MSRs are capable of burning plutonium and minor actinides from spent fuel, the liquid fuel itself presents a unique waste management challenge. The molten salt coolant requires complex chemical treatment before disposal, and the reactor's graphite moderator, when used, also becomes irradiated waste. Though MSRs achieve higher fuel burnup than LWRs, their overall waste treatment remains intricate, placing them in the *Medium* category.

Low waste streams are characteristic of Sodium-cooled Fast Reactors (SFRs) and other fast-spectrum reactors, as these designs are capable of reprocessing and reusing spent fuel, significantly reducing the overall waste volume. By recycling uranium and plutonium and potentially burning minor actinides, SFRs contribute to lower radiotoxicity and long-term waste production, making them the most favorable option in this category.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	High	Medium	Low
High	1.00	0.25	0.14
Medium	4.00	1.00	0.25
Low	7.00	4.00	1.00

Table 20. Pairwise comparison matrix of Waste Streams

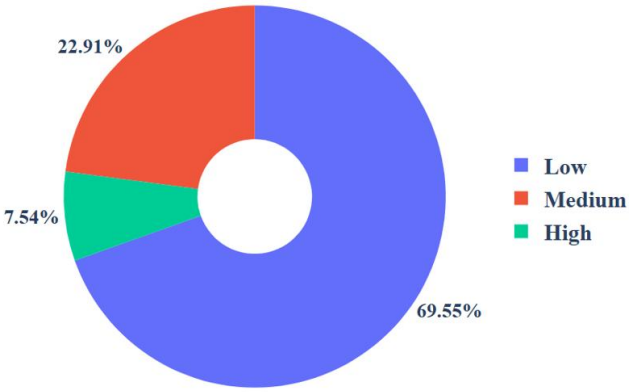


Figure 41. Intensity of each of the Ratings Scale in Waste Streams

6.1.2.5.4.3 Storage and Disposal Plan

Storage and disposal plans are essential for ensuring the safe management of radioactive waste throughout the lifecycle of a reactor, from short-term storage to final disposal. The feasibility of an effective waste storage strategy depends on existing technologies, disposal methods, and the complexity of handling specific waste materials [59]. Based on these factors, reactors are classified into three categories, ranked from most to least favorable: *High, Medium*, and *Low*.

High-rated reactors include PWRs, as their waste management technology is well-established and widely implemented. The storage and disposal of spent nuclear fuel from PWRs follows proven methodologies, with solutions such as dry cask storage and deep geological repositories already in development, like Finland's Onkalo repository [60]. This maturity in waste handling makes PWRs the most favorable option in this category.

Medium-rated reactors include HTGRs and SFRs. HTGRs benefit from higher fuel burnup and reduced radiotoxicity, which helps shorten the required disposal time. However, the need to separate coated fuel particles from the graphite matrix adds complexity to waste processing. SFRs, on the other hand, produce less long-lived waste than conventional reactors, as they burn actinides from spent fuel, significantly reducing their radiotoxicity. However, they require specialized disposal methods for sodium or lead-based coolants, which are hazardous and require further technological development.

Low-rated reactors are MSRs due to the unique challenges in storing and disposing of molten salts. Unlike solid fuel waste, fluoride and chloride salts require special containment strategies, and long-term storage solutions are still in development. Additionally, MSRs that utilize graphite moderators generate irradiated graphite waste,

which further complicates disposal. Due to these uncertainties and the need for further advancements in storage technology, MSRs rank lowest in terms of waste storage feasibility.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	High	Medium	Low
High	1.00	2.00	4.00
Medium	0.50	1.00	2.00
Low	0.25	0.50	1.00

Table 21. Pairwise comparison matrix of Storage and Disposal Plan

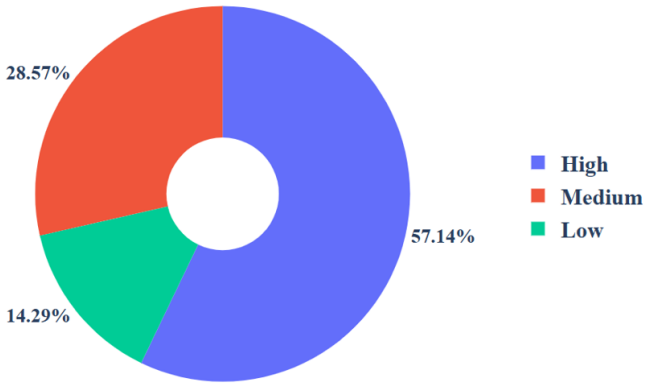


Figure 42. Intensity of each of the Ratings Scale in Storage and Disposal Plan

6.1.2.5.5 Energy Availability

6.1.2.5.5.1 Unit Capacity Factor

Unit Capacity Factor (UCF) is a key measure of a reactor's reliability and efficiency, representing the ratio of actual energy produced to the maximum possible output over a specific period. A higher UCF indicates a reactor's ability to maintain continuous operation with minimal downtime, which is essential for data centers that require a stable and uninterrupted power supply [61].

For this subcriterion, reactors are classified into two categories. The *High* category includes reactors with a UCF of 0.95 or higher, meaning they operate at or near full capacity for extended periods. These designs are the most suitable for data centers, as they ensure consistent power generation, reducing the risk of disruptions due to maintenance or operational inefficiencies.

Reactors with a UCF below 0.95 or those that have not yet defined their UCF fall into the *Standard* category. The lack of a reported UCF introduces uncertainty, as it prevents a clear assessment of the reactor's reliability and performance. As a precaution, designs with unknown UCF values have been penalized and assigned to this category. This ensures that only reactors with proven, high-performance capabilities are prioritized for continuous, mission-critical energy supply.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	High: >= 0.95	Standard: < 0.95
High: >= 0.95	1.00	3.00
Standard: < 0.95	0.33	1.00

Table 22. Pairwise comparison matrix of Unit Capacity Factor

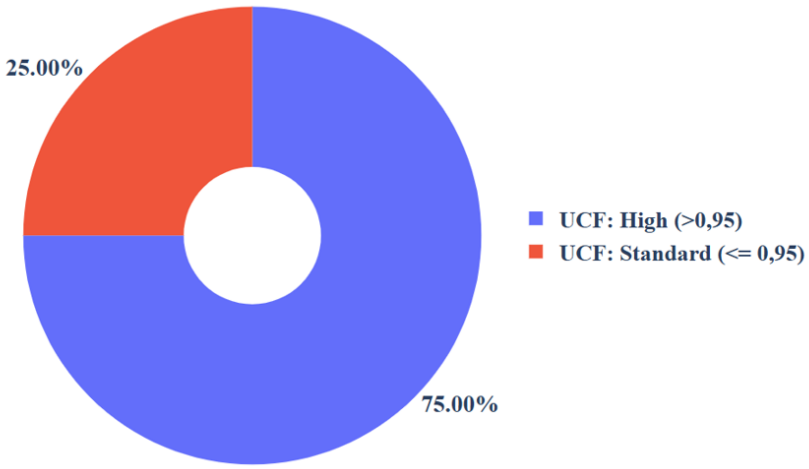


Figure 43. Intensity of each of the Ratings Scale in Unit Capacity Factor

6.1.2.5.5.2 Module Independence

Module Independence evaluates the operational configuration of multi-module SMRs, specifically whether each module functions independently or relies on shared infrastructure with other units. This subcriterion directly affects the system's ability to maintain a continuous power supply under fault or maintenance conditions, which is especially important for critical facilities like data centers that require high availability and minimal downtime.

In *Coupled configurations,* multiple reactor modules share key systems such as heat exchangers, turbines, or control infrastructure. While this can reduce overall system cost and simplify certain aspects of the design, it introduces interdependence that can compromise reliability. A failure or maintenance operation in one module may impact the performance of the entire unit, increasing the risk of partial or full outages.

In contrast, *Independent modules* are designed to operate as self-contained units, each with its own critical systems. This greatly enhances fault tolerance, as one module can be taken offline without affecting the rest. It also allows for easier scaling, phased deployment, and module-specific maintenance. For these reasons, independent operation is rated as moderately more important than coupled configurations in the context of energy availability for data centers.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Coupled Configuration	Independent Modules
Coupled Configuration	1.00	0.33
Independent Modules	3.00	1.00

Table 23. Pairwise comparison matrix of Module Independence

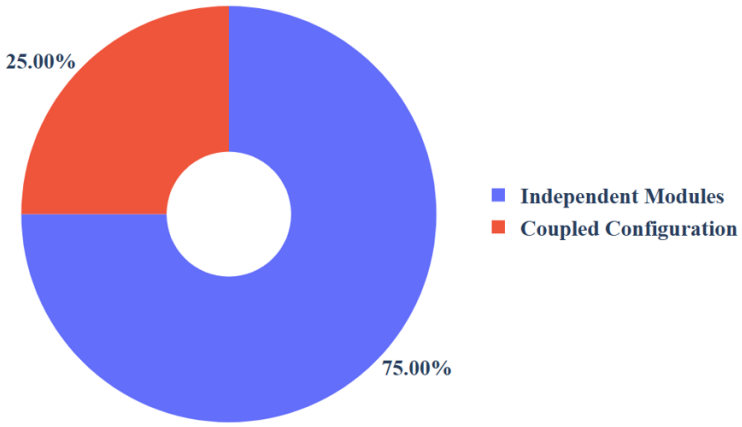


Figure 44. Intensity of each of the Ratings Scale in Module Independence

6.1.2.5.6 Land Footprint

6.1.2.5.6.1 EPZ Size

The Emergency Planning Zone (EPZ) size defines the area surrounding a nuclear reactor where protective measures must be in place in case of an accident [62]. The extent of the EPZ is influenced by factors such as fuel type, reactor design, containment system, and operating pressure, with smaller EPZs generally preferred for greater sitting flexibility, especially for applications near populated areas like data centers. However, since few SMR designs have explicitly defined their EPZ size, this criterion will instead be assessed based on operating pressure, as higher pressure increases the potential risk of radiological release.

Reactor designs are categorized into three levels based on pressure. *Extensive* EPZ applies to reactors operating at more than 10 MPa, as higher pressure can lead to greater containment risks and requires more robust emergency planning measures. *Moderate* EPZ includes reactors with operating pressures between 0.5 and 10 MPa, which present lower risk than high-pressure designs but still require significant safety precautions due to potential pressurized releases in case of failure. The most favorable category, *Minimal* EPZ, is assigned to reactors with operating pressures below 0.5 MPa, as their low-pressure systems significantly reduce the likelihood of a large-scale radiological release, making them particularly attractive for data center applications due to their increased siting flexibility.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Extensive: > 10 MPa	Moderate: 0.5-10 MPa	Minimal: < 0.5 MPa
Extensive: > 10 MPa	1.00	0.33	0.17
Moderate: 0.5-10 MPa	3.00	1.00	0.33
Minimal: < 0.5 MPa	6.00	3.00	1.00

Table 24. Pairwise comparison matrix of EPZ Size

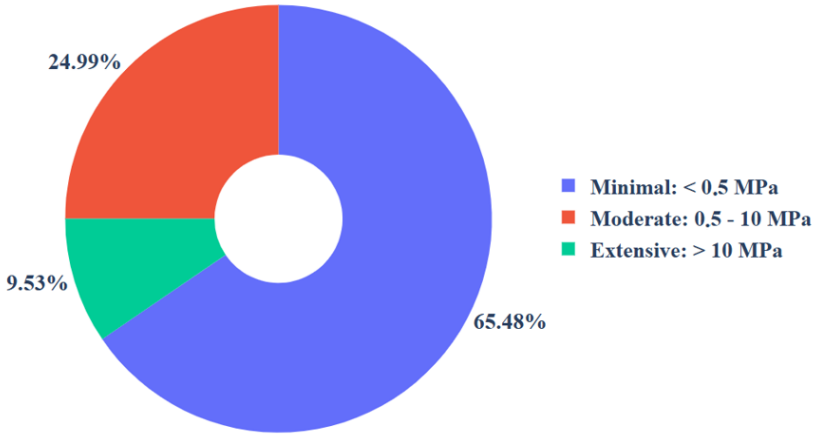


Figure 45. Intensity of each of the Ratings Scale in EPZ Size

6.1.2.5.6.2 Plant Size

The Plant Size subcriterion evaluates the total land footprint required for an SMR, considering not only the reactor building but also supporting infrastructure, such as cooling systems, waste storage, and auxiliary facilities. A smaller plant size is generally more advantageous, as it allows for greater sitting flexibility, reduces land acquisition costs, and minimizes environmental impact—key considerations for data center applications, where space efficiency is crucial.

Since some SMR designs have not clearly defined their plant size, those with unknown values are classified in the least favorable category. Based on available data, reactors are grouped into three categories according to their land use per unit of power output (acres per MWe).

Extensive plant size applies to reactors requiring more than 1 acre per 10 MWe or those with undefined land requirements. These designs demand significant space, making them less suitable for locations with land constraints. *Moderate* plant size includes reactors needing between 1 acre per 10 MWe and 1 acre per 20 MWe, representing a more balanced approach where land use is optimized without excessive spatial demands. The most favorable category, *Minimal* plant size, is assigned to reactors requiring less than 1 acre per 20 MWe, as these designs demonstrate higher power density and space efficiency, making them ideal for compact or urban installations.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Extensive	Moderate	Minimal
Extensive	1.00	0.33	0.17
Moderate	3.00	1.00	0.33
Minimal	6.00	3.00	1.00

Table 25. Pairwise comparison matrix of Plant Size

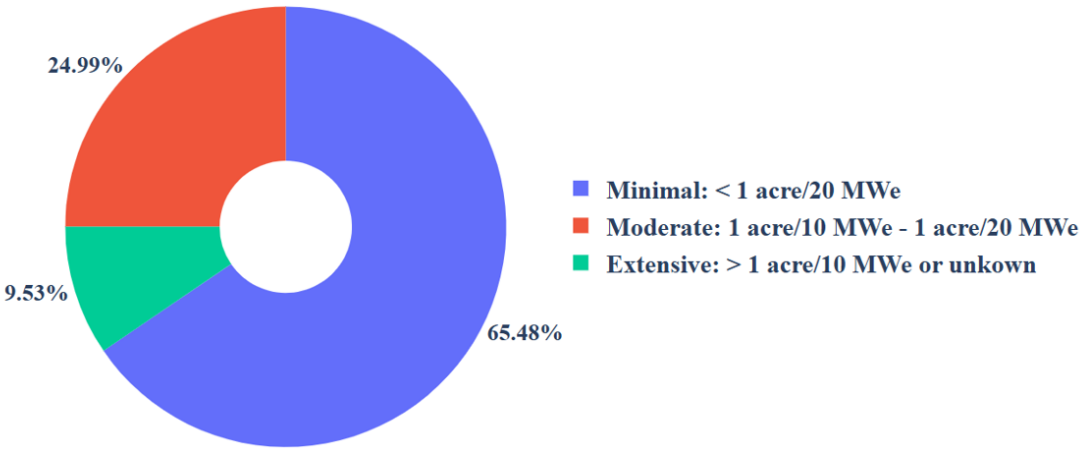


Figure 46. Intensity of each of the Ratings Scale in Plant Size

6.1.2.5.6.3 Scalability For Future Expansion

Scalability for Future Expansion assesses how easily an SMR installation can be expanded to meet increasing energy demands. As data center power requirements continue to grow, the ability to incrementally expand generation capacity is a key factor in ensuring long-term flexibility, cost-efficiency, and site optimization. Based on expansion feasibility, reactors are categorized into three levels: *Incremental Deployment, Threshold Addition,* and *New Plant*, ranked from most to least favorable.

Incremental Deployment is the most flexible approach, allowing for the addition of multiple independent units as needed. These reactors are typically modular and can be scaled up without significant infrastructure modifications, enabling continuous expansion while maintaining operational independence between units. This category is particularly well-suited for data centers with growing power needs, as it provides on-demand capacity increases with minimal disruption.

Threshold Addition refers to designs where new reactor units can be added up to a certain limit, beyond which additional infrastructure (such as shared cooling or electrical systems) becomes a constraint. Once this threshold is reached, expanding capacity further would require building an entirely new facility. While this approach offers some scalability, it limits long-term growth potential, making it less flexible than fully modular designs.

New Plant represents the least favorable option, as expansion requires constructing an entirely new facility from scratch. These reactors lack the capability for seamless incremental growth, making them less adaptable to fluctuating energy demands. Any increase in capacity would involve significant capital investment, permitting processes, and

construction timelines, making this approach less efficient for dynamic power requirements.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Incremental Deployment	Threshold Addition	New Plant
Incremental Deployment	1.00	3.00	8.00
Threshold Addition	0.33	1.00	5.00
New Plant	0.13	0.20	1.00

Table 26. Pairwise comparison matrix of Scalability for Future Expansion

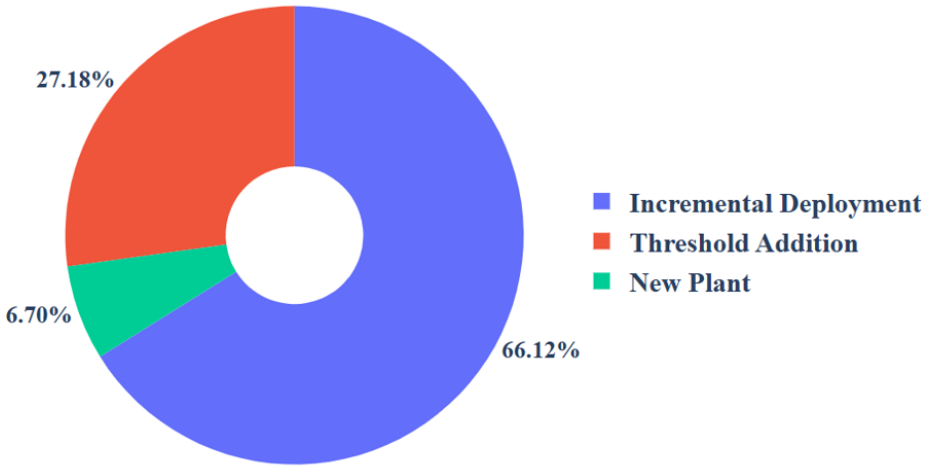


Figure 47. Intensity of each of the Ratings Scale in Scalability for Future Expansion

6.1.2.5.7 Secondary Applications

6.1.2.5.7.1 Coolant Temperature

Coolant Temperature determines the potential secondary applications of an SMR beyond electricity generation, influencing its suitability for desalination, industrial processes, and hydrogen production. While higher temperatures expand application possibilities, they also introduce engineering challenges, making moderate temperatures the most desirable.

Reactors are categorized into three levels: *Low* (<500 K), *Medium* (500 K – 900 K), and *High* (>900 K) [63]. *Low*-temperature reactors are suitable for desalination and district heating, where moderate heat is sufficient to drive large-scale thermal processes. However, reactors in the *Medium* and *High* categories can also perform these same applications, making low-temperature designs less versatile overall.

Medium-temperature reactors (500 K – 900 K) are the most favorable, as they support a wider range of industrial applications, including paper manufacturing and petroleum refining, without introducing excessive technical difficulties. Since *High*-temperature reactors can also perform these functions, *Medium*-rated designs provide the best balance between versatility and feasibility.

High-temperature reactors (>900 K) enable additional applications, such as hydrogen production, which requires extreme heat for thermochemical water splitting. However, these designs face greater engineering and material challenges, including coolant containment and structural durability. Due to these complexities, they are ranked at the same level as *Low*-temperature reactors, despite their broader application potential.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Low: < 500 K	**Medium: 500 – 900 K**	**High: > 900 K**
Low: < 500 K	1.00	0.33	1.00
Medium: 500 – 900 K	3.00	1.00	3.00
High: > 900 K	1.00	0.33	1.00

Table 27. Pairwise comparison matrix of Coolant Temperature

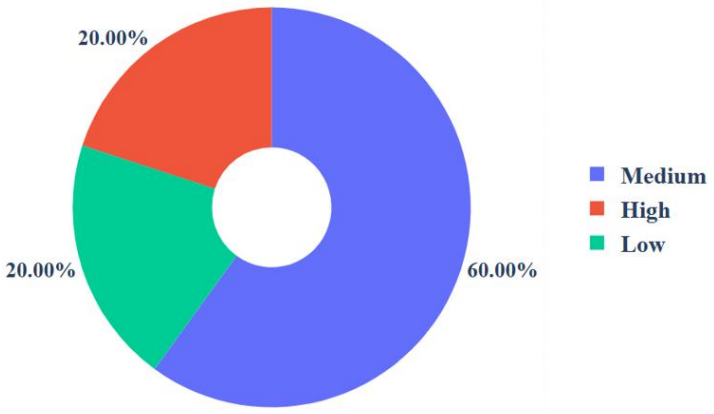


Figure 48. Intensity of each of the Ratings Scale in Coolant Temperature

6.1.2.5.8 Licensing Status

6.1.2.5.8.1 Design Status

Design Status assesses the maturity of an SMR's development, impacting licensing feasibility, deployment timelines, and project risk. Reactors are categorized into four stages: *Conceptual Design, Basic Design, Detailed Design,* and *Under Construction*, ranked from least to most favorable.

Conceptual Design is the earliest stage, where fundamental principles and system architecture are still being defined, leading to high uncertainty and development risk. *Basic Design* represents a more developed phase, with core engineering specifications established, but requiring further technical validation before regulatory approval.

Detailed Design includes reactors with finalized engineering specifications, allowing for licensing applications and prototype development, making them closer to deployment. The *Under Construction* category is the most favorable, as these reactors are already in active development, significantly reducing technical and licensing risks.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Conceptual Design	Basic Design	Detailed Design	Under Construction
Conceptual Design	1.00	0.33	0.20	0.13
Basic Design	3.00	1.00	0.33	0.20
Detailed Design	5.00	3.00	1.00	0.33
Under Construction	8.00	5.00	3.00	1.00

Table 28. Pairwise comparison matrix of Design Status

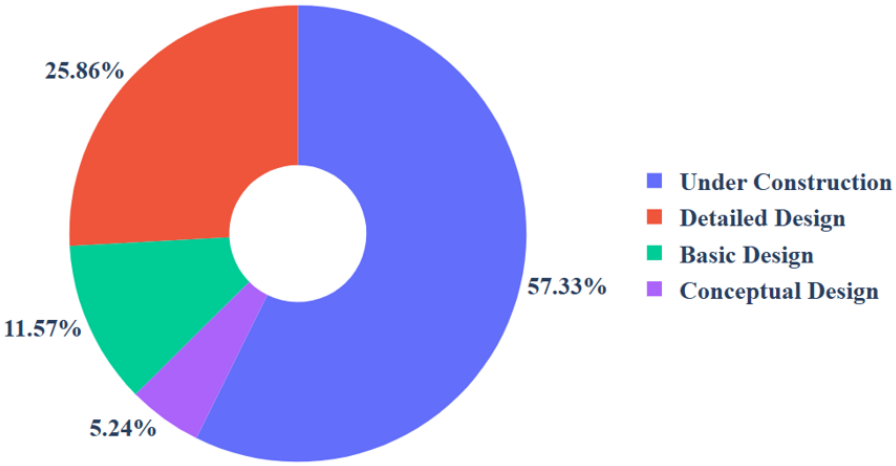


Figure 49. Intensity of each of the Ratings Scale in Design Status

6.1.2.5.8.2 Licensing Initiated

Licensing Initiated evaluates the progress of an SMR in obtaining regulatory approval, a critical step for deployment. The licensing process ensures that the reactor meets safety, environmental, and operational standards, influencing project feasibility and timeline. Reactors are classified into three categories based on their licensing status: *Approved, Work in Progress,* and *Started*, ranked from most to least favorable.

Approved reactors have successfully completed the licensing process, meaning they have regulatory clearance for construction and operation. This status significantly reduces deployment risks and delays, making these reactors the most viable options for near-term implementation.

Work in Progress applies to reactors that are actively undergoing regulatory review but have not yet received full approval. While they are progressing toward deployment, uncertainties remain regarding final certification and potential modifications required by regulators.

Started refers to reactors that have initiated the licensing process but are still in early stages of regulatory engagement. These designs face greater uncertainty as they have yet to pass key regulatory milestones, making them the least favorable in this category.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Started	Work in Progress	Approved
Started	1.00	0.25	0.13
Work in Progress	4.00	1.00	0.25
Approved	8.00	4.00	1.00

Table 29. Pairwise comparison matrix of Licensing Initiated

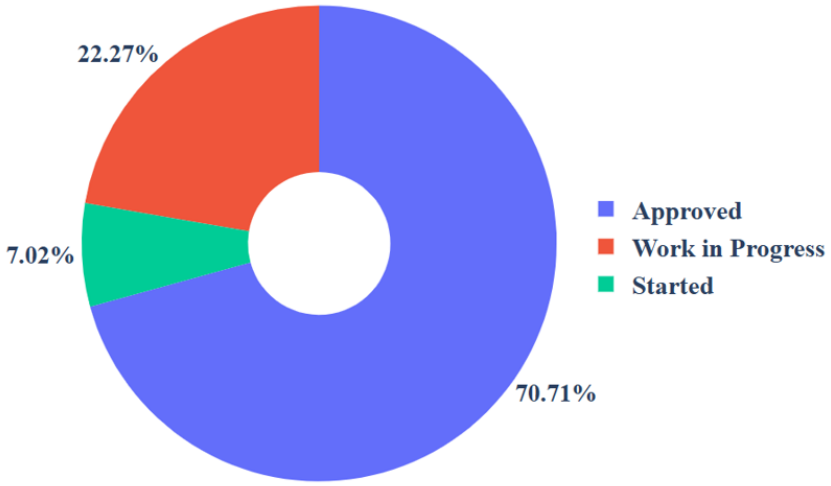


Figure 50. Intensity of each of the Ratings Scale in Licensing Status

6.1.2.5.8.3 Operational Experience

Operational Experience assesses the historical performance and accumulated knowledge from operating similar reactor technologies. A strong operational track record reduces uncertainty, improves safety, and enhances regulatory confidence, making reactors with proven designs and extensive experience more favorable for deployment. Based on historical operation, reactors are classified into four categories: *Very High, High, Medium,* and *Low*, ranked from most to least favorable.

Very High operational experience applies to PWRs, which have been the dominant reactor type worldwide for decades. Their widespread commercial deployment, extensive regulatory oversight, and well-understood behavior make them the most reliable in terms of proven performance.

High operational experience is assigned to SFRs, which have been in development since the 1950s, with early prototypes such as the EBR-I and EBR-II at Idaho National Laboratory demonstrating successful fast reactor operation [64]. Several countries, including France, Russia, and Japan, have deployed and operated SFRs at commercial and experimental scales, providing a substantial knowledge base [65].

Medium operational experience includes HTGRs, which have seen limited but successful deployment, such as the Peach Bottom and Fort St. Vrain reactors in the U.S., the HTR-10 in China, and the HTTR in Japan [66]. While they have demonstrated safe operation and high

thermal efficiency, their commercial adoption has been more limited compared to PWRs and SFRs.

Low operational experience is assigned to MSRs, as their only significant demonstration was the Molten Salt Reactor Experiment (MSRE) at Oak Ridge National Laboratory in the 1960s [67]. Despite renewed interest, modern MSRs remain in early development, with no large-scale operational track record. This lack of real-world experience makes MSRs the least favorable in this category.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Very High	High	Medium	Low
Very High	1.00	3.00	5.00	7.00
High	0.33	1.00	3.00	5.00
Medium	0.20	0.33	1.00	4.00
Low	0.14	0.20	0.25	1.00

Table 30. Pairwise comparison matrix of Operational Experience

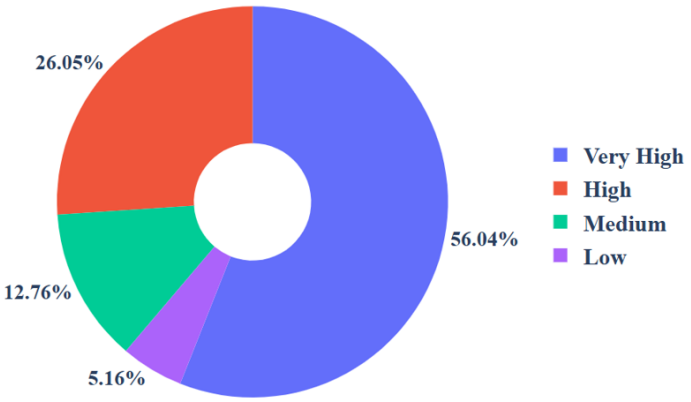


Figure 51. Intensity of each of the Ratings Scale in Operational Experience

6.1.2.5.8.4 Non-Standard Design Features

Non-Standard Design Features evaluates the extent to which a reactor departs from conventional nuclear plant design, particularly in ways that may impact licensing complexity, system reliability, or regulatory acceptance. While innovation can bring performance or cost benefits, it often requires new safety assessments, unfamiliar regulatory pathways, or additional engineering validation, which may delay deployment or introduce risk. This subcriterion helps distinguish between reactors that align closely with regulatory expectations and those that may face barriers due to their design novelty.

Standard Design refers to reactors that follow well-established design norms, such as separate containment systems, independent reactor modules, and traditional coolant and control architectures. These reactors typically require no significant adaptation from existing regulatory frameworks and are generally easier to license and integrate into current infrastructure. As a result, they are considered the most favorable in this category.

Partially Non-Standard reactors incorporate some innovative elements, such as integrated systems, shared infrastructure (e.g., a common reactor pool), or novel structural layouts. While these features may improve performance or reduce costs, they often require partial adaptation of licensing procedures or additional safety justification. These designs are still feasible, but may introduce moderate regulatory and technical complexity.

Highly Non-Standard designs present significant deviations from traditional reactor configurations. Examples include liquid-fueled reactors, non-traditional geometries, or entirely novel heat transfer systems. These reactors often lack direct regulatory precedents, and their licensing may require new evaluation frameworks or extensive safety demonstrations. While potentially transformative, their high degree of innovation introduces greater uncertainty, making them the least favorable in this subcriterion.

The intensity of each of the ratings in the scale (Table X) is obtained by pairwise comparison with the SMART software, displaying the resulting weight values in Figure X.

	Standard Design	Partially Non-Standard	Highly Non-Standard
Standard Design	1.00	3.00	6.00
Partially Non-Standard	0.33	1.00	3.00
Highly Non-Standard	0.17	0.33	1.00

Table 31. Pairwise comparison matrix of Non-Standard Design Features

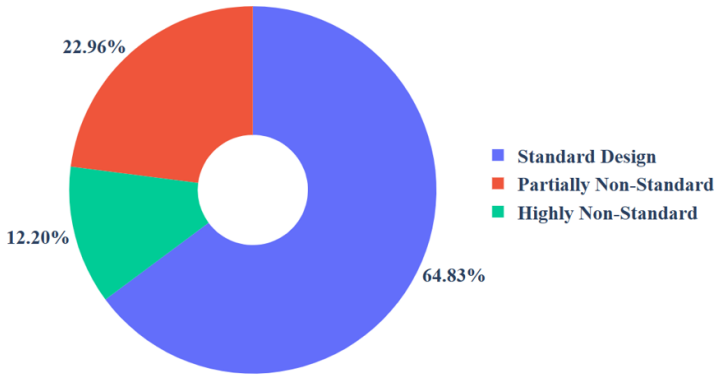


Figure 52. Intensity of each of the Ratings Scale in Non-Standard Design Features

6.1.2.6 Scoring of Reactors

To determine the final score for each selected reactor design, their specific characteristics must be evaluated using the respective rating scale for each subcriterion. This involves classifying each reactor within the appropriate rating category, as shown in Table 32 and 33. The total score for each reactor is then calculated by applying a sum-product operation, multiplying the assigned rating values by the global weight of each subcriterion. Most of the technical data used for this classification was obtained from the *IAEA SMR Booklet 2024* [40]. When certain data points were not available in that source, supplementary information was retrieved from technical reports and official publications made available by reactor developers on their websites or through institutional repositories.

	i-SMR [68]	*NuScale Power Module [69]*	*PWR-20 [70]*	*HTMR100 [71]*	*Xe-100 [72]*	*ARC-100 [73]*
Fuel Safety	Very Low	Very Low	Very Low	High	High	Low
Coolant Toxicity	Low	Low	Low	Very Low	Very Low	Medium
Source Term	Low	Low	Low	Very Low	Very Low	Medium
Configuration	Integral	Integral	Loop with Forced Convection	Loop with Forced Convection	Loop with Forced Convection	Loop with Forced Convection
Component Qualification	Low	Low	Low	High	High	Medium
Seismic Resilience	High: > 0.3 g	High: > 0.3 g	High: > 0.3 g	Standard: = 0.3g	High: > 0.3 g	Standard: = 0.3g
Refueling Time	Periodical refueling	Periodical refueling	Periodical refueling	Online refueling	Online refueling	No refueling
Enrichment	Low (0.7 – 5%)	Low (0.7 – 5%)	Low (0.7 – 5%)	Medium (5 – 10%)	High (10 – 20 %)	High (10 – 20 %)
Discharge Burnup	Low: 40 – 60 MWd/kg	Low: 40 – 60 MWd/kg	Very Low: < 40 MWd/kg	High: 80 – 100 MWd/kg	Very High: > 100 MWd/kg	Medium: 60 – 80 MWd/kg
Waste Streams	High	High	High	High	High	Low
Storage and Disposal Plan	High	High	High	Medium	Medium	Medium
Unit Capacity Factor	High: >= 0.95	High: >= 0.95	High: >= 0.95	Standard: < 0.95	High: >= 0.95	Standard: < 0.95
Module Independence	Coupled Configuration	Coupled Configuration	Independent Modules	Coupled Configuration	Coupled Configuration	Independent Modules
EPZ Size	Extensive: > 10 MPa	Extensive: > 10 MPa	Extensive: > 10 MPa	Moderate: 0.5 – 10 MPa	Moderate: 0.5 – 10 MPa	Minimal: < 0.5 MPa
Plant Size	Extensive: > 1 acre/10 MWe or unknown	Minimal: < 1 acre/20 MWe	Moderate: 1 acre/10 MWe - 1 acre/20 MWe	Minimal: < 1 acre/20 MWe	Moderate: 1 acre/10 MWe - 1 acre/20 MWe	Extensive: > 1 acre/10 MWe or unknown
Scalability for Future Expansion	Threshold Addition	Threshold Addition	Incremental Deployment	Threshold Addition	Threshold Addition	New Plant
Coolant Temperature	Medium: 500 – 900 K	Medium: 500 – 900 K	Medium: 500 – 900 K	High: > 900 K	High: > 900 K	Medium: 500 – 900 K
Design Status	Detailed Design	Detailed Design	Detailed Design	Basic Design	Basic Design	Conceptual Design
Licensing Initiated	Started	Approved	Started	Started	Work in Progress	Started
Operational Experience	Very High	Very High	Very High	Medium	Medium	High
Non-Standard Design Features	Partially Non-Standard	Partially Non-Standard	Standard Design	Partially Non-Standard	Partially Non-Standard	Partially Non-Standard

Table 32. Scoring of Reactors Part 1

	FLEX Reactor [74]	IMSR400 [75]	KP-FHR [76]	Stellarium [77]	Aurora [78]	AP-300 [79]
Fuel Safety	Medium	Medium	High	Medium	Low	Very Low
Coolant Toxicity	High	High	High	High	Medium	Low
Source Term	Medium	High	Medium	Medium	Medium	Low
Configuration	Loop with Natural Convection	Integral	Loop with Forced Convection	Loop with Forced Convection	Integral	Loop with Forced Convection
Component Qualification	Medium	High	High	High	Low	Low
Seismic Resilience	Standard: = 0.3g	Standard: = 0.3g	Standard: = 0.3g	High: > 0.3g	High: > 0.3g	Standard: = 0.3 g
Refueling Time	Periodical refueling	Periodical refueling	Online refueling	No refueling	No refueling	Periodical refueling
Enrichment	Low (0.7 – 5%)	Low (0.7 – 5%)	High (10 – 20%)	High (10 – 20%)	High (10 – 20 %)	Low (0.7 – 5%)
Discharge Burnup	Very Low: < 40 MWd/kg	Very Low: < 40 MWd/kg	Medium: 60 – 80 MWd/kg	Very High: > 100 MWd/kg	Very Low: < 40 MWd/kg	Low: 40 – 60 MWd/kg
Waste Streams	Medium	Medium	Medium	Medium	Low	High
Storage and Disposal Plan	Low	Low	Low	Low	Medium	High
Unit Capacity Factor	Standard: < 0.95	Standard: < 0.95	High: >= 0.95	Standard: < 0.95	High: >= 0.95	High: >= 0.95
Module Independence	Independent Modules	Coupled Configuration	Coupled Configuration	Coupled Configuration	Independent Modules	Independent Modules
EPZ Size	Minimal: < 0.5 MPa	Minimal: < 0.5 MPa	Minimal: < 0.5 MPa	Minimal: < 0.5 MPa	Minimal: < 0.5 MPa	Extensive: > 10 MPa
Plant Size	Minimal: < 1 acre/20 MWe	Minimal: < 1 acre/20 MWe	Extensive: > 1 acre/10 MWe or unknown	Moderate: 1 acre/10 MWe - 1 acre/20 MWe	Extensive: > 1 acre/10 MWe or unknown	Minimal: < 1 acre/20 MWe
Scalability for Future Expansion	Incremental Deployment	New Plant	Threshold Addition	New Plant	Incremental Deployment	New Plant
Coolant Temperature	High: > 900 K	High: > 900 K	High: > 900 K	High: > 900 K	Medium: 500 – 900 K	Medium: 500 – 900 K
Design Status	Basic Design	Detailed Design	Under Construction	Conceptual Design	Detailed Design	Basic Design
Licensing Initiated	Started	Work in Progress	Approved	Started	Work in Progress	Started
Operational Experience	Low	Low	Low	Low	High	Very High
Non-Standard Design Features	Highly Non-Standard	Highly Non-Standard	Highly Non-Standard	Highly Non-Standard	Highly Non-Standard	Standard Design

Table 33. Scoring of Reactors Part 2

6.2 Results

After entering all the weights of the criteria and subcriteria, their pairwise comparisons, and the characteristics of each reactor into the SMART software, the following scores and ranking have been obtained, with number 1 being the most suitable for powering data centers.

For full transparency and reproducibility, the detailed global and local weights used in the AHP process are provided in Appendix 1. These tables summarize the weight distribution and normalized performance scores that led to the final ranking.

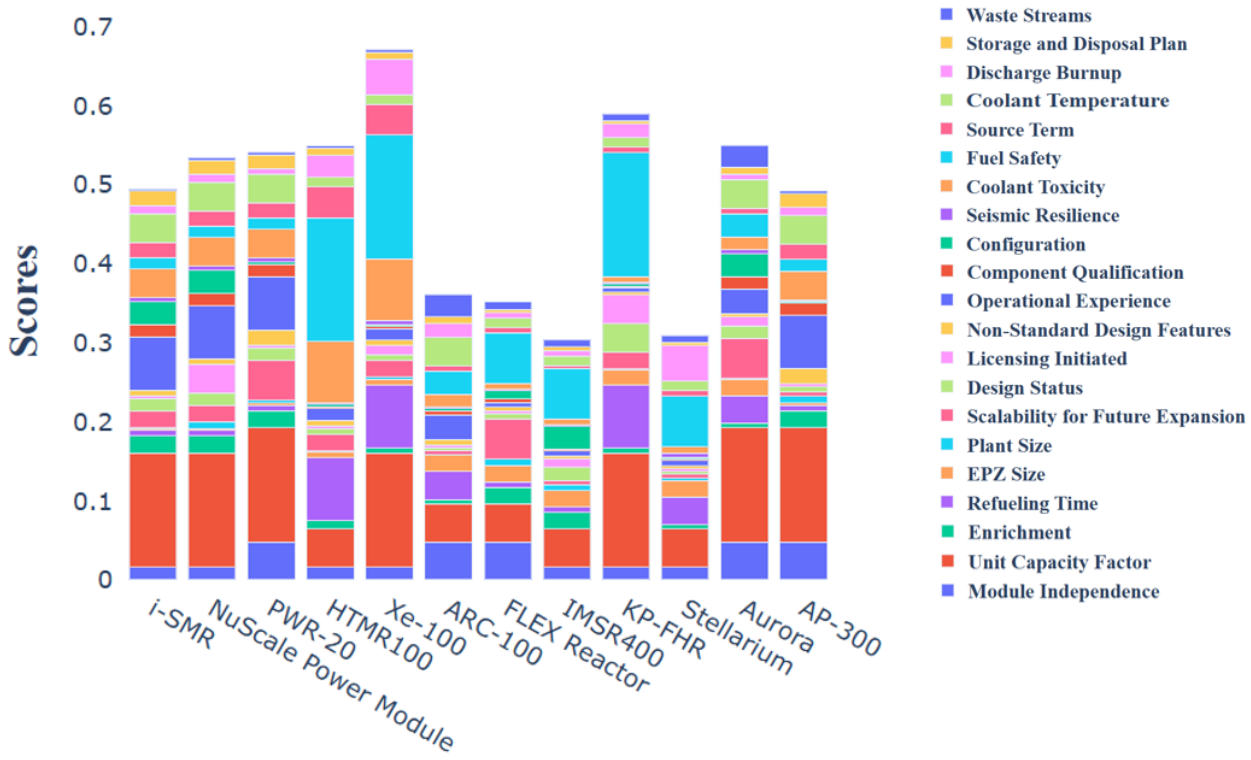


Figure 53. Scoring of Reactors

RANKING	SCORE	REACTOR	DESIGN ORGANIZATION	COUNTRY	TYPE
1	0.6708	Xe-100	X-Energy LLC	USA	HTGR
2	0.5901	KP-FHR	Kairos Power	USA	MSR
3	0.5503	Aurora Powerhouse	Oklo Inc.	USA	SFR
4	0.5493	HTMR100	STL Nuclear	South Africa	HTGR
5	0.5407	PWR-20	Last Energy	USA	PWR
6	0.5340	NuScale Power Module	NuScale Power Inc.	USA	PWR
7	0.4945	i-SMR	KHNP & KAERI	Republic of Korea	PWR
8	0.4917	AP-300	Westinghouse	USA	PWR
9	0.3610	ARC-100	ARC Nuclear Canada Inc.	Canada	SFR
10	0.3514	FLEX Reactor	MoltexFLEX	UK	MSR
11	0.3095	Stellarium	Stellaria	France	MSR
12	0.3037	IMSR400	Terrestrial Energy Inc.	Canada / USA	MSR

Table 34. Ranking of Reactors

The best three reactors which are apparently most suitable to power sustainably the data centers are: Xe-100, KP-FHR and Aurora Powerhouse, in that order.

The reactor with the highest score is Xe-100, excelling in key safety and performance criteria. Its high unit capacity factor ensures continuous and reliable power generation, a critical requirement for data centers that demand uninterrupted energy supply. This makes the Xe-100 one of the most dependable SMR options in terms of operational efficiency and uptime. Additionally, its high fuel safety, attributed to TRISO fuel, provides exceptional resistance to high temperatures and fission product containment, minimizing accident risks. The reactor's online refueling cycle further aligns with data center needs, reducing downtime and maintenance requirements. It also scores highly in low coolant toxicity, optimized discharge burnup, and source term containment, reinforcing its safety, sustainability, and waste reduction capabilities.

KP-FHR designed by Kairos Power is in second place in the ranking, demonstrating strength in fuel safety, refueling time, and licensing progress. Like the Xe-100, it benefits from TRISO fuel, ensuring robust safety and resistance to extreme conditions. Its long refueling intervals reduce operational disruptions, making it well-suited for uninterrupted data center energy supply. Furthermore, its advanced licensing status places it ahead of many competitors, reducing regulatory uncertainty and enhancing deployment feasibility. With a strong unit capacity factor, the KP-FHR offers a compelling combination of reliability and regulatory readiness.

Finally, the third with the highest score is Oklo's Aurora Powerhouse, primarily due to its strong performance in unit capacity factor, scalability, and module independence. Its ability to consistently deliver power with minimal downtime addresses one of the most critical needs of data centers. The design's scalability and modular autonomy allow flexible deployment, making it adaptable to future demand growth. Aurora also stands out for its potential in secondary applications, such as heat or hydrogen production, which, while not essential, add strategic value. Its long refueling interval reduces operational interruptions, and its proven operational experience reinforces confidence in its reliability. Combined with advanced component qualification and a compact, optimized reactor configuration, Aurora presents a robust and forward-looking option for powering energy-intensive digital infrastructure.

6.3 Sensitivity Analysis

To test the robustness of the SMR ranking obtained through the Analytic Hierarchy Process (AHP), a sensitivity analysis was performed. This analysis explores how moderate variations in the assigned weights of specific criteria affect the final reactor rankings. Since expert judgment in AHP involves subjectivity, it is essential to evaluate whether small changes in preferences lead to significant shifts in the decision outcome.

Two complementary sensitivity scenarios were conducted: (1) modifying the relative priority of top-level criteria, and (2) adjusting the weights of selected subcriteria within the same criterion.

6.3.1 Sensitivity to Criteria Weighting

To assess the robustness of the AHP-based SMR ranking, a sensitivity analysis was conducted by modifying the relative weights of the top-level evaluation criteria. The total weight distribution remained fixed at 100%, but the priority structure was altered to simulate a scenario in which stakeholder preferences shift toward deployment feasibility.

In the original model, *Safety Characteristics* held the highest weight (27.38%), followed by *Energy Availability* (19.28%) and *Licensing Status* (15.86%). This reflects a long-term, safety-driven strategic perspective commonly adopted in nuclear decision-making. In contrast, the adjusted model prioritizes *Licensing Status* (22.27%) and *Energy Availability* (20.34%), simulating the preferences of stakeholders who value regulatory maturity and rapid deployability—such as data center operators or private infrastructure developers. *Safety Characteristics* was reduced to 17.32% to reflect its relative deprioritization in this alternative scenario.

The revised pairwise comparison matrix produced the following weight distribution:

Criterion	***Original Weight***	***Adjusted Weight***
Safety Characteristics	27.38%	17.32%
Energy Availability	19.28%	20.34%
Licensing Status	15.86%	22.27%
Fuel Cycle Characteristics	11.91%	10.59%
Waste Management	9.05%	9.85%
Land Footprint	7.91%	6.87%
Reactor System Characteristics	4.99%	8.11%
Secondary Applications	3.64%	4.65%
Total	**100.00%**	**100.00%**

Table 35. Adjusted Criteria Weighting

Based on these adjusted weights, the resulting AHP ranking is presented below:

RANKING	***SCORE***	***REACTOR***	***DESIGN ORGANIZATION***	***COUNTRY***	***TYPE***
1	0.6241	NuScale Power Module	NuScale Power Inc.	USA	PWR
2	0.6145	Aurora Powerhouse	Oklo Inc.	USA	SFR
3	0.5914	PWR-20	Last Energy	USA	PWR
4	0.5841	Xe-100	X-Energy LLC	USA	HTGR
5	0.5724	i-SMR	KHNP & KAERI	Republic of Korea	PWR
6	0.5501	KP-FHR	Kairos Power	USA	MSR
7	0.5403	AP-300	Westinghouse	USA	PWR
8	0.4473	HTMR100	STL Nuclear	South Africa	HTGR
9	0.3742	ARC-100	ARC Nuclear Canada Inc.	Canada	SFR
10	0.3407	FLEX Reactor	MoltexFLEX	UK	MSR
11	0.3235	IMSR400	Terrestrial Energy Inc.	Canada / USA	MSR
12	0.2942	Stellarium	Stellaria	France	MSR

Table 36. New Reactor Ranking after Adjusted Criteria Weighting

Several trends emerge from this adjusted ranking. First, the results are more compressed, with narrower score differences between the top designs, indicating that under deployment-oriented priorities, a broader set of reactors appear comparably viable. Second, PWR-based designs have generally improved their positions, with the NuScale Power Module now in first place, PWR-20 rising to third, and AP-300 reaching seventh. This reflects the advantage of mature, licensable technologies in scenarios where regulatory progress and implementation speed are prioritized. Third, the bottom four designs remain unchanged (ARC-100, FLEX Reactor, IMSR400, and Stellarium), suggesting that these options are less sensitive to changes in criteria weighting and likely underperform across multiple dimensions.

It is also noteworthy that advanced designs such as the Aurora Powerhouse, Xe-100, and KP-FHR maintain strong performance despite the adjusted priorities, demonstrating technical resilience. These findings highlight the value of conducting sensitivity analyses in multi-criteria decision-making, particularly in applications like SMR selection, where different stakeholders may emphasize different dimensions of performance and feasibility.

6.3.2 Sensitivity to Subcriteria Weighting

While the previous sensitivity analysis explored variations at the level of top-level criteria, this section evaluates the robustness of the ranking with respect to changes in the relative weights of specific subcriteria. In multi-criteria decision-making processes such as AHP, the distribution of importance within each criterion can significantly influence the final ranking, especially when certain subcriteria heavily favor specific reactor technologies.

To this end, three distinct sensitivity scenarios were tested. Each scenario modifies the weight of a strategically relevant subcriterion, either reducing or increasing its influence within its corresponding criterion. The goal is to understand how such changes affect the ranking outcome, and to identify which reactors are more or less sensitive to specific aspects of the evaluation framework.

6.3.2.1 Reduction of Fuel Safety Weight

In this scenario, the weight of the *Fuel Safety* subcriterion within the *Safety Characteristics* category was reduced from 57.14% to 40%. The remaining weight was redistributed between *Coolant Toxicity* and *Source Term*, which increased to 40% and 20% respectively. This adjustment reflects a more balanced approach to evaluating safety, where coolant properties and source term containment are given equal or greater relevance.

Subcriterion	***Original Weight***	***Adjusted Weight***
Fuel Safety	57.14%	40.00%
Coolant Toxicity	28.57%	40.00%
Source Term	14.29%	20.00%
Total	**100.00%**	**100.00%**

Table 37. Adjusted Safety Characteristics Weight

Based on this new distribution, the following ranking was obtained:

RANKING	SCORE	REACTOR	DESIGN ORGANIZATION	COUNTRY	TYPE
1	0.6708	Xe-100	X-Energy LLC	USA	HTGR
2	0.5584	PWR-20	Last Energy	USA	PWR
3	0.5517	NuScale Power Module	NuScale Power Inc.	USA	PWR
4	0.5506	Aurora Powerhouse	Oklo Inc.	USA	SFR
5	0.5493	HTMR100	STL Nuclear	South Africa	HTGR
6	0.5489	KP-FHR	Kairos Power	USA	MSR
7	0.5122	i-SMR	KHNP & KAERI	Republic of Korea	PWR
8	0.5094	AP-300	Westinghouse	USA	PWR
9	0.3614	ARC-100	ARC Nuclear Canada Inc.	Canada	SFR
10	0.3379	FLEX Reactor	MoltexFLEX	UK	MSR
11	0.2960	Stellarium	Stellaria	France	MSR
12	0.2888	IMSR400	Terrestrial Energy Inc.	Canada / USA	MSR

Table 38. New Reactor Ranking after Adjusted Safety Characteristics Weight

Xe-100 remains the clear top performer, maintaining a strong lead despite the reduced emphasis on fuel-specific safety. The bottom four positions are unchanged, confirming their consistent underperformance. Notably, several designs in the middle of the ranking are tightly grouped.

6.3.2.2 Increase of Refueling Time Weight

This scenario analyzes the effect of giving more weight to the *Refueling Time* subcriterion within the *Fuel Cycle Characteristics* category. The revised distribution increases the weight of *Refueling Time* to 80%, reducing *Enrichment* to 20%, in order to simulate the importance of long refueling intervals in applications that require uninterrupted power, such as data centers.

Subcriterion	Original Weight	Adjusted Weight
Refueling Time	66.67%	80.00%
Enrichment	33.33%	20.00%
Total	**100.00%**	**100.00%**

Table 39. Adjusted Fuel Cycle Characteristics Weight

The updated ranking is shown below:

RANKING	SCORE	REACTOR	DESIGN ORGANIZATION	COUNTRY	TYPE
1	0.6844	Xe-100	X-Energy LLC	USA	HTGR
2	0.6037	KP-FHR	Kairos Power	USA	MSR
3	0.5609	HTMR100	STL Nuclear	South Africa	HTGR
4	0.5550	Aurora Powerhouse	Oklo Inc.	USA	SFR
5	0.5335	PWR-20	Last Energy	USA	PWR
6	0.5269	NuScale Power Module	NuScale Power Inc.	USA	PWR
7	0.4874	i-SMR	KHNP & KAERI	Republic of Korea	PWR
8	0.4846	AP-300	Westinghouse	USA	PWR
9	0.3658	ARC-100	ARC Nuclear Canada Inc.	Canada	SFR
10	0.3442	FLEX Reactor	MoltexFLEX	UK	MSR
11	0.3142	Stellarium	Stellaria	France	MSR
12	0.2965	IMSR400	Terrestrial Energy Inc.	Canada / USA	MSR

Table 40. New Reactor Ranking after Fuel Cycle Characteristics Weight

The overall ranking remains stable, with Xe-100 and KP-FHR holding the top positions. The only change occurs between HTMR100 and Aurora Powerhouse, which swap third and fourth places. This shift reflects the greater importance now assigned to extended refueling cycles, favoring reactors like HTMR100 that score strongly on this feature. The bottom four designs remain unchanged, indicating that their relative performance is not significantly affected by this subcriterion adjustment.

6.3.2.3 Increase of Module Independence Weight

This scenario explores the effect of assigning equal importance to *Module Independence* and *Unit Capacity Factor* within the *Energy Availability* criterion. Originally, *Unit Capacity Factor* accounted for 75% of the criterion's weight, while *Module Independence* represented only 25%. In this revised configuration, both subcriteria were weighted equally at **50%**, reflecting a context in which the flexibility and operational autonomy of modular SMR units are considered as critical as their overall capacity factor.

Subcriterion	*Original Weight*	*Adjusted Weight*
Unit Capacity Factor	75.00%	50.00%
Module Independence	25.00%	50.00%
Total	**100.00%**	**100.00%**

Table 41. Adjusted Energy Availability Weight

The updated AHP ranking is shown below:

RANKING	*SCORE*	*REACTOR*	*DESIGN ORGANIZATION*	*COUNTRY*	*TYPE*
1	0.6386	Xe-100	X-Energy LLC	USA	HTGR
2	0.5580	KP-FHR	Kairos Power	USA	MSR
3	0.5503	Aurora Powerhouse	Oklo Inc.	USA	SFR
4	0.5493	HTMR100	STL Nuclear	South Africa	HTGR
5	0.5407	PWR-20	Last Energy	USA	PWR
6	0.5019	NuScale Power Module	NuScale Power Inc.	USA	PWR
7	0.4917	AP-300	Westinghouse	USA	PWR
8	0.4624	i-SMR	KHNP & KAERI	Republic of Korea	PWR
9	0.3932	ARC-100	ARC Nuclear Canada Inc.	Canada	SFR
10	0.3835	FLEX Reactor	MoltexFLEX	UK	MSR
11	0.3095	Stellarium	Stellaria	France	MSR
12	0.3037	IMSR400	Terrestrial Energy Inc.	Canada / USA	MSR

Table 42. New Reactor Ranking after Adjusted Energy Availability Weight

The ranking remained highly stable under this adjustment, with Xe-100 retaining its leading position. The only change was a swap between AP-300 and i-SMR, which reflects the moderate advantage gained by designs that prioritize modular autonomy. Additionally, many reactors scored similarly, suggesting that the model became more balanced and less sensitive to small differences when *Module Independence* was given greater importance.

This confirms that while *Unit Capacity Factor* remains a strong differentiator, the ability to operate independently and flexibly in modular units can have a measurable impact on reactor selection, especially in distributed or scalable infrastructure scenarios.

6.3.2.4 Increase of Licensing Initiated Weight

In this scenario, the weight of the *Licensing Initiated* subcriterion within the *Licensing Status* category was increased to reflect its importance in enabling short- and medium-term deployment. This subcriterion captures whether the reactor has formally begun the licensing process—an essential factor for project feasibility in commercial settings.

The revised distribution increased *Licensing Initiated* from 22.70% to 46.73%, reducing the weights of the other subcriteria proportionally. This adjustment simulates a context in which regulatory maturity is prioritized above conceptual development or operational legacy.

Subcriterion	*Original Weight*	*Adjusted Weight*
Licensing Initiated	22.70%	46.73%
Operational Experience	42.36%	27.72%
Design Status	22.70%	16.01%
Non-Standard Design	12.23%	9.54%
Total	**100.00%**	**100.00%**

Table 43. Adjusted Licensing Status Weight

The updated AHP ranking is as follows:

RANKING	*SCORE*	*REACTOR*	*DESIGN ORGANIZATION*	*COUNTRY*	*TYPE*
1	0.6738	Xe-100	X-Energy LLC	USA	HTGR
2	0.6146	KP-FHR	Kairos Power	USA	MSR
3	0.5459	Aurora Powerhouse	Oklo Inc.	USA	SFR
4	0.5441	HTMR100	STL Nuclear	South Africa	HTGR
5	0.5426	NuScale Power Module	NuScale Power Inc.	USA	PWR
6	0.5122	PWR-20	Last Energy	USA	PWR
7	0.4688	i-SMR	KHNP & KAERI	Republic of Korea	PWR
8	0.4659	AP-300	Westinghouse	USA	PWR
9	0.3515	ARC-100	ARC Nuclear Canada Inc.	Canada	SFR
10	0.3501	FLEX Reactor	MoltexFLEX	UK	MSR
11	0.3094	Stellarium	Stellaria	France	MSR
12	0.3079	IMSR400	Terrestrial Energy Inc.	Canada / USA	MSR

Table 44. New Reactor Ranking after Adjusted Licensing Status Weight

The results show that NuScale overtakes PWR-20, benefitting from its more advanced licensing progress. Apart from this, the overall ranking remains highly stable, with Xe-100 and KP-FHR continuing to lead. This confirms that *Licensing Initiated* is a meaningful but not dominant driver within the AHP structure. The bottom four positions remain unchanged, reflecting consistent underperformance across the board.

6.3.3 Conclusions

The sensitivity analysis conducted in this chapter has demonstrated the overall robustness and consistency of the AHP-based SMR ranking framework. By introducing controlled changes in both the top-level criteria weights and the subcriteria distributions, it was possible to simulate realistic variations in stakeholder priorities and evaluate their impact on the decision-making outcome.

In the criteria-level sensitivity (Section 6.3.1), shifting the priority toward deployment feasibility—emphasizing *Licensing Status* and *Energy Availability*—led to a more compressed ranking where mature PWR designs such as NuScale and PWR-20 moved upward. However, advanced designs like Xe-100 and KP-FHR maintained strong performance, highlighting their resilience across strategic scenarios.

In the subcriteria-level sensitivity (Section 6.3.2), four targeted scenarios were tested by adjusting the internal weighting within key criteria (*Safety Characteristics*, *Fuel Cycle Characteristics*, *Energy Availability*, and *Licensing Status*). These experiments revealed that while localized changes in the ranking occurred—such as swaps between AP-300 and i-SMR, or between NuScale and PWR-20—the overall structure of the ranking remained stable. In every scenario, Xe-100 consistently ranked first, and the bottom four reactors (ARC-100, FLEX Reactor, Stellarium, IMSR400) remained in the lowest positions, confirming their systematic underperformance.

This analysis confirms that the model is not overly sensitive to individual assumptions and that its conclusions are robust under a range of plausible preference structures. Moreover, the observed local variations underscore the flexibility of the AHP framework in adapting to different stakeholder priorities, making it a valuable tool for multi-criteria decision-making in complex, high-stakes environments like SMR deployment for data centers.

7 ECONOMIC FEASIBILITY AND COST ANALYSIS

7.1 Introduction

7.1.1 Objective of the analysis

This economic analysis aims to assess the financial viability of SMRs as an energy source for data centers, comparing their costs against conventional power sources (conventional nuclear, natural gas, coal) and renewables (solar, wind).

Data centers are among the most energy-intensive infrastructures, consuming massive amounts of electricity for computing operations, cooling systems, and security. As cloud computing, AI, and digital services grow, energy demand rises, making stable, cost-effective, and sustainable power solutions essential.

The key research question is: *How do the capital and operational costs of SMR-powered data centers compare with those powered by renewable or conventional energy sources?*

To address this, the analysis will compare the Total Cost of Ownership (TCO) and the Levelized Cost of Electricity (LCOE) of SMRs with alternative power sources, considering both short-term and long-term financial implications.

7.1.2 Key Metrics

To evaluate the economic feasibility of different energy solutions for data centers, this analysis applies a set of key financial metrics that reflect both initial investment requirements and long-term operational performance:

- **CAPEX (Capital Expenditure):** The total upfront cost required to build and commission a project, including equipment, construction, engineering, and financing during construction. It represents the initial investment before operation begins.
- **OPEX (Operational Expenditure):** The ongoing costs associated with running and maintaining a facility, including fuel, labor, maintenance, and regulatory compliance. It reflects the day-to-day financial burden of operation.
- **LCOE (Levelized Cost of Electricity):** A metric that represents the average cost per unit of electricity generated over the lifetime of a power system, taking into account CAPEX, OPEX, financing, and energy output. It enables cost comparisons across different technologies.
- **TCO (Total Cost of Ownership):** The total lifecycle cost of a project, including capital, operating, and decommissioning expenses, discounted over time. It provides a comprehensive view of long-term economic viability.
- **EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization):** A profitability indicator that measures the gross operational cash flow of a project before accounting for financing and accounting costs. It reflects the project's core financial performance.

- **NPV (Net Present Value):** The difference between the present value of expected future cash flows and the initial investment, using a specific discount rate. A positive NPV indicates a financially viable project.
- **IRR (Internal Rate of Return):** The discount rate at which the NPV of a project becomes zero. It reflects the expected annual return of the investment and is used to compare project profitability.

7.1.3 Economic Advantages and Disadvantages of SMRs

SMRs exhibit a unique economic profile shaped by their modular design, lower total capital requirements, and suitability for scalable, resilient power supply—especially relevant for applications such as data centers.

SMRs present several economic strengths that distinguish them from both conventional nuclear and intermittent renewable technologies:

- **Modular design and factory construction** enable shorter construction times and improved cost control, reducing the financial uncertainty typically associated with nuclear projects.
- Their **lower total capital outlay** makes SMRs more accessible to a broader range of investors, especially those unable to finance gigawatt-scale projects.
- **Scalability** allows operators to phase deployment according to demand growth, avoiding overcapacity and optimizing financial efficiency.
- Their **smaller project scope and standardized designs** reduce the likelihood of delays and cost overruns.
- SMRs offer **continuous, dispatchable power**, potentially eliminating the need for costly backup or storage systems required by intermittent renewables.

However, these benefits are counterbalanced by economic challenges that reflect the early-stage nature and scale limitations of SMRs:

- Because SMRs produce fewer megawatts per unit, they **lose the economies of scale** that make conventional nuclear more cost-effective on a per-kilowatt basis.
- **Initial deployments face FOAK cost penalties**, including licensing, regulatory navigation, and site-specific development.
- Despite their lower absolute cost, SMRs can have **long payback periods**, which may not align with investor expectations for shorter return cycles.
- **Regulatory uncertainty and limited market experience** further complicate financing and risk assessment, potentially inflating both capital and operational expenditures.

7.2 Capital Expenditures (CAPEX)

7.2.1 Definition and Relevance

The Capital Expenditure (CAPEX) represents the total upfront investment required to deploy a SMR as a dedicated power source for a data center. This includes both the overnight capital cost (OVC)—which accounts for construction, equipment, engineering, and licensing—and the interest during construction (IDC), which captures the cost of financing the project over the construction period [80]:

$$CAPEX = OVC + IDC$$

7.2.2 Overnight Capital Cost (OVC)

According to the U.S. Department of Energy (DOE) and the International Atomic Energy Agency (IAEA), the estimated overnight capital cost for current SMR designs ranges between 4,000 and 7,000 USD/kWe, based on 2019 U.S. dollars [81]. Adjusting for approximately 19% of cumulative inflation between 2019 and 2023 [82], this range increases between 4,760 and 8,330 USD/kWe in 2023 dollars. This estimate, which corresponds to a BOAK (between first and Nth of a kind) deployment scenario, is based on public data from major SMR development programs which anticipate reduced construction costs through modularization and standardization.

The following table provides a detailed breakdown of overnight capital costs for four advanced nuclear reactor technologies—PWR, SFR, HTGR, and MSR—expressed as a percentage of total investment. This comparison highlights how each reactor type allocates resources across major cost categories such as direct construction, indirect services, reactor equipment, and preconstruction activities [81]:

COST CATEGORY	*PWR*	*SFR*	*HTGR*	*MSR*
Preconstruction Costs	2.35%	5.30%	1.19%	2.74%
Direct Costs	39.65%	49.55%	40.25%	39.41%
Structures	10.33%	11.03%	5.66%	9.59%
Reactor Equipment	12.50%	23.95%	14.79%	14.51%
Energy Conversion System	8.76%	3.88%	9.13%	8.16%
Electrical Equipment	3.92%	5.56%	8.11%	4.55%
Heat Rejection System	1.85%	3.31%	2.56%	1.61%
Miscellaneous Equipment	2.29%	1.81%	0.00%	1.46%
Indirect Costs	44.16%	34.33%	44.83%	43.90%
Owner Costs	12.57%	9.77%	12.76%	12.50%
Supplementary Costs	1.28%	1.05%	0.96%	1.45%

Table 45. Breakdown of OVC for PWR, SFR, HTGR and MSR

The distribution of overnight capital costs across PWR, SFR, HTGR, and MSR technologies reveals clear differences in how investment is allocated. While direct costs remain relatively consistent for most reactor types—around 39–50% of the total—SFRs show the highest direct cost share, indicating a more construction-intensive profile.

Indirect costs, which include engineering, project management, and contingencies, vary significantly: HTGRs and MSRs present the highest shares (above 43%), suggesting greater design complexity and support requirements, possibly linked to their lower commercialization levels.

Reactor equipment and energy conversion systems account for a substantial portion of total cost across all technologies, while owner's costs and preconstruction phases remain relatively stable.

Overall, the cost structure indicates that while basic construction components are comparable, emerging reactor types still face higher indirect overheads, reflecting early-stage deployment and the need for specialized systems and project governance.

7.2.3 Interest During Construction (IDC)

When evaluating the total capital investment required for deploying SMRs, it is essential to account not only for the OVC, but also for the financial cost associated with project financing over the construction period. This component, known as IDC, represents the accumulated return owed to lenders or investors during the non-operational phase of the project.

IDC corresponds to the interest charges incurred on borrowed capital (or expected return on equity) between the start of construction and the plant's entry into commercial operation. According to guidelines from the IEA and NEA, IDC should be considered as part of the total CAPEX, and is especially relevant in nuclear projects where construction timelines are long and financing needs substantial [83].

To allow for fair comparisons across technologies and countries, it is recommended to calculate IDC at different standard interest rates, without including country-specific tax or subsidy structures. In this analysis, four interest rate scenarios are considered: 5%, 10%, 15% and 20%.

For the analysis, the next assumptions are made:

- A construction duration of 4 years
- Four interest rate scenarios: 5%, 10%, 15% and 20%.
- OVC values ranging from 4,760 to 8,330 USD/kWe
- Uniform capital expenditures distributed evenly across each year of construction (a standard assumption in preliminary analyses)

To calculate the IDR, the next formula will be used:

$$IDC = \sum_{j=1}^{T} C_j \cdot [(1+r)^{T-j} - 1]$$

Where:

- C_j: capital invested in year j
- r: interest rate
- T: total construction period

Assuming equal annual capital investment, the following annual cash flows are obtained:

- For OVC = 4,760 $/kWe → C_j = 1,190 $/kWe
- For OVC = 8,330 $/kWe → C_j = 2,082.5 $/kWe

Based on these annual flows and the selected interest rates, the following table summarizes the resulting IDC values:

Interest Rate	*OVC ($/kWe)*	*IDC ($/kWe)*
5%	4,760	498.19
	8,330	867.49
10%	4,760	1,032.35
	8,330	1,806.61
15%	4,760	1,612.21
	8,330	2,821.37
20%	4,760	2,237.62
	8,330	3,915.83

Table 46. IDC values for different Interest Rates and OVCs

Based on this method, the estimated IDC for SMRs ranges from 498 to 3,916 USD/kWe, depending on the cost base and financing terms. This component typically accounts for 10%–30% of the total capital investment, and must be included in any comprehensive economic evaluation. As construction durations shorten and financing structures improve (e.g., through government-backed loans), the IDC burden may be significantly reduced, improving the overall competitiveness of SMRs in the energy market.

7.2.4 CAPEX Summary Table

As mentioned earlier, the total CAPEX is defined as the sum of the Overnight Capital Cost (OVC) and the Interest During Construction (IDC). Based on the calculations above, the following table summarizes the minimum and maximum CAPEX values obtained under the assumed financing conditions for each interest rate:

Interest Rate	*OVC ($/kWe)*	*IDC ($/kWe)*	*CAPEX ($/kWe)*
5%	4,760	498	**5,258**
	8,330	867	**9,197**
10%	4,760	1,032	**5,792**
	8,330	1,807	**10,137**
15%	4,760	1,612	**6,372**
	8,330	2,821	**11,151**
20%	4,760	2,238	**6,998**
	8,330	3,916	**12,246**

Table 47. CAPEX values for different Interest Rates and OVCs

While the values above represent generalized cost estimates, additional considerations must be addressed when SMRs are deployed specifically for powering data center infrastructure.

7.2.5 Data Center-Specific Considerations

Although no publicly available studies provide quantified CAPEX adjustments specifically for SMRs integrated into data centers, a structured estimate can be derived by evaluating individual cost drivers and potential synergies. The table below outlines key factors expected to influence the capital expenditure of SMR deployments when tailored for high-reliability digital infrastructure:

Cost Driver	*Justification*	*Estimated Impact*
Tier III/IV power redundancy	Dual electrical feeds, N+1/2N redundancy, and resilient control interfaces	+2% to +5%
Enhanced cybersecurity and physical security	Site surveillance, access control, and nuclear-grade digital infrastructure	+1% to +4%
Regulatory and licensing complexity	Additional planning, urban siting procedures, and safety reviews	+1% to +3%
Subtotal – Additional Requirements		**+4% to +12%**
Cogeneration potential (thermal reuse)	Use of reactor waste heat for absorption cooling or heating systems	−1% to −3%
Proximity to load / on-site deployment	Reduced transmission infrastructure and land acquisition costs	−1% to −2%
Subtotal – Infrastructure Synergies		**−2% to −5%**
Net Estimated CAPEX Adjustment		**+2% to +10%**

Table 48. Adjusted CAPEX for Data Centers

Assuming a CAPEX of $10,000 per kilowatt, the total unadjusted amount for a 100 MW SMR would be $1 billion. Applying the estimated net adjustment range of +2% to +10%, the adjusted value for a data center-integrated SMR would fall between $1.02 and $1.10 billion.

These values reflect the cumulative effect of cost increases related to high-reliability and regulatory requirements, offset in part by infrastructure and thermal integration synergies. While this estimation remains indicative, it provides a transparent and defensible range suitable for preliminary economic assessments. Further cost modeling based on detailed design and siting conditions would be necessary to validate these figures in a real-world deployment scenario. However, these variations will not be considered in the LCOE calculations, as doing so would introduce context-specific assumptions beyond the generalized scope of this thesis.

7.3 Operational Expenditures (OPEX)

7.3.1 Definition and Scope

Operating Expenditures (OPEX) represent the total ongoing costs required to operate a nuclear reactor throughout its commercial life. These include expenses such as staffing, maintenance, refueling, consumables, insurance, and regulatory compliance. It is represented as a consolidated cost per unit of energy produced (USD/MWh).

7.3.2 Total OPEX Estimates

Based on values reported in international literature and advanced reactor economic assessments, the total OPEX for SMRs is commonly estimated to range between 15 and 35 USD/MWh, expressed in 2019 U.S. dollars [81]. This range includes the full spectrum of operational costs, including staffing, maintenance, refueling, insurance, and regulatory compliance, and serves as a benchmark for early-stage cost modeling.

To reflect current economic conditions, these figures have been adjusted for inflation using a cumulative rate of approximately 19% between 2019 and 2023. Applying this correction results in an updated OPEX range of approximately 17.85 to 41.65 USD/MWh, in 2023 dollars. This adjusted range will be used as a reference input for lifecycle cost models and comparative assessments, including LCOE calculations and technology benchmarking.

7.3.3 Breakdown of OPEX by reactor type

Although the analysis adopts a single-value range for OPEX in USD/MWh, a detailed breakdown of operational cost categories is useful for understanding the internal composition of expenses across different reactor types. The table below shows the relative share of each category as a percentage of total OPEX for four reactor technologies: [81].

COST CATEGORY	*PWR*	*SFR*	*HTGR*	*MSR*
Operating Staff Costs	95.98%	58.14%	55.62%	98.18%
O&M staff	35.54%	24.61%	20.21%	30.12%
O&M staff	21.77%	17.70%	12.27%	12.58%
Offsite technical support	13.77%	6.90%	7.94%	17.54%
Management staff	20.90%	13.70%	12.62%	17.68%
Onsite management staff	10.34%	8.41%	6.53%	4.23%
Other admin and general expenses	10.56%	5.29%	6.09%	13.45%
Salary-related costs	9.63%	4.83%	5.55%	12.27%
Utilities, supplies, and consumables	23.40%	11.73%	13.49%	29.81%
Taxes and insurance	6.52%	3.27%	3.76%	8.31%
Annualized fuel cost	4.02%	41.86%	44.38%	1.82%
Refueling operations	0.00%	0.00%	0.00%	0.00%
Nuclear fuel	2.01%	29.46%	36.16%	0.51%
Fuel processing charges	2.01%	12.40%	9.22%	1.31%
Special nuclear materials	0.00%	0.00%	0.00%	0.00%

Table 49. Breakdown of OPEX for PWR, SFR, HTG and MSR

This breakdown highlights significant differences in operational strategies. For example, MSRs show very low fuel costs and high labor intensity, whereas HTGRs and SFRs allocate a larger share of their OPEX to fuel and refueling processes.

Despite the internal variability between reactor types, for modeling purposes this analysis applies a unified OPEX value range, as previously stated.

7.3.4 Considerations for Data Centers

The integration of SMRs into data center environments introduces unique operational challenges and opportunities that may influence operating expenditures. Factors such as increased staffing, stricter maintenance regimes, and cybersecurity requirements can raise costs, while benefits like stable load profiles, thermal reuse, and co-location may reduce them.

The following estimates are not drawn from a single source, but rather represent a reasoned approximation based on qualitative insights and patterns found in nuclear cost literature. They are intended to support comparative modeling under different deployment conditions.

OPEX Factor (Operational)	***Description***	***Estimated Impact***
Increased staffing load	Continuous staffing or shift coverage to meet 24/7 SLA	+5% to +10%
Enhanced preventive maintenance	Shorter maintenance cycles to avoid downtime	+3% to +6%
Ongoing cybersecurity monitoring	Licensing, patching, threat detection, audits	+2% to +4%
Back-up system O&M (if required)	Diesel, battery or hybrid systems require testing and upkeep	+1% to +3%
Higher insurance and compliance	Critical infrastructure status may increase regulatory monitoring burden	+1% to +2%
Flat, high-capacity utilization	Reduces ramping, simplifies load-following, improves thermal stability	−3% to −5%
Co-location & simplified logistics	Shorter distances for inspection, easier access to equipment	−2% to −3%
Thermal integration savings	Less cooling or HVAC energy needed if heat is reused	−1% to −2%
→ Net estimated adjustment		**−3% to +12%**

Table 50. Adjusted OPEX for Data Centers

From a practical standpoint, these operational factors represent a balance of upward and downward pressures on OPEX. On one hand, the mission-critical nature of data center operations often necessitates higher personnel availability, stricter maintenance intervals,

and robust cybersecurity systems. These factors are especially relevant for First-of-a-Kind (FOAK) SMR deployments, where operational conservatism may be higher.

On the other hand, SMRs powering data centers benefit from an exceptionally flat and predictable load profile, which allows for efficient scheduling, minimal load-following requirements, and thermal stability. Co-location with the energy consumer reduces transmission and distribution complexity, while the potential for reusing thermal output—for instance, in absorption cooling systems—can marginally lower auxiliary power demands.

Overall, these conditions justify an estimated OPEX adjustment range of −3% to +12% relative to base values. In economic modeling, this range can be used to capture the potential variability in operational costs depending on site-specific design, regulatory environment, and operational maturity.

While the specific operational context of data centers may lead to minor variations in SMR operating expenditures, this analysis adopts the general OPEX range of 17.85–41.65 USD/MWh as the standard input for all economic modeling. This decision ensures consistency with widely referenced values in international literature and facilitates comparison across different energy technologies. The potential adjustments discussed in the previous section are presented solely for illustrative purposes and are not incorporated into the core financial calculations of this study.

7.4 Cost Adjustment Factors

7.4.1 FOAK vs BOAK Coast Premiums

When estimating the capital costs of SMRs, it is essential to distinguish between different levels of technological maturity. First-of-a-kind (FOAK) reactors, which represent initial commercial or demonstration units, typically face significantly higher costs than more mature designs. These cost increases stem from a combination of factors including limited construction experience, design finalization, non-standardized components, extended licensing processes, and supply chain immaturity.

In contrast, Between-of-a-kind (BOAK) projects refer to subsequent units that benefit from partial standardization and early learning effects, but have not yet reached full economies of scale typical of nth-of-a-kind (NOAK) deployments. BOAK estimates are often used as a practical reference point for assessing realistic near-term deployment scenarios.

According to a review of multiple cost studies [81], FOAK SMRs can be expected to incur a cost premium of 30% to 110% over BOAK equivalents. These ratios reflect the variability across reactor designs and contexts, with literature-based multipliers commonly grouped into three categories:

- **Low estimate:** 1.3× (30% increase)
- **Medium estimate:** 1.6× (60% increase)
- **High estimate:** 2.1× (110% increase)

To quantify the potential impact of FOAK-related cost escalation, the table below applies the identified multipliers—1.3× (low), 1.6× (medium), and 2.1× (high)—to the base BOAK CAPEX range (5,258–12,246 USD/kWe). These adjusted values provide a practical reference for evaluating early-stage SMR deployments under varying levels of project maturity and technological readiness.

FOAK Premium Level	*Min CAPEX (USD/kWe)*	*Max CAPEX (USD/kWe)*
Low (1.3×)	6,835	15,920
Medium (1.6×)	8,413	19,594
High (2.1×)	11,042	25,717

Table 51. FOAK CAPEX values

This analysis adopts the **medium multiplier of 1.6** as the base case for FOAK CAPEX projections. While it is possible that actual premiums could be lower for simplified designs or higher for novel configurations, the 1.6× factor provides a balanced assumption for early-stage economic modeling.

FOAK premiums are applied directly to the inflation-adjusted BOAK CAPEX values discussed in Section 7.2. These adjusted figures serve as the basis for evaluating the lifecycle cost metrics (TCO and LCOE) of initial SMR deployments in data center applications.

7.4.2 Learning Rates

While FOAK deployments typically result in high capital costs due to development complexity, regulatory delays, and first-time implementation challenges, these initial estimates do not fully reflect the long-term economic potential of SMR technologies. As subsequent units are constructed, learning effects play a critical role in cost reduction, driven by improvements in project management, design optimization, manufacturing standardization, and supply chain maturity.

This progressive reduction in cost is quantified through the concept of a learning rate (LR), which represents the percentage decrease in cost for each doubling of cumulative deployment. For example, a learning rate of 10% implies that when the number of deployed units doubles, the cost per unit drops to 90% of its previous value. Mathematically, the cost of the Nth unit can be estimated as [81]:

$$Cost_N = FOAK \cdot (1 - LR)^{\log_2 N}$$

Where:

- $Cost_N$: capital cost of the Nth unit
- *FOAK*: capital cost of the first unit
- *LR*: learning rate (as a decimal)
- *N*: cumulative number of units deployed

While direct learning rate data for advanced SMRs is limited—particularly in markets with low reactor construction volume such as the United States—estimates can be drawn from related industries and international nuclear experience. Literature reviews suggest learning rates in the range of 5% to 15%, depending on the technology, policy environment, and supply chain readiness.

Scenario	*Learning Rate (LR)*
Low	5%
Medium	10%
High	15%

Table 52. Learning Rates Scenarios

These learning rates can be used in scenario modeling to simulate the impact of cumulative deployment on capital costs over time. This is especially relevant for SMRs intended for replicated applications such as fleet deployment across multiple data centers or industrial campuses, where standardization and repetition can yield significant cost savings beyond FOAK premiums.

7.4.3 Multi-unit deployment advantages

A key economic advantage of SMRs lies in their modularity, which allows multiple reactor units to be deployed at the same site, forming a larger power plant with shared infrastructure. This configuration enables economies of scale, particularly in capital and operational expenditures, as components such as turbine systems, auxiliary buildings, control rooms, and maintenance staff can be shared across units.

In capital cost modeling, this benefit is typically captured using a scaling relationship that reflects diminishing marginal costs per unit when multiple reactors are installed together. The capital cost per unit in a multi-module plant can be estimated using a power-law formula [81]:

$$OVC_n = \frac{1}{n} \cdot OVC_1 \cdot n^{MUE}$$

Where:

- OVC_n: adjusted overnight capital cost per unit in a plant with *n* reactor modules
- OVC_1: overnight capital cost for a single-unit configuration
- *MUE*: multi-unit exponent (typically between 0.80 and 0.85)

While this reduction in OVC also lowers the total CAPEX indirectly—by decreasing the capital base used to compute interest during construction (IDC)—the scaling adjustment itself is applied solely to the OVC component. This approach enables more accurate modeling of capital efficiency in modular reactor deployments, particularly in large-scale infrastructure like data centers.

On the operational side, multi-unit plants also benefit from shared staffing, coordinated maintenance, and centralized monitoring. These efficiencies can be captured using a correction factor, known as a multi-unit OPEX multiplier (MOM), which adjusts the baseline OPEX of a single-unit plant downward to reflect multi-unit operation. Estimated MOM values typically range from 0.50 to 0.70, depending on the degree of integration and the number of modules deployed.

Scaling Level	*CAPEX Exponent (MUE)*	*OPEX Multiplier (MOM)*
Low Benefit	0.850	0.700
Medium Benefit	0.825	0.624
High Benefit	0.800	0.500

Table 53. CAPEX Exponents (MUE) and OPEX Multipliers (MOM)

To illustrate the cost reduction potential of multi-unit deployments, the following example assumes a four-unit SMR configuration. The table presents the adjusted overnight capital cost (OVC) and operating expenditures (OPEX) under three scaling scenarios, based on representative multi-unit exponents (MUE) and OPEX multipliers (MOM).

Scenario	*Adjusted OVC (Min) USD /kWe*	*Adjusted OVC (Max) USD /kWe*	*Adjusted OPEX (Min) USD /MWh*	*Adjusted OPEX (Max) USD /MWh*
High Benefit	3,607.92	6,313.85	8.93	20.82
Medium Benefit	3,735.15	6,536.51	11.14	25.99
Low Benefit	3,866.87	6,767.02	12.50	29.15

Table 54. Adjusted OVC and OPEX for different Deployment Scenarios

These results illustrate the economic potential of multi-module SMR configurations. With capital cost reductions of up to ~24% and OPEX savings nearing 50% in high-benefit scenarios, modular deployments offer a compelling path to improve SMR competitiveness—particularly in high-demand, mission-critical applications such as data centers.

7.5 Levelized Cost of Electricity (LCOE)

7.5.1 LCOE as a Comparative Metric

The Levelized Cost of Electricity (LCOE) is a widely used metric in energy economics that represents the average cost per unit of electricity generated over the lifetime of a power plant. It incorporates all relevant costs, including capital expenditures (CAPEX), operating expenditures (OPEX), fuel costs, and decommissioning, discounted to present value. The LCOE provides a standardized basis for comparing different power generation technologies on an equivalent basis, regardless of their size, lifetime, or operational characteristics.

By expressing costs in terms of USD per megawatt-hour (USD/MWh), the LCOE enables stakeholders, policymakers, and investors to assess the relative economic competitiveness of technologies such SMRs, renewables, and fossil fuel plants. It captures not only the upfront investment but also ongoing expenses and fuel price volatility, thus providing a holistic view of total generation costs.

In this analysis, the LCOE will be calculated for SMR deployment scenarios including FOAK, BOAK, and multi-unit configurations, using consistent assumptions on cost inputs, financing, capacity factors, and operational lifetimes. The resulting LCOE values will be benchmarked against those of alternative energy sources to evaluate the economic feasibility of SMRs for powering data centers.

7.5.2 Input Parameters and Assumptions

The calculation of the Levelized Cost of Electricity (LCOE) relies on key technical and economic inputs that describe the characteristics of the Small Modular Reactor (SMR) scenarios analyzed. These inputs include:

- **Capital Expenditure (CAPEX):** Inflation-adjusted overnight capital costs, including Interest During Construction (IDC) and decommissioning, expressed in USD per kilowatt electric (USD/kWe).
- **Operating Expenditure (OPEX):** Annual operational costs, covering staffing, maintenance, fuel, insurance, and regulatory compliance, expressed in USD per megawatt-hour (USD/MWh).
- **Capacity Factor (CF):** The ratio of actual electricity generated to the maximum possible output, assumed at 92%, reflecting continuous operation aligned with data center power needs.
- **Plant Lifetime (n):** Assumed to be 60 years, consistent with projections for advanced nuclear technologies.
- **Discount Rate (r):** Four rates are considered—5%, 10%, 15% and 20%—to capture a range of financing conditions and risk profiles.

The LCOE is calculated as the present value of all costs over the plant's lifetime divided by the present value of total electricity generated, using the formula [84]:

$$LCOE = \frac{\sum_{t=0}^{n} \frac{CAPEX_t + OPEX_t}{(1+r)^t}}{\sum_{t=0}^{n} \frac{Electricity\ Output_t}{(1+r)^t}}$$

Where:

- $CAPEX_t$ is the capital expenditure at year t, generally concentrated at $t=0$.
- $OPEX_t$ is the operational expenditure in year t.
- $Electricity\ Output_t$ is the electricity generated at year t (MWh).
- r is the discount rate.
- n is the operational lifetime in years.

In this analysis, decommissioning costs are assumed to be incorporated within the CAPEX and are not considered as separate cash flows.

To provide a comprehensive assessment of financial uncertainty and risk, this study evaluates the LCOE under four discount rate scenarios: 5%, 10%, 15%, and 20%. This range captures a spectrum of potential financing conditions, from favorable institutional support or public-private partnerships (5%) to high-risk, commercially financed deployments (20%). By incorporating these multiple scenarios, the analysis enables a more robust sensitivity assessment of how discount rates affect the economic feasibility of SMRs powering data centers.

7.5.3 LCOE Calculation for Different Deployment Scenarios

The economic viability of SMRs can vary significantly depending on the stage of deployment, scale of installation, and financial conditions. To capture this variability, the LCOE is calculated across a range of representative scenarios reflecting both technology maturity and multi-unit deployment. Each scenario is analyzed under four discount rate assumptions, 5%, 10%, 15% and 20%, to account for favorable and conservative financing environments.

The four scenarios considered are:

- **BOAK Single Unit:** Represents a mature design with moderate cost reductions from initial deployments, reflecting typical near-term commercial plants operating as standalone units.
- **BOAK with 4 Units:** Models the cost efficiencies and operational synergies arising from deploying four SMR units at a single site, applying multi-unit scaling effects to both capital and operational expenditures.
- **FOAK Single Unit:** Captures the higher costs and risks associated with initial demonstration units, including design finalization, licensing, and supply chain challenges, resulting in significant FOAK premiums applied to the BOAK baseline.
- **FOAK with 4 Units:** Combines the elevated FOAK costs with multi-unit scale benefits, reflecting the potential for cost reductions even within first deployment clusters.

Each scenario's LCOE is computed using the adjusted CAPEX and OPEX values, discounted over the assumed 60-year plant lifetime at different discount rates. This yields a total of sixteen distinct LCOE values per scenario, enabling a comprehensive assessment of SMR economics under varying maturity and scale conditions.

7.5.3.1 BOAK Single Unit

The following table summarizes the LCOE for the BOAK Single Unit across different discount rates, CAPEX and OPEX:

Discount Rate	*CAPEX ($/kWe)*	*OPEX ($/MWh)*	*LCOE ($/MWh)*
5%	5,258	17.85	51.88
		41.65	75.68
	9,197	17.85	78.12
		41.65	101.92
10%	5,792	17.85	89.92
		41.65	113.72
	10,137	17.85	143.05
		41.65	167.85
15%	6,372	17.85	136.45
		41.65	160.25
	11,151	17.85	225.40
		41.65	249.20
20%	6,998	17.85	191.52
		41.65	215.32
	12,246	17.85	321.76
		41.65	345.56

Table 55. BOAK Single Unit LCOE values

The LCOE results for the BOAK single-unit SMR scenario with a 92% capacity factor demonstrate the substantial influence of capital expenditure, operational costs, and financing conditions on overall economics. Under the lower CAPEX assumption of 5,258 USD/kWe and a 5% discount rate, the LCOE ranges from approximately **52 to 76 USD/MWh**, depending on OPEX. When the discount rate increases to 10%, the LCOE rises to between **90 and 114 USD/MWh** for a CAPEX of 5,792 USD/kWe.

For higher CAPEX scenarios, such as 9,197 and 10,137 USD/kWe, LCOE estimates span from **78 to 102 USD/MWh** (at 5%) and **143 to 168 USD/MWh** (at 10%), respectively. As the discount rate increases to 15% and 20%, LCOE values escalate significantly—reaching up to **345 USD/MWh** under the most conservative assumptions (12,246 USD/kWe at 20% discount rate and high OPEX). These results highlight the high sensitivity of LCOE to both the cost of capital and operational efficiency, emphasizing the critical role of financing structures, OPEX control, and deployment strategies in determining the economic viability of SMRs for data center applications.

7.5.3.2 BOAK with 4 Units

The following table summarizes the LCOE for the BOAK with 4 Units across different discount rates, CAPEX and OPEX:

Discount Rate	*CAPEX ($/kWe)*	*OPEX ($/MWh)*	*LCOE ($/MWh)*
5%	4,025	11.14	37.52
		25.99	52.37
	7,043	11.14	57.31
		25.99	72.16
10%	4,334	11.14	65.09
		25.99	79.94
	7,584	11.14	105.55
		25.99	120.40
15%	4,663	11.14	97.94
		25.99	112.79
	8,160	11.14	163.05
		25.99	177.90
20%	5,013	11.14	135.54
		25.99	150.39
	8,772	11.14	228.83
		25.99	243.68

Table 56. BOAK with 4 units LCOE values

The LCOE results indicate that deploying SMRs as a cluster of four units significantly improves cost efficiency due to shared infrastructure and operational synergies. For the lower CAPEX estimate of 4,025 USD/kWe, the LCOE ranges from approximately 37.52 to 52.37 USD/MWh at a 5% discount rate, increasing to 65.09 to 79.94 USD/MWh at 10%. For the higher CAPEX estimate of 7,043 USD/kWe, the LCOE rises to 57.31–72.16 USD/MWh at 5%, and to 105.55–120.40 USD/MWh for 7,584 USD/kWe at 10%. At higher discount rates, LCOE values continue to rise significantly: for example, 8,772 USD/kWe at a 20% discount rate results in LCOEs of 228.83 to 243.68 USD/MWh.

These figures reflect a medium-benefit multi-unit scenario, representing a balanced outlook between optimistic and conservative assumptions on cost reductions from modular deployment. This analysis highlights how multi-unit deployment can enhance the economic attractiveness of SMRs, especially for applications like data centers that require scalable and reliable power solutions.

7.5.3.3 FOAK Single Unit

The following table summarizes the LCOE for the FOAK Single Unit across different discount rates, CAPEX and OPEX:

Discount Rate	CAPEX ($/kWe)	OPEX ($/MWh)	LCOE ($/MWh)
5%	8,413	17.85	72.99
		41.65	96.79
	14,715	17.85	114.31
		41.65	138.11
10%	9,267	17.85	133.22
		41.65	157.02
	16,219	17.85	219.76
		41.65	243.56
15%	10,195	17.85	207.65
		41.65	231.45
	17,842	17.85	350.01
		41.65	373.81
20%	11,197	17.85	295.72
		41.65	319.52
	19,594	17.85	504.11
		41.65	527.91

Table 57. FOAK Single Unit LCOE values

The LCOE results for the FOAK single-unit SMR scenario show a clear increase in cost relative to the BOAK scenario. With a CAPEX of 8,413 USD/kWe, the LCOE ranges from 72.99 to 96.79 USD/MWh at a 5% discount rate, depending on operational costs. As the discount rate increases to 10%, the LCOE rises to between 133.22 and 157.02 USD/MWh. For a higher CAPEX scenario of 16,219 USD/kWe, LCOE values increase significantly, ranging from 219.76 to 243.56 USD/MWh at 10%, and up to 350.01 to 373.81 USD/MWh at a 15% discount rate. In the most conservative case, with a CAPEX of 19,594 USD/kWe and a 20% discount rate, the LCOE reaches between 504.11 and 527.91 USD/MWh. These values reflect the higher initial costs and the financial risks associated with FOAK deployments, which are more expensive than BOAK scenarios. As more units are built and the design matures, it is expected the LCOE to decrease as a result of learning effects and economies of scale.

7.5.3.3 FOAK with 4 Units

The following table summarizes the LCOE for the FOAK with 4 Units across different discount rates, CAPEX and OPEX:

Discount Rate	*CAPEX ($/kWe)*	*OPEX ($/MWh)*	*LCOE ($/MWh)*
5%	6,440	11.14	53.35
		25.99	68.20
	11,269	11.14	85.01
		25.99	99.86
10%	6,934	11.14	97.46
		25.99	112.31
	12,134	11.14	162.20
		25.99	177.05
15%	7,461	11.14	150.04
		25.99	164.89
	13,056	11.14	254.20
		25.99	269.05
20%	8,020	11.14	210.17
		25.99	225.02
	14,035	11.14	359.44
		25.99	374.29

Table 58. FOAK with 4 units LCOE values

The LCOE results for the FOAK with 4 units scenario illustrate a clear impact of both CAPEX and OPEX on the cost of electricity. With a CAPEX of 6,440 USD/kWe, the LCOE ranges from 53.35 to 68.20 USD/MWh at a 5% discount rate, increasing to 97.46 to 112.31 USD/MWh at a 10% discount rate.

For the higher CAPEX case of 12,134 USD/kWe, LCOE increases substantially, ranging from 162.20 to 177.05 USD/MWh at 10%, and for the most conservative scenario of 14,035 USD/kWe and a 20% discount rate, it reaches values between 359.44 and 374.29 USD/MWh.

These results clearly demonstrate that while multi-unit deployments provide economies of scale and shared infrastructure, the initial higher costs associated with FOAK technologies still result in significantly higher LCOE values compared to BOAK configurations.

7.6 Total Cost of Ownership (TCO)

7.6.1 TCO Definition and Scope

The Total Cost of Ownership (TCO) represents the full, lifetime financial commitment associated with deploying and operating an energy generation system. In the context of this thesis, TCO is used to evaluate the long-term economic viability of SMRs, both as standalone units and when integrated directly with data centers.

TCO encompasses all capital expenditures, ongoing operational expenditures, and other lifecycle-related costs such as maintenance, staffing, insurance, regulatory compliance, and fuel. While decommissioning costs are not always reported consistently across studies, in this analysis they are assumed to be embedded within the CAPEX estimates used.

TCO provides a more holistic view than LCOE alone, as it reflects not just the cost of producing electricity but the total financial burden of owning and operating the asset over time. This is particularly relevant for data center applications, where energy reliability and infrastructure integration can significantly influence long-term costs.

This chapter presents two distinct TCO analyses:

- A **baseline scenario**, in which the SMR is deployed as a traditional, standalone energy asset.
- A **co-location scenario**, where the SMR is specifically designed to power a large-scale data center, incorporating any synergies or additional costs related to integration.

By comparing these two configurations, the analysis aims to quantify the cost impact of adapting SMR deployments to meet the unique energy, reliability, and operational requirements of data centers.

7.6.2 Methodology and Formula

To calculate the TCO for SMR-based power solutions, this study adopts a discounted cash flow approach. The TCO formula accounts for both upfront capital investment and all future operational costs, adjusted for the time value of money using an appropriate discount rate. This methodology enables direct comparison between different energy supply configurations over a 60-year project horizon.

The TCO is calculated using the following formula [80]:

$$TCO = CAPEX + \sum_{t=1}^{n} \frac{OPEX_t}{(1+r)^t}$$

Where:

- **CAPEX:** Capital Expenditures include the costs of constructing and installing the SMR plant, along with any grid infrastructure and safety systems. Decommissioning costs are already included in this figure.
- **OPEX:** Operating Expenditures cover staffing, maintenance, insurance, fuel, regulatory compliance, and other ongoing costs.
- **r:** Discount rate, reflecting the cost of capital and risk over time.
- **n:** The expected lifespan of the reactor (usually 60 years).

For simplicity, this analysis assumes a constant annual OPEX value in real terms, adjusted using the chosen discount rate. This assumption is in line with standard practice in long-term infrastructure cost modeling, particularly for technologies with predictable operating profiles like nuclear.

The following discount rates are considered to reflect a range of financial environments:

- **5%**: Represents a favorable financing scenario (e.g., public-private partnerships, government guarantees).
- **10%**, **15%**, and **20%**: Represent increasingly conservative or high-risk scenarios, capturing variability in capital markets and project risk perception.

This approach allows the TCO to serve as a complementary metric to LCOE, especially when assessing infrastructure that may offer non-economic value—such as reliability, proximity to load, or carbon neutrality. It also provides a practical framework for evaluating the economic implications of integrating SMRs directly with hyperscale data centers, where infrastructure synergies and reliability requirements can significantly alter long-term cost dynamics.

7.6.3 Case A: SMR Standalone Configuration

To assess the long-term economic viability of deploying SMRs in data center applications, this study evaluates sixteen distinct TCO scenarios. These scenarios are built from combinations of two deployment types—BOAK and FOAK—two plant configurations (a single 100 MW unit or a cluster of four 100 MW units), and four discount rates: 5%, 10%, 15%, and 20%. The goal is to understand how scale, technological maturity, and financial conditions influence the full cost profile of SMR deployments.

The inputs for CAPEX and OPEX are taken as average values from previous results for each configuration. For example, the OPEX used for single-unit plants is fixed at 29.75 USD/MWh, while for multi-unit clusters it is 18.57 USD/MWh, reflecting economies of scale. The plant lifespan is assumed to be 60 years with a 92% capacity factor.

To illustrate the calculation process, consider the case of a BOAK single-unit deployment with a 5% discount rate. The plant has a capacity of 100 MW, a CAPEX of 7,228 USD/kW, and an OPEX of 29.75 USD/MWh. Over one year, the reactor produces approximately 805,920 MWh. This results in an annual operational cost of roughly 23.96 million USD. Using a

present value annuity factor corresponding to 60 years at a 5% discount rate (approximately 18.256), the lifetime present value of OPEX is about 97.78 million USD. When added to the total capital investment of 722.8 million USD, the TCO reaches 820.58 million USD. Expressed per unit, this equates to 8,205.84 USD/kW and 24.37 USD/MWh.

The following table summarizes the TCO values for all cases in three key units: total USD, USD/kW, and USD/MWh. These values facilitate direct comparison across deployment strategies and financial conditions.

Scenario	*Discount Rate*	*TCO (USD/kW)*	*TCO (USD/MWh)*	*TCO (USD billion)*
BOAK Single Unit (1x100 MW)	5%	8,205.84	24.37	0.821
	10%	10,670.71	22.07	1.067
	15%	12,704.71	20.98	1.270
	20%	15,199.12	20.42	1.520
BOAK with 4 Units (4X100 MW)	5%	8,454.47	23.43	3.382
	10%	8,998.67	21.90	3.599
	15%	9,577.78	20.92	3.831
	20%	10,193.35	20.41	4.077
FOAK Single Unit (1x100 MW)	5%	13,110.14	38.93	1.311
	10%	15,491.32	36.83	1.549
	15%	17,992.83	35.31	1.799
	20%	20,664.89	34.23	2.066
FOAK with 4 Units (4X100 MW)	5%	11,589.40	23.96	4.636
	10%	10,944.48	22.63	4.378
	15%	11,191.24	23.14	4.476
	20%	11,776.86	24.36	4.711

Table 59. SMR Standalone TCO values for different Scenarios

The TCO results reveal clear patterns across the 16 analyzed scenarios. First-of-a-kind (FOAK) deployments, both single-unit and multi-unit, consistently present higher costs than their BOAK counterparts due to increased capital investments tied to early-stage development, licensing, and engineering risk. However, deploying SMRs as multi-unit clusters reduces the TCO substantially in both FOAK and BOAK cases, highlighting the economic advantages of modularity, infrastructure sharing, and economies of scale.

While higher discount rates may result in lower TCO values when expressed in USD/MWh, this does not reflect better financial performance. Rather, it is a result of the mathematical effect of discounting: higher rates disproportionately reduce the present value of long-term OPEX, which can make total costs appear lower. This effect can mask the fact that higher discount rates represent more expensive or riskier capital, which negatively impacts the real-world feasibility of a project.

In conclusion, TCO is a valuable metric for comparing lifecycle costs, but it must be interpreted alongside financial risk and project context. For SMR-powered data centers, BOAK multi-unit deployments operating under moderate discount rates (10–15%) appear to offer the best balance between cost-efficiency and implementation viability.

7.6.4 Case B: SMR + Data Center Co-location

In addition to evaluating standalone SMR deployments, a second TCO case was developed to assess the financial implications of co-locating an SMR with a data center. This scenario captures both cost synergies—such as reduced transmission infrastructure and thermal reuse—and additional expenditures tied to higher reliability standards, regulatory complexity, and enhanced cybersecurity or physical protection. To model this configuration more accurately, the TCO was computed as the sum of two differentiated components: one corresponding to the SMR system tailored for data center integration, and the other reflecting the infrastructure and operations of a conventional hyperscale data center.

For the SMR component, the CAPEX and OPEX values were adjusted based on integration-related assumptions discussed in Sections 7.2.5 and 7.3.4, where cost impacts of co-location were estimated through both upward and downward modifiers. Specifically, the CAPEX is estimated to increase by a range of +2% to +10% over the general SMR baseline. Meanwhile, the OPEX may vary between −3% and +12%, depending on the balance between operational synergies and added complexity in staffing, monitoring, or compliance.

The following table summarizes the adjusted CAPEX and OPEX ranges for each scenario:

Scenario	*Discount Rate*	*CAPEX General (USD/kW)*	*CAPEX Data Center (USD/kW)*	*OPEX General (USD/MWh)*	*OPEX Data Center (USD/MWh)*
BOAK Single Unit (1x100 MW)	5%	7,228	7,373 – 7,951	29.75	28.86 – 33.32
	10%	7,965	8,124 – 8,762		
	15%	8,762	8,937 – 9,638		
	20%	9,622	9,814 – 10,584		
BOAK with 4 Units (1x100 MW)	5%	5,534	5,645 – 6,087	18.57	18.01 – 20.80
	10%	5,959	6,078 – 6,555		
	15%	6,412	6,540 – 7,053		
	20%	6,893	7,031 – 7,582		
FOAK Single Unit (1x100 MW)	5%	11,564	11,795 – 12,720	29.75	28.86 – 33.32
	10%	12,743	12,998 – 14,017		
	15%	14,019	14,299 – 15,421		
	20%	15,396	15,704 – 16,936		
BOAK with 4 Units (4x100 MW)	5%	8,555	9,032 – 9,740	18.57	18.01 – 20.80
	10%	9,534	9,725 – 10,487		
	15%	10,259	10,464 – 11,285		
	20%	11,028	11,249 – 12,131		

Table 60. Adjusted CAPEX and OPEX for Data Centers for different Scenarios

Based on industry benchmarks for Tier III–IV hyperscale facilities, the capital expenditure (CAPEX) for a 100 MW IT-load data center typically ranges between 10,000 and 15,000 USD per kilowatt. These figures reflect the high infrastructure, redundancy, and reliability standards required at this scale. On the operational side, annual operating expenditures (OPEX)—which include energy, cooling, maintenance, and staffing—generally fall between 11 and 28 USD per megawatt-hour. These values provide a realistic cost reference for modeling the TCO of a data center when evaluating co-location with an SMR [85].

Table X presents the total TCO results for all 16 scenarios, combining SMR and data center costs across different deployment configurations and discount rates:

Scenario	Discount Rate	CAPEX Total (USD/kW)	OPEX Total (USD/MWh)	TCO (USD/kW)	TCO (USD/MWh)	TCO (billion USD)
BOAK Single Unit (1x100 MW)	5%	17,373 – 22,951	39.86 – 61.32	18,391 – 24,517	38.03 – 50.70	1.839 – 2.452
	10%	18,124 – 23,762		20,058 – 26,737	41.48 – 55.29	2.006 – 2.674
	15%	18,937 – 24,638		21,829 – 29,087	45.14 – 60.15	2.183 – 2.909
	20%	19,814 – 25,584		23,669 – 31,514	48.95 – 65.17	2.367 – 3.151
BOAK with 4 Units (4x100 MW)	5%	15,645 – 21,087	29.01 – 48.80	15,386 – 22,334	31.82 – 46.19	6.154 – 8.933
	10%	16,078 – 21,555		17,485 – 23,923	36.16 – 49.47	6.994 – 9.569
	15%	16,540 – 22,053		18,645 – 25,593	38.56 – 52.93	7.458 – 10.237
	20%	17,031 – 22,582		19,837 – 27,302	41.02 – 56.46	7.935 – 10.921
FOAK Single Unit (1x100 MW)	5%	21,795 – 27,720	39.86 – 61.32	22,813 – 29,286	47.18 – 60.57	2.281 – 2.929
	10%	22,998 – 29,017		24,932 – 31,992	51.56 – 66.16	2.493 – 3.199
	15%	24,299 – 30,421		27,191 – 34,870	56.23 – 72.11	2.719 – 3.487
	20%	25,704 – 31,936		29,559 – 37,866	61.13 – 78.31	2.956 – 3.787
FOAK with 4 Units (4x100 MW)	5%	19,032 – 24,740	29.01 – 48.80	19,773 – 25,987	40.89 – 53.74	7.909 – 10.395
	10%	19,725 – 25,487		21,132 – 27,855	43.70 – 57.60	8.453 – 11.142
	15%	20,464 – 26,285		22,569 – 29,825	46.67 – 61.68	9.027 – 11.930
	20%	21,249 – 27,131		24,055 – 31.851	49.75 – 65.87	9.622 – 12.740

Table 61. SMR + Data Center TCO values for different Scenarios

These results confirm that co-locating SMRs with data centers introduces a new set of trade-offs that significantly influence the total cost of ownership. While integration can lead to valuable synergies—such as thermal reuse and reduced grid infrastructure—these are often offset by increased capital and operational demands driven by high-reliability standards, cybersecurity protocols, and complex regulatory interfaces.

Across all scenarios, TCO values are consistently higher than in standalone SMR deployments, particularly under FOAK conditions. Single-unit configurations remain the most expensive, whereas multi-unit BOAK scenarios show the best cost performance, with TCO ranging from approximately 31,082 to 49,472 USD/kW and 36.16 to 50.70 USD/MWh. As in the previous case, higher discount rates reduce TCO values in MWh terms due to the present value discounting of long-term OPEX, but they reflect greater financing difficulty and investor risk.

Ultimately, these findings suggest that while co-location strategies offer potential long-term efficiency gains, they require careful cost-benefit balancing and are more economically attractive in mature, lower-risk deployment environments. For applications where energy reliability, physical security, and integration flexibility are paramount—such as hyperscale or mission-critical data centers—SMR co-location remains a promising but capital-intensive solution.

7.7 Comparative Analysis with Alternative Energy Sources

7.7.1 Selected Technologies: Coal, Gas, Solar PV, Wind, Grid Supply

When assessing the economic and operational viability of Small Modular Reactors (SMRs) for powering data centers, it is essential to benchmark against a range of alternative energy sources commonly utilized in this sector. The selected technologies for comparison include:

- **Natural Gas:** A prevalent fossil fuel source characterized by relatively low capital expenditures and high operational flexibility. Gas plants can efficiently provide baseload and peaking power. However, they are subject to fuel price volatility and emit significant amounts of greenhouse gases, which may incur carbon pricing or regulatory costs.
- **Coal:** A traditional energy source with high capital and operational costs, substantial greenhouse gas emissions, and increasing regulatory constraints. While its role in many energy systems is declining, coal-fired generation remains significant in some regions, thus relevant for economic and environmental benchmarking.
- **Conventional Nuclear:** A stable, low-emission baseload power source with high capital costs and extended construction periods. It provides reliable 24/7 electricity, making it well-suited for data centers, but faces challenges related to regulatory hurdles, public acceptance, and nuclear waste management.
- **Wind Power:** A renewable energy source with low operating costs and zero direct emissions. Its intermittency requires integration with energy storage or backup power to ensure reliable supply for continuous data center operations. Wind resource availability varies widely with location.
- **Solar Photovoltaic (PV):** A rapidly growing renewable source offering clean, scalable electricity with near-zero fuel costs. Like wind, solar power is variable and typically requires storage systems to maintain uninterrupted supply, especially critical for data centers demanding 24/7 power.

This suite of technologies covers a broad spectrum of economic, operational, and environmental profiles, providing a comprehensive context to evaluate the potential of SMRs in powering data centers.

7.7.2 CAPEX and OPEX Benchmark Ranges

This section compares the average CAPEX and OPEX of SMR deployments—across BOAK and FOAK, single and multi-unit configurations—with benchmark values for other energy technologies, such as CCGT, coal, solar PV, and wind. Due to the strong influence of financing conditions, the comparison is structured by discount rate: 5%, 10%, 15%, and 20%. Four tables summarize the results, and each is briefly introduced and analyzed below.

The following table presents representative values for conventional and renewable energy sources [86]:

Energy Source	CAPEX (USD/ kWe)		OPEX (USD/MWh)	
	Low	High	Low	High
Natural Gas	700	1,150	15	40
Coal	3,200	6,775	8	23
Conventional Nuclear	8,475	13,925	21	28
Wind (Onshore)	1,025	1,700	10	15
Solar PV	700	1,400	5	15

Table 62. CAPEX and OPEX values for different Energy Sources

7.7.2.1 5% Discount Rate

The table bellow summarizes the values for BOAK and FOAK SMR configurations under a 5% discount rate:

SMR	CAPEX (USD/ kWe)		OPEX (USD/MWh)	
	Low	High	Low	High
BOAK Single Unit	5,258	9,197	17.85	41.65
BOAK with 4 Units	4,025	7,043	11.14	25.99
FOAK Single Unit	8,413	14,715	17.85	41.65
FOAK with 4 Units	6,440	11,269	11.14	25.99

Table 63. CAPEX and OPEX values for different SMR Scenarios with a 5% Discount Rate

The following charts compare the CAPEX and OPEX of SMRs with those of conventional energy sources under a 5% discount rate:

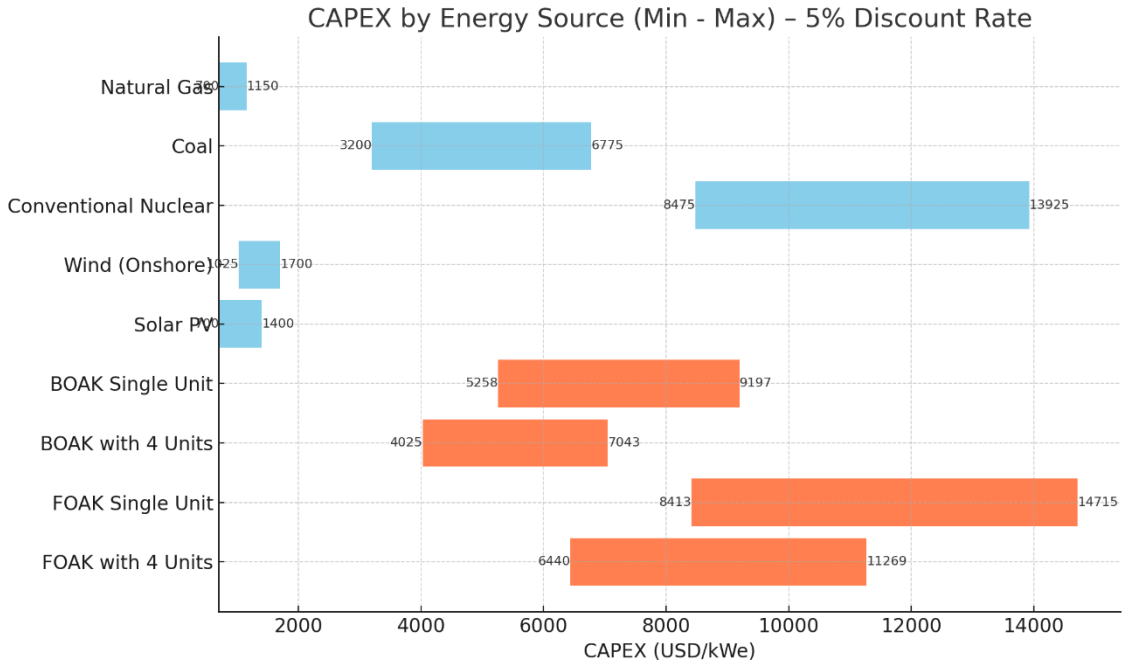


Figure 54. CAPEX by Energy Source at 5% Discount Rate

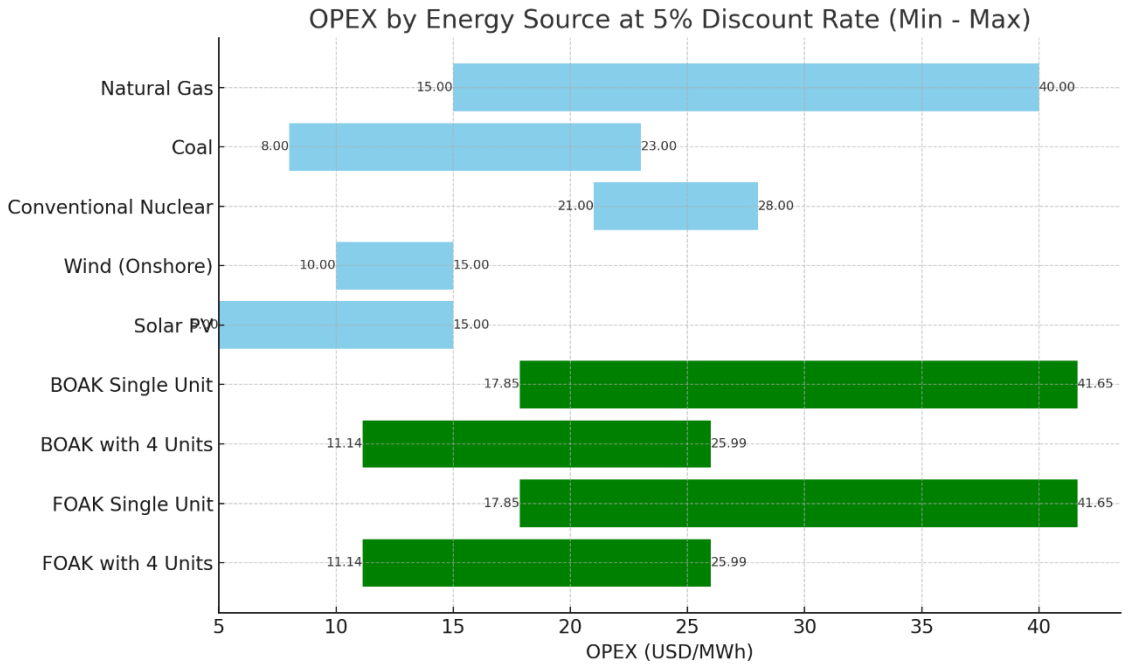


Figure 55. OPEX by Energy Source at 5% Discount Rate

The CAPEX and OPEX comparisons at a 5% discount rate reveal several important insights. While SMRs—especially FOAK deployments—show significantly higher capital costs compared to other energy sources, their operational costs are relatively competitive, particularly in BOAK and multi-unit configurations. This highlights the potential long-term value of SMRs in stable, high-capacity applications like data centers, despite their upfront investment requirements.

Notably, the 5% discount rate represents a favorable financing environment that reduces the impact of high CAPEX on lifecycle economics. Under these conditions, the cost competitiveness of SMRs improves, particularly in BOAK scenarios with shared infrastructure. However, compared to renewables like onshore wind and solar PV, SMRs still face a substantial cost gap in both CAPEX and OPEX at this rate, underscoring the importance of continued innovation and deployment experience to close this differential.

7.7.2.2 10% Discount Rate

The table bellow summarizes the values for BOAK and FOAK SMR configurations under a 10% discount rate:

SMR	***CAPEX (USD/ kWe)***		***OPEX (USD/MWh)***	
	Low	High	Low	High
BOAK Single Unit	5,792	10,137	17.85	41.65
BOAK with 4 Units	4,334	7,584	11.14	25.99
FOAK Single Unit	9,267	16,219	17.85	41.65
FOAK with 4 Units	6,934	12,134	11.14	25.99

Table 64. CAPEX and OPEX values for different SMR Scenarios with a 10% Discount Rate

The following chart compares the CAPEX and of SMRs with those of conventional energy sources under a 10% discount rate. Note that the OPEX values remain unchanged across discount rate scenarios, so the OPEX chart is identical to the one shown for the 5% case.

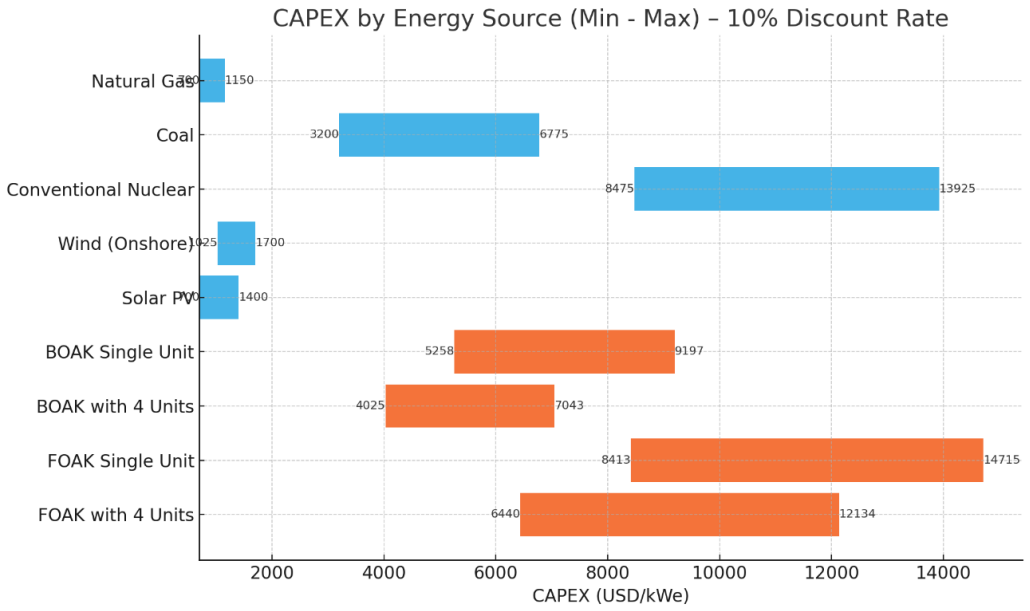


Figure 56. CAPEX by Energy Source at 10% Discount Rate

At a 10% discount rate, CAPEX values for all SMR configurations increase compared to the 5% scenario, as expected due to the higher cost of capital impacting the total investment. FOAK deployments remain the most capital-intensive options, especially the FOAK with 4 Units case, which reaches up to 16,219 USD/kWe. However, even BOAK SMRs—particularly the 4-unit deployment—remain relatively competitive, with values still lower than those of conventional nuclear.

When comparing with traditional energy sources, SMRs sit above wind, solar, coal, and gas in terms of capital cost, but below conventional nuclear in many cases. This underlines the persistent challenge of high upfront investment for SMRs, even under more conservative financial assumptions. The rising CAPEX with increasing discount rates reinforces the importance of favorable financing mechanisms for SMR viability.

7.7.2.3 15% Discount Rate

The table bellow summarizes the values for BOAK and FOAK SMR configurations under a 15% discount rate:

SMR	***CAPEX (USD/ kWe)***		***OPEX (USD/MWh)***	
	Low	High	Low	High
BOAK Single Unit	6,372	11,151	17.85	41.65
BOAK with 4 Units	4,663	8,160	11.14	25.99
FOAK Single Unit	10,195	17,842	17.85	41.65
FOAK with 4 Units	7,461	13,056	11.14	25.99

Table 65. CAPEX and OPEX values for different SMR Scenarios with a 15% Discount Rate

The following chart compares the CAPEX and of SMRs with those of conventional energy sources under a 15% discount rate. Note that the OPEX values remain unchanged across discount rate scenarios, so the OPEX chart is identical to the one shown for the 5% and 10 % cases.

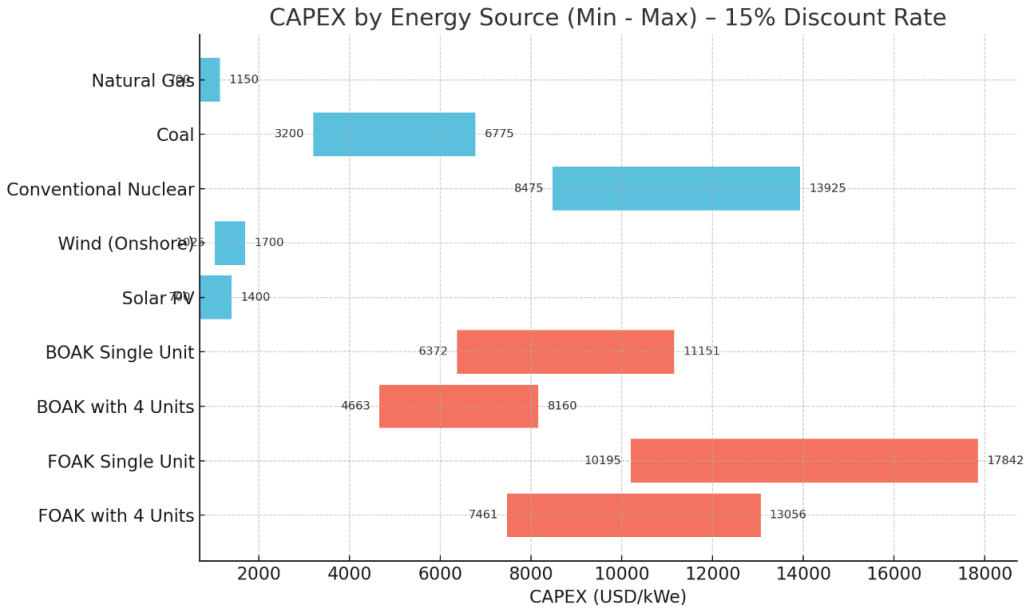


Figure 57. CAPEX by Energy Source at 15% Discount Rate

Compared to lower discount rates, the 15% discount rate scenario shows a consistent increase in CAPEX values across all SMR cases. This reflects the higher cost of capital and perceived project risk. For instance, FOAK 4-unit configurations rise from a maximum of around 11,269 USD/kWe at 5% to 13,056 USD/kWe at 15%, emphasizing the financial burden imposed by steeper discounting. While the relative ranking among SMR types remains unchanged—BOAK 4-unit being the least capital-intensive and FOAK single-unit the most—the overall gap with conventional technologies like coal and natural gas widens. This demonstrates how higher discount rates penalize capital-heavy technologies such as SMRs more severely, reinforcing the importance of favorable financing conditions to enhance SMR competitiveness.

7.7.2.4 20% Discount Rate

The table bellow summarizes the values for BOAK and FOAK SMR configurations under a 15% discount rate:

SMR	***CAPEX (USD/ kWe)***		***OPEX (USD/MWh)***	
	Low	High	Low	High
BOAK Single Unit	6,998	12,246	17.85	41.65
BOAK with 4 Units	5,013	8,772	11.14	25.99
FOAK Single Unit	11,197	19,594	17.85	41.65
FOAK with 4 Units	8,020	14,035	11.14	25.99

Table 66. CAPEX and OPEX values for different SMR Scenarios with a 20% Discount Rate

The following chart compares the CAPEX and of SMRs with those of conventional energy sources under a 20% discount rate. Note that the OPEX values remain unchanged across discount rate scenarios, so the OPEX chart is identical to the one shown for the 5%, 10% and 15% cases.

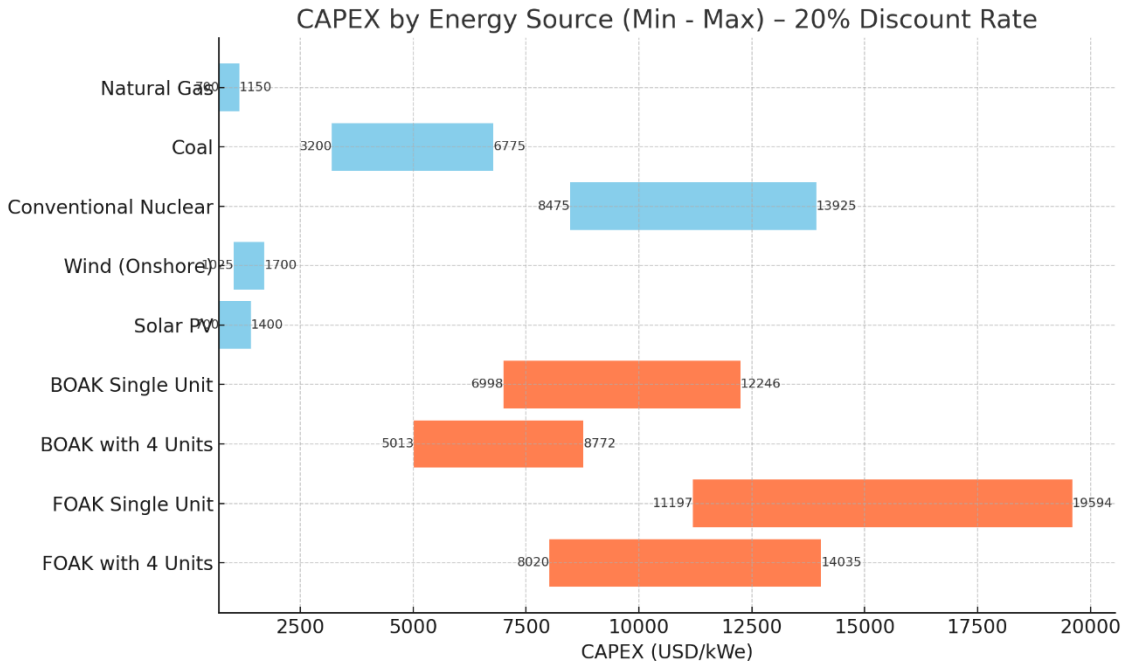


Figure 58. CAPEX by Energy Source at 20% Discount Rate

At a 20% discount rate, CAPEX values for all SMR configurations increase notably, with FOAK designs—especially multi-unit ones—reaching some of the highest capital cost ranges across all scenarios. These elevated figures illustrate how high discount rates amplify the financial burden of early-stage nuclear projects, pushing SMRs further from the cost range of conventional energy sources like natural gas, wind, or even coal.

Compared to 5%, 10%, and 15% discount rate scenarios, the 20% case underscores the growing disadvantage of FOAK SMRs in high-risk financing environments. Although BOAK and multi-unit configurations still offer some cost mitigation, maintaining discount rates in the 5%–10% range appears essential to preserving the economic viability of SMR deployments.

7.7.3 LCOE Comparison: SMR vs Alternative Sources

This section presents a comparative analysis of LCOE values for SMR deployments against representative LCOE ranges from alternative energy sources. Recognizing the critical impact of financing costs, the analysis is segmented by discount rate: 5%, 10%, 15%, and 20%. Four summary tables are provided, each corresponding to a different discount rate. These are introduced individually below, followed by a focused discussion of the trends and relative performance of SMRs under varying financial assumptions.

The following table presents representative values for conventional and renewable energy sources [87]:

Energy Source	LCOE (USD/MWh)	
	Low	High
Natural Gas	115	221
Coal	68	166
Conventional Nuclear	141	221
Wind (Onshore)	24	75
Solar PV	24	96

Table 67. LCOE values for different Energy Sources

7.7.3.1 5% Discount Rate

The table bellow summarizes the values for BOAK and FOAK SMR configurations under a 5% discount rate:

SMR	LCOE (USD/MWh)	
	Low	High
BOAK Single Unit	51.88	101.92
BOAK with 4 Units	37.52	72.16
FOAK Single Unit	72.99	138.11
FOAK with 4 Units	53.35	99.86

Table 68. LCOE values for different SMR Scenarios with a 5% Discount Rate

The following chart compares the LCOE with those of conventional and renewable energy sources under a 5% discount rate.

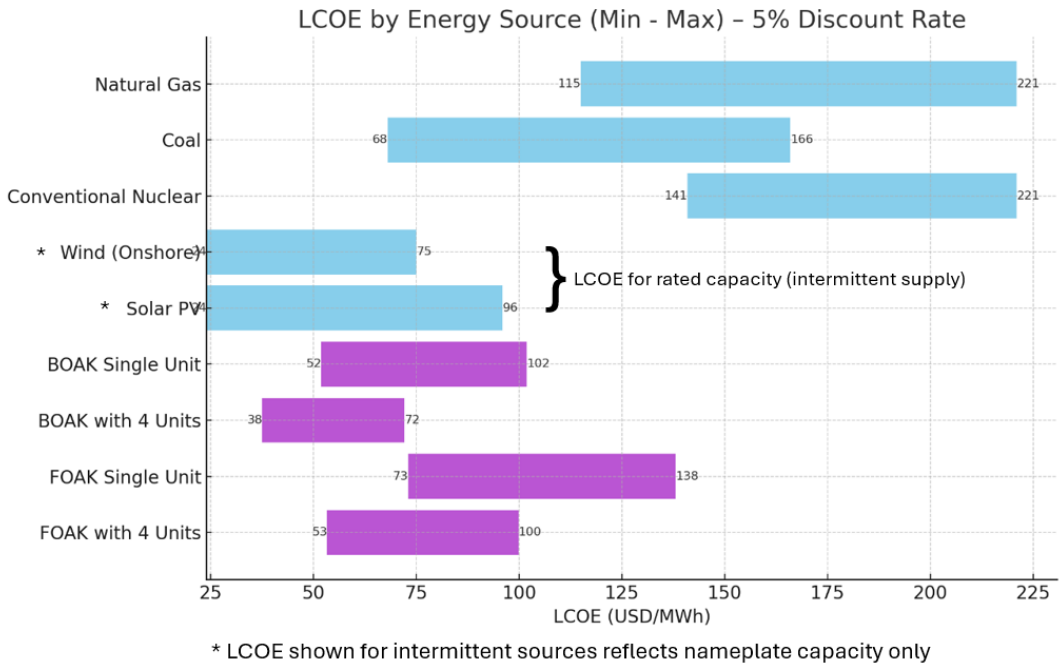


Figure 59. LCOE by Energy Source at 5% Discount Rate

Under a 5% discount rate, SMRs—particularly BOAK deployments with four units—exhibit competitive LCOE values, ranging from approximately 38 to 72 USD/MWh. These values are

lower than those of conventional nuclear and fossil fuel sources, such as coal and natural gas, and approach the mid-range costs of wind and solar PV. FOAK SMRs still present higher LCOEs due to elevated capital costs, but clustering into multi-unit configurations helps significantly narrow the gap. Overall, the results highlight that with favorable financing, SMRs can offer cost-effective baseload generation, especially when deployed in modular clusters and beyond the first-of-a-kind stage.

7.7.3.2 10% Discount Rate

The table bellow summarizes the values for BOAK and FOAK SMR configurations under a 10% discount rate:

SMR	***LCOE (USD/MWh)***	
	Low	High
BOAK Single Unit	89.92	167.85
BOAK with 4 Units	65.09	120.40
FOAK Single Unit	133.22	243.56
FOAK with 4 Units	97.46	177.05

Table 69. LCOE values for different SMR Scenarios with a 10% Discount Rate

The following chart compares the LCOE with those of conventional and renewable energy sources under a 10% discount rate.

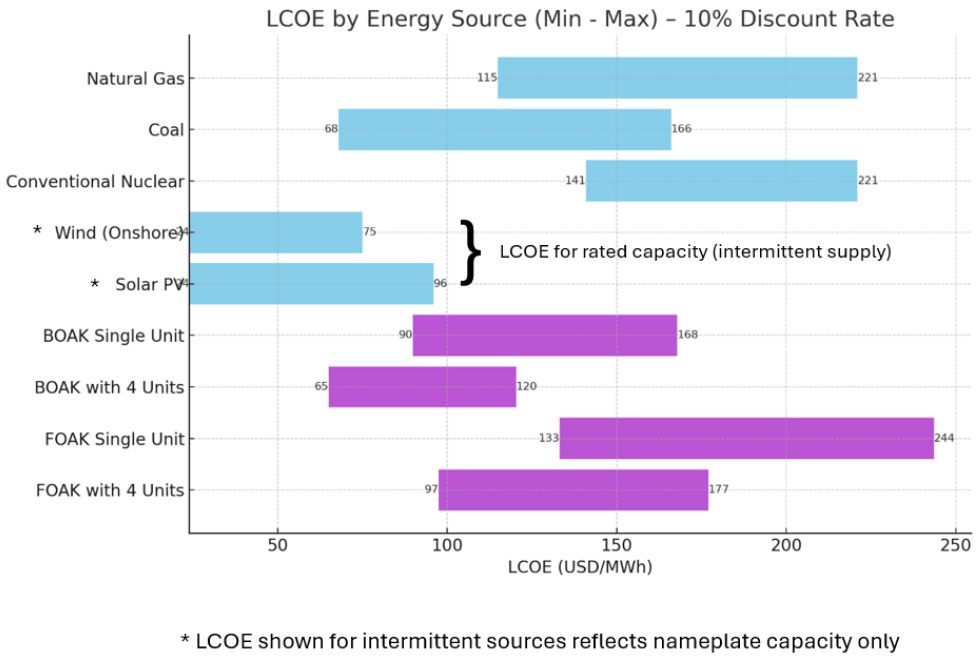


Figure 60. LCOE by Energy Source at 10% Discount Rate

At a 10% discount rate, LCOE values for all SMR configurations increase significantly compared to the 5% case. Most notably, FOAK deployments now approach or exceed the upper cost bounds of natural gas and coal, particularly for single-unit implementations.

BOAK deployments with four units still maintain competitive LCOE values, staying below conventional nuclear and overlapping with fossil fuel ranges.

This trend highlights the sensitivity of SMR economics to financing conditions. As the cost of capital rises, the relative advantage of early-stage and modular nuclear projects diminishes, reinforcing the importance of minimizing financial risk and achieving BOAK status to remain cost-competitive.

7.7.3.3 15% Discount Rate

The table bellow summarizes the values for BOAK and FOAK SMR configurations under a 15% discount rate:

SMR	***LCOE (USD/MWh)***	
	Low	High
BOAK Single Unit	136.45	249.20
BOAK with 4 Units	97.94	177.90
FOAK Single Unit	207.65	373.81
FOAK with 4 Units	150.04	269.05

Table 70. LCOE values for different SMR Scenarios with a 15% Discount Rate

The following chart compares the LCOE with those of conventional and renewable energy sources under a 15% discount rate.

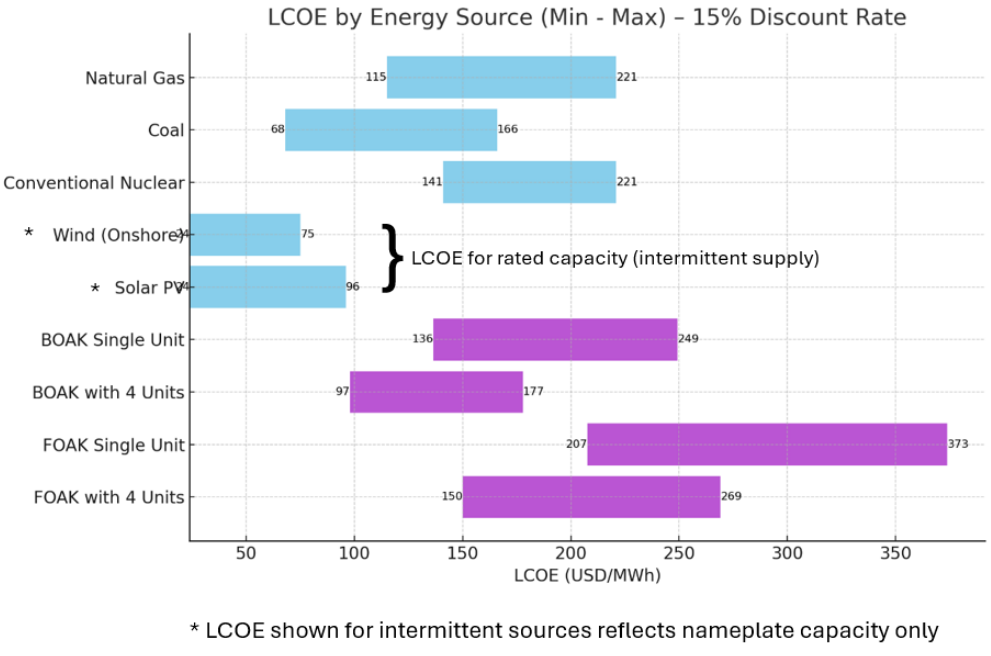


Figure 61. LCOE by Energy Source at 15% Discount Rate

Under a 15% discount rate, the Levelized Cost of Electricity (LCOE) for SMRs increases substantially across all configurations. BOAK single units range from $136/MWh to $249/MWh, while FOAK single units rise to between $208/MWh and $374/MWh. These values reflect a noticeable escalation from the 5% and 10% cases, driven primarily by the

higher cost of capital, which disproportionately impacts capital-intensive technologies like nuclear.

Compared to conventional energy sources, most SMR configurations at 15% exceed the upper bounds of solar, wind, and even coal. Only the BOAK with 4 units remains marginally competitive, staying close to coal's upper LCOE range. This trend underscores the sensitivity of SMRs to financing conditions and highlights the critical importance of securing low-cost capital to improve competitiveness.

7.7.3.4 20% Discount Rate

The table bellow summarizes the values for BOAK and FOAK SMR configurations under a 20% discount rate:

SMR	***LCOE (USD/MWh)***	
	Low	High
BOAK Single Unit	191.52	345.56
BOAK with 4 Units	135.54	243.68
FOAK Single Unit	295.72	527.91
FOAK with 4 Units	210.17	374.29

Table 71. LCOE values for different SMR Scenarios with a 20% Discount Rate

The following chart compares the LCOE with those of conventional and renewable energy sources under a 20% discount rate.

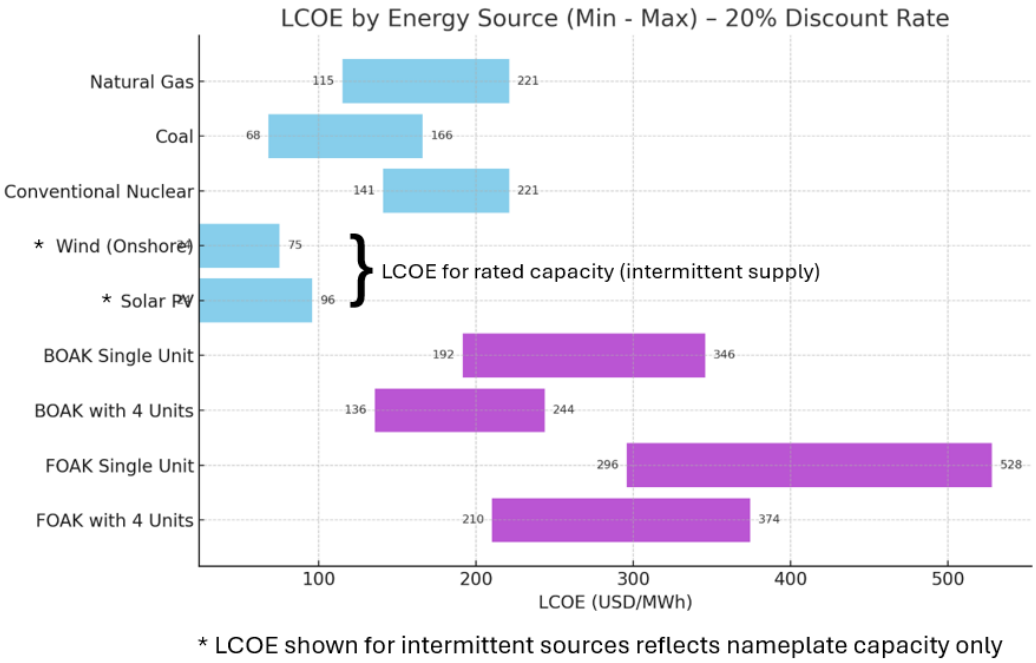


Figure 62. LCOE by Energy Source at 20% Discount Rate

At a 20% discount rate, SMRs show a significant increase in LCOE across all configurations. FOAK designs are particularly affected, with single-unit FOAK reaching a high of over 520 USD/MWh. Even BOAK multi-unit deployments, which are typically the most cost-competitive among SMRs, see their LCOE range widen to 136–244 USD/MWh, approaching the upper bounds of conventional generation technologies.

Compared to lower discount rates (5–15%), the cost penalty of higher financing assumptions becomes clearly evident. While at 5% BOAK multi-units had LCOEs comparable to wind and solar, at 20% they lose that competitive edge entirely. This illustrates how crucial favorable financing conditions are for SMR competitiveness—particularly for FOAK projects, whose high capital intensity makes them disproportionately sensitive to discount rate assumptions.

7.7.4 Economic Implications of Resilience and Reliability

In the context of powering data centers, resilience and reliability are not just technical attributes, but also significant economic factors. The ability of a power generation source to consistently deliver electricity under varying conditions, including peak demand, grid failures, or extreme weather events, can have profound implications for capital and operational expenditures.

To illustrate the magnitude of these costs, consider a baseline scenario where a 100 MW data center is supplied by intermittent renewables such as solar PV. In this case, maintaining 12 hours of backup capacity would require approximately 1,200 MWh of battery storage. At a representative installed cost of ~$350 per kWh (based on Lazard and NREL estimates) [87], this alone implies a CAPEX of about $420 million. In addition, redundant grid interconnections and substation infrastructure are typically required to ensure reliability and satisfy Tier III–IV standards, which can increase the electrical CAPEX by approximately 10–15% of the generation investment [88]. For a system of this scale, this translates into an estimated $20–40 million in extra capital cost, here conservatively represented as $20 million.

SMRs, by contrast, deliver high capacity factors above 90% and do not depend on weather conditions or time of day. They inherently avoid these costs by providing continuous, dispatchable baseload power. Black-start capability further reduces dependence on external grid support, shortening recovery times and eliminating the need for complex re-energization procedures.

Beyond capital investment, operational losses due to downtime must be considered. According to the Ponemon Institute (2016), the average cost of unplanned outages in Tier III–IV data centers exceeds $9,000 per minute [89]. For mission-critical facilities, a single one-hour interruption can result in losses surpassing half a million dollars, excluding reputational damage or SLA penalties. Over a 60-year project horizon, even a handful of such events can accumulate millions of dollars in lost value.

When these resilience-related costs are accounted for, the higher CAPEX of SMR deployments becomes more competitive in total cost of ownership (TCO) terms. The table

below compares the indicative lifecycle costs of an SMR-based system (using a representative OPEX of $15 million per year) with those of a solar PV plus battery configuration, highlighting the economic value of inherent reliability.

Concept	*SMR (100 MW)*	*Solar PV + Batteries (100 MW)*
CAPEX generation	$1,000 million	$400M (solar) + $420M (batteries) = **$820 million**
OPEX per year (estimate)	$15 million	$9 million
Net Present Value of downtime costs	$0	~$3 million
Additional grid redundancy costs	$0	~$20 million
Indicative TCO over 60 years	~$1,900 million	~$2,050 million

Table 72. TCO comparison for SMR vs Solar PV + Batteries

This comparison shows that while SMRs require a higher upfront investment, their predictable operating costs and high inherent reliability offset much of the additional infrastructure and contingency spending required for renewable alternatives. When downtime and resilience-related investments are included, the TCO gap is effectively neutralized or reversed in favor of SMRs.

The following chart illustrates the cumulative cost components over a 60-year project horizon, comparing a SMR configuration with a solar PV plus battery storage system supplying a 100 MW data center.

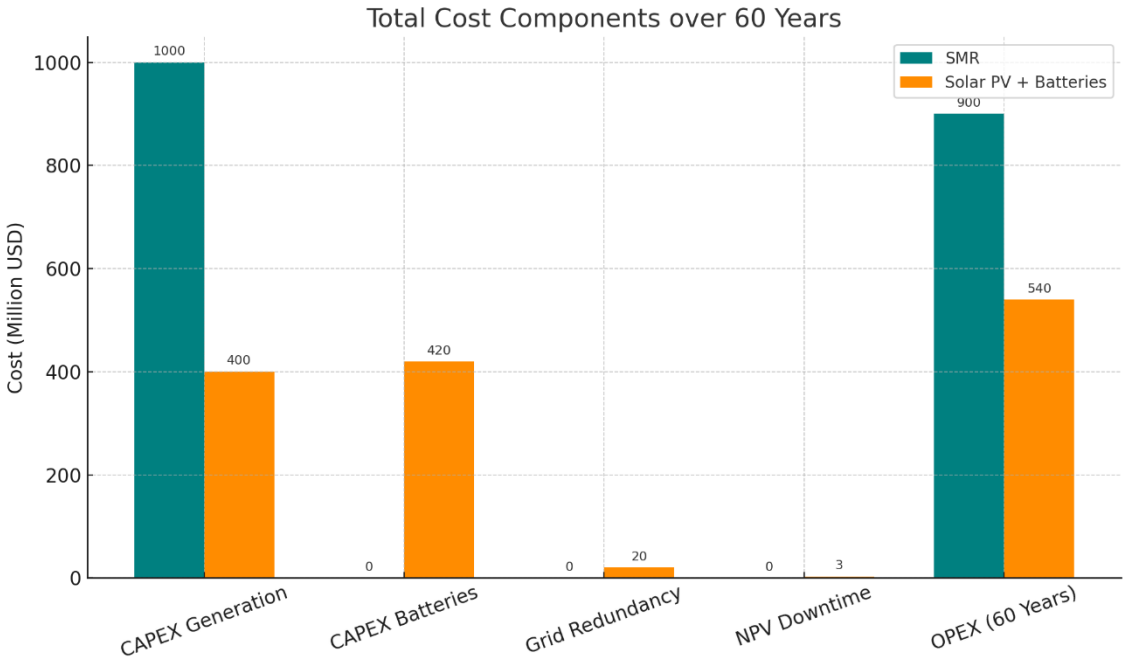


Figure 63. Cost of Components SMR vs Solar PV + batteries

7.7.5 Emissions-Related Economic Impacts

While SMRs and other advanced nuclear technologies are primarily evaluated on technical and capital cost metrics, the economic implications of carbon emissions and environmental regulation play an increasingly critical role in energy investment decisions—particularly in sectors like data centers that seek long-term cost predictability and compliance stability.

Fossil-based power sources such as natural gas and grid-supplied electricity are subject to variable and potentially rising costs associated with greenhouse gas emissions. This exposure arises through carbon pricing mechanisms, including direct carbon taxes and emissions trading schemes (ETS), which are being adopted or tightened in many jurisdictions [90]. These regulatory instruments can substantially increase the effective cost of carbon-intensive electricity generation over time.

For instance, under a carbon tax of $50 per metric ton of CO_2, a modern natural gas plant emitting approximately 0.4–0.6 tons/MWh would face an additional cost of $20–30/MWh [81]. Coal plants, with higher emission intensities, may incur even greater penalties. These costs are often not included in base LCOE calculations but are material in long-term operational budgets, particularly for energy-intensive consumers like data centers.

In contrast, SMRs produce negligible direct CO_2 emissions during operation and thus are effectively insulated from these financial risks. This emissions advantage translates into a stable and predictable cost profile, especially attractive for long-term infrastructure planning. Moreover, selecting a low- or zero-carbon power source can help operators pre-empt future regulatory tightening, qualify for tax incentives or emissions credits, and align with green procurement standards increasingly mandated by hyperscale clients and government contracts.

Additionally, emissions-related compliance often entails indirect economic burdens—reporting requirements, environmental audits, permit renewals, and investment in monitoring technologies—all of which add to the lifecycle cost of fossil-based generation. SMRs, by virtue of their low-emissions profile, reduce exposure to these secondary regulatory costs.

In summary, the emissions footprint of a power generation source has a direct and growing impact on its economic competitiveness. While natural gas and grid electricity may appear cost-effective today, their susceptibility to carbon pricing and regulatory risk may undermine their long-term viability. SMRs, with their zero-emission output and regulatory stability, offer a compelling economic case for data centers prioritizing cost predictability, risk mitigation, and compliance with emerging decarbonization policies.

7.8 Profitability Assessment

This section evaluates the potential financial performance of deploying SMR in a fully integrated configuration with a 100 MW hyperscale data center. Unlike the preceding TCO and LCOE analyses, which focused on lifecycle costs and unit economics, this assessment centers on project profitability using standard investment appraisal metrics: EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization), Net Present Value (NPV), and Internal Rate of Return (IRR).

To illustrate financing risk, NPV and IRR are computed for four discount rates: 5% (favorable institutional support), 10% (commercial debt), 15% (high risk) and 20% (very high risk). Each scenario is also analyzed for three CAPEX levels: $1.5 billion (optimistic low-cost build), $3 billion (moderate cost) and $5 billion (high cost).

This approach provides a clearer understanding of whether such an asset could achieve sustainable returns over its operational lifetime, particularly relevant to investors and data center operators exploring zero-carbon, resilient energy solutions.

7.8.1 Revenue Model and EBITDA Estimation

In line with industry practice for colocation and cloud data centers, revenues were estimated assuming a hosting price of $250 per kW per month—a representative value reflecting Tier III–IV facilities offering high-reliability uptime and secure infrastructure [91].

At full utilization, the annual revenue is:

$$Annual\ Revenue = 100{,}000\ kW \times 250 \frac{USD}{kW \cdot month} \times 12 \frac{months}{year} = 300{,}000{,}000 \frac{USD}{year}$$

This gross income represents the total potential revenue from hosting contracts, regardless of the power source.

Operating costs—including nuclear operations, maintenance, staffing, and data center expenses—are estimated at $15 million/year. This yields an annual EBITDA of:

$$EBITDA = \$300M - \$15M = \$285M/year$$

7.8.2 Net Present Value (NPV)

To assess the financial attractiveness over a 60-year project lifetime, the Net Present Value (NPV) was calculated by discounting annual EBITDA at different discount rates, using the next standard annuity formula:

$$PV = EBITDA \times \frac{1 - (1 + r)^{-n}}{r}$$

Where:

- EBITDA: $285 million/year,
- n = 60 years,

- r = discount rate.

The next table summarizes the PV for the different discount rates:

Discount Rate	*PV (billion $)*
5%	5.395
10%	2.840
15%	1.900
20%	1.425

Table 73. Present Values (PV) for different Discount Rates

The next table summarizes the NPV for the different discount rates and CAPEX values:

Discount Rate	*CAPEX (billion $)*	*NPV (billion $)*
5%	1.5	3.895
	3	2.395
	5	0.395
10%	1.5	1.340
	3	-0.160
	5	-2.160
15%	1.5	0.400
	3	-1.100
	5	-3.100
20%	1.5	-0.075
	3	-1.575
	5	-3.575

Table 74. NPV values for different Discount Rates

7.8.3 Internal Rate of Return (IRR)

The IRR represents the discount rate at which the project's NPV equals zero—effectively the breakeven return required to justify the initial investment.

The next table summarizes the IRR for the different CAPEX values:

CAPEX (billion $)	*IRR (%)*
1.5	19.00
3	9.46
5	5.47

Table 75. IRR values for different CAPEX values

7.8.4 Interpretation and Discussion

These updated results reinforce how strongly financing conditions shape the viability of SMR deployments:

At a 5% discount rate, all CAPEX scenarios show positive NPVs, even the highest-cost case ($5B), with the most favorable case ($1.5B CAPEX) reaching an NPV of nearly $3.9B. This demonstrates that low-cost financing can make even capital-intensive projects attractive over a 60-year lifetime.

At 10%, the breakeven point is just below $3B CAPEX. The $3B case delivers a near-zero NPV (~-$160M), suggesting that only lower-cost designs are likely to achieve acceptable economic returns without support mechanisms.

At 15%, only the lowest-cost configuration ($1.5B CAPEX) remains viable, generating a modest NPV of $400M. All other scenarios turn clearly uneconomical, with IRRs below the discount rate.

At 20%, even the most efficient scenario becomes unfeasible, as NPVs across all CAPEX levels turn negative, showing that high-risk or commercially priced financing effectively eliminates the business case, despite strong operational cash flow.

The IRR results further illustrate the dominance of capital costs:

- With $1.5B CAPEX, the IRR reaches 19.0%,
- But drops to 9.46% at $3B and just 5.47% at $5B—barely above the lowest discount rate.

This underscores how quickly profitability erodes with rising upfront costs, even when long-term EBITDA is strong and consistent.

7.8.5 Conclusion

This analysis confirms that co-locating SMRs with data centers can generate substantial, stable cash flow (EBITDA = $285M/year), but long-term feasibility depends almost entirely on controlling CAPEX and securing low-cost financing.

- With concessional or government-backed financing (5%), even large-scale SMRs ($5B) can achieve cost recovery and moderate returns.
- Under market or commercial financing (≥10%), only low-CAPEX designs remain economically viable.

For policymakers and investors, this sensitivity underscores the importance of:

- Structuring innovative financing instruments (e.g., green bonds, loan guarantees).
- Pursuing modular or phased SMR deployments to reduce upfront costs.
- Partnering with data center operators willing to pay a premium for ultra-reliable, low-carbon power under long-term contracts.

In short, while SMRs hold strong potential to power next-generation digital infrastructure, ensuring profitability requires deliberate alignment between technology, financing, and end-user strategy.

7.9 Risk of Budget Overruns in Nuclear Projects

One of the most critical considerations in assessing the economic feasibility of SMR deployments is the historical tendency for nuclear projects to exceed initial budget estimates. Unlike conventional thermal or renewable power plants, nuclear projects have repeatedly experienced substantial cost escalations due to a combination of technical complexity, regulatory scrutiny, and extended construction timelines.

A comprehensive review published by the OECD Nuclear Energy Agency and analyses by the World Nuclear Association show that budget overruns in nuclear projects are not the exception but the norm [83]. Empirical data indicates that:

- Large conventional nuclear power plants have historically experienced cost overruns averaging 96% over initial estimates [92].
- The IEA reported that many nuclear projects in the OECD exceed their planned budgets by 50–100%, with some cases reaching overruns of 150–200%, depending on design novelty and regulatory environment [83].
- For example, the Olkiluoto 3 EPR in Finland ended up costing nearly three times its original budget, and the Vogtle 3 and 4 AP1000 units in the United States faced cost escalations of over 100% compared to early projections [93-94].
- Even in more recent efforts to modularize construction, the inherent complexity and learning curve of nuclear technology continue to create significant financial uncertainty.

While SMRs are specifically designed to mitigate these problems by using standardized modules and reducing on-site assembly, the reality is that FOAK deployments still carry high uncertainty.

As a general guideline, it is reasonable to assume cost contingency reserves of around 30–50% for mature BOAK (Built Once As Known) deployments, and 60–100% for FOAK (First Of A Kind) projects. These margins account for potential design changes, supply chain uncertainties, and regulatory adaptations typically encountered during nuclear construction:

Project Type	***Recommended Contigency Multiplier***
BOAK SMR	x 1.3 – 1.5 (30-50% cost overrun)
FOAK SMR	x 1.6 – 2.0 (60-100% cost overrun)
Large Conventional NPP	x 1.5 –21.5 (50-150% cost overrun)

Table 76. Contingency Multiplier for different Project Types

This means that a BOAK SMR project with an expected CAPEX of $3 billion could realistically end up costing $3.9–4.5 billion, while a FOAK project could rise to $4.8–6 billion.

These budget escalation risks have several direct consequences:

- Even projects that look profitable in static TCO or EBITDA scenarios may become economically infeasible if costs grow by 50–100%.
- High discount rates (10–20%) amplify the negative impact of budget overruns by raising the cost of financing.
- The break-even point in years or cumulative EBITDA lengthens, increasing exposure to market and regulatory changes.

For example, in the profitability scenarios modeled in section 7.8, a CAPEX of $3 billion under favorable financing conditions can sustain a positive NPV. But if costs increase by 50% (to $4.5 billion), NPV becomes negative in all but the most optimistic discount rate scenarios.

Based on historical data and current market experience, budget overruns must be considered a structural risk in nuclear projects, even for SMRs. To mitigate this, several measures are recommended:

- Include realistic contingency reserves in early economic models (at least +30–50% for BOAK, +60–100% for FOAK).
- Structure financing with flexible tranches or milestone-based disbursements to avoid liquidity crises if costs rise.
- Consider progressive deployment strategies (e.g., build one or two units before scaling to clusters of four) to limit exposure to FOAK risk.

Ultimately, while SMRs hold significant promise for reliable, decarbonized power, investors and policymakers must recognize that cost overruns are an intrinsic challenge in nuclear infrastructure, and proactive contingency planning is essential to maintaining financial feasibility.

7.10 Conclusions

This comprehensive economic assessment has evaluated the viability of SMRs as a primary energy solution for large-scale data centers, benchmarked against conventional and renewable generation technologies. The analysis has integrated a wide range of considerations, including capital expenditures, operating costs, financing scenarios, emissions factors, and profitability metrics. Several key conclusions emerge from this evaluation.

First, SMRs consistently exhibit higher capital expenditures than both fossil fuel and renewable alternatives, particularly in First-of-a-Kind (FOAK) configurations. Even under favorable financing scenarios (5% discount rate), FOAK single-unit deployments can exceed $14,000 per installed kilowatt—substantially above the CAPEX of wind, solar PV, or combined-cycle gas generation. However, this disadvantage is partly mitigated by economies of scale and standardization in multi-unit deployments. BOAK configurations with four units show CAPEX reductions of approximately 20–30%, narrowing the gap relative

to conventional nuclear and coal and offering a more balanced cost profile as deployment experience increases.

Second, operational expenditures for SMRs remain notably stable and competitive across scenarios. OPEX was estimated between $18 and $42 per megawatt-hour, comparable to conventional nuclear plants and significantly less exposed to volatility than fossil fuels. Levelized Cost of Electricity (LCOE) calculations further highlighted how sensitive SMRs are to financing conditions: while BOAK multi-unit configurations can achieve LCOEs in the range of $37–$72/MWh at a 5% discount rate, these figures can escalate to more than $200/MWh in FOAK cases at 15–20% discount rates. This underlines that securing affordable, low-risk capital is essential for SMR competitiveness.

Third, the analysis of Total Cost of Ownership (TCO) revealed that discount rates exert a complex influence on lifecycle economics. While higher discount rates mathematically reduce the present value of long-term OPEX (thus lowering TCO in MWh terms), they simultaneously reflect a higher cost of capital and greater financing risk. This means that while apparent cost per megawatt-hour may decline in high-discount scenarios, the real-world feasibility of funding such projects becomes more challenging. Overall, BOAK multi-unit deployments operating under moderate discount rates (10–15%) demonstrated the most promising balance of cost efficiency and implementation feasibility.

Fourth, resilience and reliability considerations emerged as critical economic differentiators. Unlike variable renewables, SMRs provide dispatchable baseload power without requiring large-scale storage or redundant backup systems. In the comparative modeling of a 100 MW data center powered by SMRs versus a solar PV plus battery configuration, the inclusion of resilience-related costs—such as battery storage, grid interconnections, and the economic impact of downtime—effectively closed the CAPEX advantage of renewables. Over a 60-year horizon, the stable output of SMRs translates into more predictable cash flows and reduced operational risk, particularly in mission-critical data center environments.

Fifth, emissions-related financial impacts were shown to reinforce the economic case for SMRs. As carbon pricing becomes more prevalent, fossil-based power sources face escalating compliance costs. In contrast, SMRs—producing negligible operational CO_2 emissions—offer protection against carbon taxes and emissions trading obligations. For data center operators with decarbonization targets and contractual sustainability commitments, this emissions stability is a strong differentiator.

Sixth, the profitability analysis—using EBITDA as a proxy—demonstrated that SMR co-location scenarios can generate substantial annual cash flows when paired with hyperscale data centers. For example, under a representative pricing model of $250/kW per month, annual revenues can reach $300 million for a 100 MW deployment. Even assuming high OPEX and reserve provisions, EBITDA margins remained positive in most configurations, suggesting that the combined data center and SMR model is economically viable over the long term, provided that CAPEX does not exceed $3–5 billion and financing costs are contained. However, higher discount rates and FOAK premiums can significantly erode

profitability, underscoring the importance of mature deployment pathways and targeted incentives.

Finally, the analysis acknowledged the historical tendency of nuclear projects to exceed budgets—sometimes by 50–100% or more. While SMRs are designed to reduce complexity and improve cost predictability, early-stage deployments remain exposed to construction delays, supply chain immaturity, and regulatory uncertainty. These risks can amplify capital costs well beyond initial estimates, potentially undermining otherwise promising business cases. This reality reinforces the need for conservative financial planning, robust contingency allowances, and a focus on learning curves to bring costs down over time.

In summary, the economic feasibility of SMRs for powering data centers is defined by a complex interplay of upfront investment, operational expenditures, financing terms, resilience benefits, and emissions considerations. While SMRs are not yet universally cost-competitive with renewables in terms of headline CAPEX or LCOE, their unique value proposition—anchored in reliability, emissions stability, and lifecycle predictability—makes them a compelling option for mission-critical applications. As markets evolve and deployment experience accumulates, SMRs are well positioned to play a strategic role in enabling secure, sustainable, and economically resilient digital infrastructure.

8 ENVIRONMENTAL IMPACT ANALYSIS

8.1 Introduction

The rapid expansion of digital infrastructure has led to a significant increase in the energy consumption of data centers worldwide. As these facilities continue to grow in number and capacity, their environmental impact—particularly in terms of greenhouse gas (GHG) emissions—has become a pressing concern. Traditionally powered by electricity derived from fossil fuels, data centers contribute substantially to global carbon emissions, challenging the sustainability of the digital economy.

SMRs have emerged as a promising alternative to conventional power sources. These next-generation nuclear reactors offer low operational emissions, compact design, and high reliability, making them a viable solution for energy-intensive applications such as data centers. However, assessing their true environmental benefit requires a comprehensive evaluation beyond operational emissions alone.

This chapter aims to analyze the environmental impact of powering data centers with SMRs by applying a life-cycle assessment (LCA) methodology. The focus is placed on quantifying the potential reduction in carbon footprint compared to conventional energy sources, while also considering other key environmental factors such as land use, water consumption, and nuclear waste generation.

8.2 Methodology: Life-Cycle Assessment (LCA)

8.2.1 Objective and Scope

The primary objective of this life-cycle assessment (LCA) is to quantify the environmental impact—specifically, the carbon footprint—of supplying power to a large-scale data center using SMRs. The assessment focuses initially on evaluating the full life-cycle greenhouse gas emissions associated with SMR-based electricity generation. This serves as the baseline scenario.

In subsequent sections, the SMR scenario will be compared to alternative energy sources, including:

- Conventional grid electricity (average mix)
- Natural gas
- Coal
- Onshore wind
- Utility-scale solar photovoltaic (PV)

The analysis aims to identify the relative emissions intensity of each option and the potential benefits of SMR adoption in reducing the carbon footprint of high-demand digital infrastructure.

8.2.2 Functional Unit

The functional unit of this LCA is defined as the total amount of electricity required to power a hyperscale data center operating at a constant load of 100 megawatts (MW) over the course of one year. This corresponds to 876 gigawatt-hours (GWh) of electricity annually:

$$100\ MW\ \cdot 8760\ hours/year = 876\ GWh/year$$

This energy demand is representative of large-scale commercial data centers operated by major technology firms and cloud providers.

8.2.3 System Boundaries

The system boundaries for the SMR scenario encompass the entire life cycle of the power generation system, including [95]:

- **Upstream:** Extraction and processing of raw materials (e.g., uranium, steel, concrete), fuel enrichment, and reactor component manufacturing.
- **Construction:** SMR fabrication, site preparation, and installation.
- **Operation:** Routine operation of the reactor, including fuel use and maintenance.
- **Waste Management:** Interim and long-term storage of spent nuclear fuel and low-level radioactive waste.
- **End-of-Life:** Reactor decommissioning and site remediation.

Transmission and distribution impacts are excluded under the assumption that the SMR is deployed on-site or adjacent to the data center, minimizing losses and infrastructure needs. Similarly, the environmental impacts of the data center itself (e.g., servers, cooling) are outside the scope of this assessment.

8.2.3.1 Upstream: Material Extraction and Fuel Processing

This phase includes the extraction and processing of raw materials necessary for both the reactor infrastructure and nuclear fuel. Key components are:

- Mining and milling of uranium ore, which generates GHG emissions through diesel usage and chemical processing.
- Uranium enrichment, often energy-intensive, particularly if older gas diffusion methods are used (modern centrifuge-based systems are more efficient).
- Fabrication of fuel rods, typically involving ceramic sintering and zirconium cladding, which also contribute to embedded emissions.
- Additionally, the extraction and manufacturing of structural materials (e.g., high-grade steel, concrete, control systems) fall under this category.

These upstream processes are critical as they account for a significant portion of total life-cycle emissions in nuclear systems, particularly when fuel cycles are open and not recycled.

8.2.3.2 Construction Phase

The construction of an SMR includes modular manufacturing, transportation of units, site preparation, and assembly. The modular nature of SMRs generally reduces on-site construction time and labor compared to conventional reactors, but emissions from the production and transport of high-mass materials (steel, reinforced concrete) are substantial.

Construction emissions stem from:

- Cement production, a major source of CO_2.
- Steel fabrication, involving high-temperature processes.
- Transportation of modules and equipment, which may involve long-distance freight.
- Construction machinery and on-site energy use.

Though construction is a one-time impact amortized over the reactor's lifespan, it can represent a large share of front-loaded emissions.

8.2.3.3 Operation Phase

SMR operation generates negligible direct GHG emissions during electricity production, as nuclear fission does not involve combustion. However, minor indirect emissions can arise from:

- Routine maintenance and component replacement.
- Support systems such as water pumping, reactor monitoring, and cooling systems.
- Use of backup diesel generators in certain operational scenarios.

The compact, passive safety features of most SMR designs also reduce the need for complex active systems, which may lower operational energy demands relative to traditional nuclear plants.

8.2.3.4 Waste Management

Spent nuclear fuel and radioactive waste generated during SMR operation require long-term management. This includes:

- On-site interim storage in spent fuel pools or dry casks.
- Transportation to centralized repositories or reprocessing facilities.
- Final disposal, potentially in deep geological storage (not yet operational in most countries).

Although the volume of waste is low, the radiotoxicity and required containment period are high, and fuel cycle activities contribute to life-cycle emissions via storage infrastructure, shielding materials, and energy use in handling.

8.2.3.5 Decommissioning Phase

At the end of its service life (typically 40–60 years), the SMR must be safely dismantled and the site restored. This phase includes:

- Disassembly of reactor components, with potential radioactive contamination.
- Waste classification, packaging, and removal.
- Site remediation to meet environmental and regulatory standards.

Decommissioning emissions are relatively minor compared to other stages but must be accounted for, especially when considering the large mass of shielding and containment structures that require disposal or recycling.

8.2.4 Data Sources

This study relies entirely on secondary data from established literature, databases, and international energy agencies. The quantification of life-cycle greenhouse gas emissions is based on previously published LCAs SMRs and other electricity generation technologies. Original modeling or detailed process-based calculations are beyond the scope of this thesis.

Emission factors—expressed in grams of CO_2-equivalent per kilowatt-hour (gCO_2-eq/kWh)—have been sourced from peer-reviewed LCA studies, reports from the Intergovernmental Panel on Climate Change (IPCC), the International Atomic Energy Agency (IAEA), the National Renewable Energy Laboratory (NREL), and relevant academic publications [95][96][97][98]. When multiple sources were available, median or harmonized values were selected to ensure consistency and comparability across technologies.

While this approach limits the precision and context-specific accuracy of the analysis, it enables a valid comparative assessment of carbon footprint across energy scenarios, in alignment with the objective of this thesis.

8.3 Results: Carbon Footprint Comparison

8.3.1 SMR Emissions by Life-Cycle Phase

Based on a synthesis of recent literature and international databases, the estimated life-cycle greenhouse gas (GHG) emissions associated with powering a 100 MW data center for one year (876 GWh) SMR are presented below [94]:

Life-Cycle Phase	*Emission Factor (gCO_2-eq/kWh)*	*Annual Emissions (tCO_2-eq)*	*Share (%)*
Upstream (fuel and materials)	2.0 – 4.0	1,752 – 3,504	~40%
Construction	1.5 – 3.0	1,314 – 2,628	~30%
Operation	1.0 – 2.0	876 – 1,752	~20%
Waste Management	0.4 – 0.8	350 – 701	~8%
Decommissioning	0.1 – 0.2	88 – 175	~2%
Total	**5.0 – 10.0**	**4,380 – 8,760**	**100%**

Table 77. Greenhouse Gas (GHG) Emissions for every SMR Life-Cycle Phase

These estimates represent average values based on current literature for modern SMR designs. However, it is important to note that life-cycle emissions can vary considerably depending on the specific SMR technology and, in particular, its fuel cycle characteristics.

For SMRs that use a short refueling cycle—typically every 1.5 to 2 years, as seen in most light-water SMR concepts—the fuel cycle (mining, enrichment, and fabrication) can contribute to over 80% of the total life-cycle emissions. This is due to the cumulative energy intensity of repeated fuel processing and replacement.

In contrast, advanced SMRs with extended refueling intervals—for example, 10 to 15 years or more, as in molten salt reactors or fast neutron designs—can significantly reduce the relative impact of the fuel cycle. In these cases, construction and structural material production tend to dominate the carbon footprint, while fuel-related emissions become a much smaller fraction of the total.

As such, while this study assumes a representative average of 5-10 gCO_2-eq/kWh, further differentiation by SMR type is critical for precise environmental assessments. Future LCA research should account for the diversity in fuel strategies and technological maturity across different SMR classes.

8.3.2 Comparison with Other Electricity Sources

To contextualize the life-cycle carbon footprint of SMR-based electricity supply, the table below compares it with other conventional and renewable power generation technologies, based on published emission factors [97].

Energy Source	*Emission Factor (gCO_2-eq/kWh)*	*Annual Emissions (tCO_2-eq)*
SMR	5 – 10	4,380 – 8,760
Conventional Nuclear (LWR)	12	10,512
Coal	820	718,320
Natural Gas (CCGT)	490	428,240
Solar PV (utility-scale)	45	39,420
Wind (onshore)	12	10,512
Grid electricity (global avg*)	600	525,600

Table 78. Emission Factor and Annual Emissions for different Energy Sources

* Based on IEA's global electricity generation mix (2023).

While SMRs show a clear advantage over fossil-based electricity sources in terms of life-cycle greenhouse gas emissions, they also perform competitively when compared to low-carbon alternatives. Their carbon footprint is comparable to or slightly lower than that of conventional nuclear and onshore wind, and significantly lower than that of utility-scale solar photovoltaic (PV).

Based on published life-cycle assessment studies, SMRs typically emit between 5 and 10 grams of CO_2-equivalent per kilowatt-hour (gCO_2-eq/kWh). This places them among the lowest-emission electricity sources currently available.

Compared to conventional energy sources, the potential emissions reduction from adopting SMRs is substantial:

- Up to 98–99% lower than coal (820 gCO_2-eq/kWh),
- Around 97–98% lower than natural gas (490 gCO_2-eq/kWh),
- Approximately 98% lower than the global average electricity mix (600 gCO_2-eq/kWh).

Even when compared with utility-scale solar PV, SMRs can reduce life-cycle emissions by up to 78–89%, depending on the values considered. Relative to conventional nuclear and onshore wind (both around 12 gCO_2-eq/kWh), SMRs offer a modest reduction of around 17–58%.

These results highlight the significant potential of SMRs to decarbonize large-scale, continuous power demands such as those found in data centers, particularly where grid decarbonization or renewable deployment is limited by land use, intermittency, or infrastructure constraints.

8.4 Other Environmental Factors

While carbon emissions are a central metric in evaluating the environmental impact of power generation technologies, they do not capture the full range of ecological considerations. SMRs, like all energy systems, interact with the environment in multiple ways beyond greenhouse gas emissions. These include their use of natural resources such as water and land, as well as the generation and handling of long-lived radioactive waste. Assessing these additional factors is essential to develop a holistic understanding of the sustainability of SMRs in the context of data center energy supply.

8.4.1 Water Use

Water use is a critical environmental factor for many power generation technologies, particularly thermal systems. In electricity generation, water is primarily used for cooling purposes, which varies greatly depending on the technology and cooling method. The two main types of water metrics are:

- **Withdrawal**: the total amount of water drawn from a source.
- **Consumption**: the portion of water that is evaporated or otherwise not returned.

According to data from the IEA the water consumption for major electricity sources is as follows [99]:

Energy Source	*Water Consumption (gallons/MWh)*	*Water Consumption (liters/MWh)*
Coal (once-through)	300	~1,135
Natural Gas (combined cycle)	250	~946
Nuclear (once-through)	400	~1,514
Solar PV	20	~76
Wind	<1	<4
SMR	125–250	~475–946

Table 79. Water Use by Energy Source

SMRs, while not yet widely deployed at commercial scale, are expected to have lower water consumption than conventional nuclear reactors, primarily due to their smaller core size, passive safety systems, and potential for air-cooled or hybrid cooling configurations. Based on current estimates and comparisons with small-scale thermal plants, SMRs are expected to consume approximately 125–250 gallons per megawatt-hour (475–946 liters/MWh), depending on the design and cooling system implemented.

This positions SMRs as significantly more water-efficient than coal and conventional nuclear, and somewhat more efficient than natural gas. However, they still consume more water than solar PV and wind, which are nearly water-free in operation.

For data centers, especially those located in water-stressed regions, the water efficiency of SMRs can be a relevant sustainability advantage. Selecting low-water-use SMR designs or integrating dry cooling technologies may further reduce operational water demands and enhance environmental compatibility.

8.4.2 Nuclear Waste Management

Radioactive waste generation is one of the most distinctive environmental concerns associated with nuclear power. Like conventional nuclear reactors, SMRs produce several categories of radioactive waste, including:

- **High-level waste (HLW):** primarily spent nuclear fuel, which remains highly radioactive and thermally hot for thousands of years.
- **Intermediate- and low-level waste (ILW/LLW):** contaminated equipment, resins, filters, and structural materials.

Despite their smaller size, SMRs are expected to generate a comparable amount of spent fuel per unit of electricity generated as large nuclear plants, particularly for designs based on traditional LWR technology [100]. However, some advanced SMR designs may allow for higher fuel burnup and reduced waste volumes, depending on the reactor type and fuel cycle strategy.

Spent fuel is initially stored on-site in cooling pools and later transferred to dry cask storage. Long-term solutions such as deep geological repositories are being developed in several countries but are not yet operational at commercial scale. These facilities aim to isolate HLW from the biosphere for timeframes exceeding 100,000 years.

From a climate and air quality perspective, nuclear waste is a trade-off for zero operational carbon emissions and the absence of other pollutants like sulfur dioxide, nitrogen oxides, or particulate matter. Unlike fossil fuels, nuclear waste is contained, monitored, and regulated throughout its life cycle.

Importantly, waste management practices differ significantly depending on the SMR design and fuel type [58]. PWR-based SMRs produce standard spent fuel and benefit from mature, well-established storage and disposal technologies, though they still generate large volumes of long-lived waste. HTGRs, which use TRISO fuel embedded in graphite, present additional challenges due to the generation of irradiated graphite that is difficult to process and dispose of. MSRs, with their liquid fuel and coolant salts, require complex chemical treatment prior to disposal, and their long-term waste behavior is still under active investigation. In contrast, SFRs can recycle and reuse nuclear fuel, significantly reducing the quantity and radiotoxicity of waste, although handling and disposal of sodium coolant require specialized infrastructure.

Nonetheless, public perception, regulatory complexity, and the absence of final disposal infrastructure remain challenges. Any widespread deployment of SMRs for data centers or other applications would need to be paired with robust and transparent waste management strategies, supported by long-term policy planning.

8.4.3 Land Use

Land use is another key environmental factor when evaluating power generation technologies, particularly for applications like data centers, where spatial constraints and land costs are significant. SMRs offer a substantial advantage in this regard due to their exceptionally high power density and compact physical footprint.

A typical SMR installation, even when including security perimeters and support infrastructure, can require as little as 0.01–1 acres per megawatt (MW) of capacity. In contrast [101]:

- Solar PV requires approximately 5–10 acres per MW, depending on system design and location.
- Onshore wind occupies 30–60 acres per MW when including spacing between turbines, though much of that land can be co-used (e.g., for agriculture).
- Large nuclear plants fall between 1–5 acres per MW, but require longer construction timelines and larger exclusion zones.

- Natural gas combined-cycle plants typically require 0.5–1 acre per MW, although additional land may be needed for pipelines, cooling infrastructure, and buffer zones.
- Coal-fired power plants use around 1–3 acres per MW, including space for coal storage, ash disposal, and flue gas treatment facilities.

SMRs can often be co-located with the facilities they power, such as data centers, reducing or eliminating the need for long-distance transmission infrastructure. This proximity not only lowers land use impacts but also improves energy efficiency and system resilience.

In land-constrained regions or urban environments, the compact nature of SMRs presents a clear environmental and logistical benefit over most renewable and conventional energy sources.

8.4.4 Site Selection and Environmental Constraints

While the compact footprint of SMRs offers clear advantages in terms of land efficiency and flexibility, selecting an appropriate site for deployment involves more than spatial considerations alone. Several environmental constraints must be carefully evaluated to ensure the safety, sustainability, and long-term viability of SMR installations [102]—particularly when colocated with critical infrastructure like data centers.

One of the most important siting factors is cooling water availability. Many SMR designs rely on water, molten salts, or liquid metals as coolants, requiring a stable and abundant thermal sink for effective heat rejection. As such, proximity to rivers, lakes, or coastal zones can be advantageous in terms of thermal performance, passive safety system effectiveness, and operational reliability. Conversely, siting in arid or water-stressed regions may pose challenges for both reactor efficiency and environmental compliance.

Seismic and geological risks also play a critical role in siting decisions. In regions with high tectonic activity, selecting geologically stable locations is essential to ensure the structural integrity of containment systems, safeguard against natural hazards, and comply with nuclear regulatory requirements. Some advanced SMR designs emphasize inherent seismic resilience; however, siting on stable ground remains a foundational criterion for risk mitigation.

Additional environmental considerations include population density, existing infrastructure, exclusion zone requirements, and proximity to protected ecosystems. In some cases, brownfield redevelopment—installing SMRs on the sites of retired coal or gas plants—offers a strategic opportunity to minimize land disturbance and leverage pre-existing grid connections.

By integrating environmental constraints into the site selection process from the earliest planning stages, SMR projects can reduce licensing and construction risks, improve public acceptance, and align more closely with broader sustainability goals. For data center applications in particular, co-location on environmentally suitable and technically compatible sites will be critical to realizing the full benefits of modular nuclear energy.

8.5 Conclusions

This chapter has evaluated the environmental impact of using SMRs as a power source for large-scale data centers, considering a range of sustainability dimensions such as life-cycle greenhouse gas emissions, water consumption, radioactive waste generation, land use, and site selection constraints. The findings reaffirm the environmental viability of SMRs in this application and highlight several areas where they provide meaningful advantages over both fossil-based and renewable alternatives.

In terms of carbon emissions, SMRs exhibit exceptionally low life-cycle values, typically ranging from 5 to 10 grams of CO_2-equivalent per kilowatt-hour (gCO_2-eq/kWh). This places them among the cleanest energy sources currently available, comparable to wind and slightly below conventional nuclear. SMRs can reduce emissions by approximately 97–99% compared to fossil fuels like coal and natural gas, and by around 98% relative to the global average electricity mix. Even when compared to solar PV, which is often seen as the benchmark for clean energy, SMRs can achieve emissions reductions of 78–89%, depending on design and fuel cycle strategy. These characteristics make SMRs highly attractive for continuous, high-demand loads such as hyperscale data centers, particularly in regions where grid decarbonization is incomplete or land availability limits renewable deployment.

Water use, while not negligible, is significantly lower for SMRs compared to conventional nuclear or coal-fired power plants. Current estimates suggest consumption in the range of 125–250 gallons per megawatt-hour (475–946 liters/MWh), which is lower than both natural gas and nuclear, though higher than wind and solar. Many SMR designs allow for air-cooled or hybrid cooling configurations, offering an additional advantage in water-scarce environments.

Radioactive waste management remains one of the most debated environmental aspects of nuclear power. While SMRs are expected to generate comparable waste volumes per unit of electricity as traditional nuclear plants, the nature of that waste varies substantially by reactor type. PWR-based SMRs produce standard spent fuel and benefit from mature handling technologies. HTGRs, on the other hand, generate irradiated graphite that is difficult to process. MSRs introduce unique waste challenges due to their use of liquid fuel and coolant salts, which require advanced chemical treatment and containment strategies. SFRs stand out for their ability to recycle actinides, thereby reducing the volume and radiotoxicity of long-lived waste. These differences reinforce the need to align reactor selection with robust and design-specific waste strategies. Although interim storage is well-established, the lack of operational deep geological repositories remains a systemic gap that must be addressed to fully realize the environmental benefits of SMRs.

Regarding land use, SMRs present a clear logistical and environmental advantage. With power densities enabling installation footprints as low as 0.01–1 acre per megawatt, they are far more spatially efficient than wind or solar technologies. This compact footprint enables co-location with energy-consuming infrastructure like data centers, reduces the

need for long-distance transmission, and allows for brownfield redevelopment, which further minimizes environmental disturbance.

Finally, site selection plays a decisive role in determining the broader environmental compatibility of SMR deployment. While SMRs are more flexible in siting due to their modular nature, environmental constraints such as cooling water availability, seismic risk, and proximity to populated or protected areas must be carefully considered. Integrating these parameters early in the planning process can reduce licensing hurdles, enhance public acceptance, and ensure safe and sustainable operation over the long term.

In summary, SMRs represent a promising low-impact power source for the digital infrastructure sector. Their combination of ultra-low carbon emissions, moderate water use, manageable waste (when supported by appropriate strategies), and compact siting requirements offers a compelling path toward sustainable, resilient, and scalable energy for data centers. However, long-term environmental success will depend on continued advancements in waste management, careful site selection, and transparent policy development aligned with the unique characteristics of each SMR technology.

9 PROJECT PLANNING

The development of this Research Project was structured into a series of sequential and interrelated tasks, spanning both the industrial and academic phases of the project. Careful planning was essential to ensure a coherent progression from the initial literature review to the final techno-economic and environmental analyses. This section first describes the main tasks carried out throughout the project, followed by a Gantt chart illustrating their timeline and interdependencies.

9.1 Description of Project Tasks

The execution of the project was divided into the following key tasks, carried out across two distinct phases: an industrial phase at IDOM (Spain) and an academic phase at Illinois Institute of Technology (USA).

1. **Preliminary Literature Review**: A comprehensive review of existing literature, technical reports, and official databases was performed to understand the current landscape of SMRs, data center energy demands, and previous feasibility studies. This task laid the foundation for all subsequent analyses.
2. **Definition of Objectives and Methodology**: The project scope was formally defined, including the selection of evaluation criteria, exclusion filters, and the decision to apply the Analytic Hierarchy Process (AHP) for SMR selection. The structure and methodology were aligned with the research goals of both institutions involved.
3. **Selection and Filtering of SMR Designs**: An extensive filtering of over 80 SMR concepts listed in the IAEA database was conducted using exclusion criteria relevant to data center applications. This allowed for a focused analysis of viable reactor designs.
4. **AHP Model Development and Implementation**: The AHP method was applied to prioritize and rank the most suitable SMRs. This task involved defining the hierarchy of criteria and sub-criteria, performing pairwise comparisons, consistency checks, and running the analysis in dedicated software tools.
5. **Economic Feasibility and Cost Analysis**: This task involved evaluating the economic viability of using SMRs to power data centers. It included the estimation of key financial metrics such as CAPEX, OPEX, LCOE, and TCO, as well as a profitability assessment using NPV and IRR. The analysis also compared SMRs to conventional energy sources under different deployment and discount rate scenarios, considering both standalone and co-located configurations.
6. **Environmental Impact Assessment**: The environmental performance of SMRs was evaluated in terms of emissions, waste generation, land use, and lifecycle footprint, with comparisons against renewable and fossil energy sources.
7. **Drafting and Revision of the Final Report**: The results of each phase were compiled and structured into the final written document, with iterative review and feedback from both the industry supervisor and academic advisor.

9.2 Gantt Chart

To visualize the chronological organization of the project, the following Gantt chart outlines the timeline of the main tasks described above. It reflects both the industrial and academic phases of the thesis, highlighting the duration and overlap of each activity throughout the nine-month development period.

Task	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6	Month 7	Month 8	Month 9
Industrial Phase - IDOM (Spain)									
Literature Review									
Definition of Objectives and Methodology									
SMR Selection and Filtering									
AHP Model Development and Analysis									
Initial Report Writing									
Academic Phase - Illinois Tech (USA)									
Continued Literature Review									
AHP Corrections									
Economic Analysis									
Environmental Impact Analysis									
Final Report Writing and Revisions									

Figure 64. Project's Gantt Diagram

10 Project Budget Description

This section outlines the financial expenses associated with the development of the project. Three main cost categories are identified: labor costs, hardware costs, and software costs.

Although the project is fundamentally grounded in a bibliographic review of existing SMR designs and data, it involves the application of decision-making methodologies (such as the Analytic Hierarchy Process), detailed economic modeling, and environmental impact analysis. Therefore, while no experimental work or material resources were required, the study includes applied and technical components developed through software tools and quantitative assessments. As a result, no additional categories of expenses have been considered.

10.1 Labor Costs

The labor cost has been divided into two distinct phases:

Phase 1 – Collaboration with Industry (IDOM, Spain)

In the initial stage, the project was developed at the engineering firm IDOM, under the supervision of a senior nuclear engineer. The following labor costs are based on typical Spanish engineering wages, converted to USD:

Task	*Student (h)*	*Senior Engineer (h)*
Literature review	120	-
Meetings and technical review	10	15
Tool training and setup	50	20
Report writing	30	-
AHP analysis and implementation	210	-
TOTAL (h)	420	35
Labor Cost ($/h)	24	52
Labor Cost ($)	10,080	1,820
TOTAL LABOR COST PHASE 1 ($)	**11,900**	

Table 80. Labor Costs during Phase 1 of the Project

Phase 2 – Academic Work at Illinois Tech

The second phase was carried out under the supervision of a faculty advisor at Illinois Institute of Technology. The rates reflect typical U.S. academic wages for graduate research work:

Task	*Student (h)*	*Faculty Advisor (h)*
Continued literature review	70	–
Meetings and revisions	10	15
Methodology development (AHP)	30	10
Economic modeling	100	-
Environmental analysis	50	-
Report writing	180	-
TOTAL (h)	440	25
Labor Cost ($/h)	20	60
Labor Cost ($)	8,800	1,500
TOTAL LABOR COST PHASE 2 ($)	**10,300**	

Table 81. Labor Costs during Phase 2 of the Project

Combining both phases, the total labor cost amounts to $22,200, of which $18,880 corresponds to the student and $3,320 to the supervisors (industry and academic).

Total Labor Cost: $22,200

10.2 Hardware Cost

The student used a personal laptop during both phases of the project. A straight-line depreciation model was applied:

- Purchase value: $800
- Estimated useful life: 5 years (60 months)
- Time used for the project: 9 months → 15% depreciation
- Depreciated cost: 15% × $800 = $120.00

Total hardware cost: $120.00

10.3 Software Costs

All software tools were provided via academic or institutional licenses: Microsoft Office 365 and SMART.

Total software cost: $0

10.4 Additional Costs

In addition to labor, hardware, and software, the project incurred minor operational expenses related to electricity usage and internet access. Throughout the nine-month development period, the student used a personal laptop for approximately three hours per day. Based on an estimated power consumption of 250 W and a standard U.S. electricity rate of $0.15 per kWh, the total electricity cost was calculated. Additionally, the internet service, necessary for accessing literature, running software tools, and maintaining communication with advisors, was estimated at $60 per month.

The following table summarizes the additional operational costs:

Item	*Unit Cost*	*Quantity*	*Total ($)*
Electricity (250 W, 810 h)	$0.15 / kWhh	202.5 kWh	$30.38
Internet Service	$60 / month	9 months	$540
TOTAL			**$570.38**

Table 82. Additional Costs during the Project

These values bring the overall additional costs to $570.38, which are included in the final budget summary.

Total additional cost: $570.38

10.5 Total Cost

The total cost of the project is the sum of all previously detailed components: labor, hardware, software, and additional operational expenses. The following table presents the complete breakdown of the project's financial estimation:

Cost Category	*Amount ($)*
Labor cost	$22,200.00
Hardware depreciation	$120.00
Software cost	$0.00
Additional costs	$570.38
Total	**$22,890.38**

Table 83. Total Costs for the Project

This total of **$22,890.38** reflects the full estimated cost of carrying out the project under both industrial and academic supervision, using personal computing resources and institutionally provided software licenses.

11 CONCLUSIONS

This project explores the feasibility of SMRs as a sustainable and reliable power source for large-scale data centers. The work focused on evaluating the suitability of SMRs from three key perspectives: technical selection through a structured decision-making process, economic feasibility, and environmental impact. The combined analysis of these dimensions allows for a comprehensive understanding of the challenges and opportunities of SMR-powered data centers in the context of the global energy transition and digital infrastructure growth.

The first part of the study was devoted to identifying the most technically viable SMR designs for data center applications. A two-stage approach was applied. In Stage 1, exclusion criteria were used to filter more than 80 designs listed in the IAEA database, narrowing the pool to those compatible with the operational and spatial requirements of data centers. Criteria included technology readiness level, power output, compatibility with USA and Europe and modular redundancy capabilities.

Stage 2 consisted of a multi-criteria evaluation using the AHP. A detailed weighting system was developed to prioritize factors such as safety features, energy availability, licensing progress, fuel cycle characteristics, waste management and deployment flexibility. The final ranking revealed several top-performing SMR designs, primarily based on advanced reactor technologies such as HTGR, MSR, and SFR, rather than conventional PWR.

Among the evaluated designs, the top three SMRs most suitable for sustainably powering data centers are the Xe-100, KP-FHR, and Aurora Powerhouse, in that order. The Xe-100 stood out for its exceptional safety features, high capacity factor, online refueling capability, and robust TRISO fuel performance, making it highly aligned with the continuous power demands of data centers. The KP-FHR, developed by Kairos Power, ranked second thanks to its strong licensing progress, long refueling intervals, and high reliability. In third place, Oklo's Aurora Powerhouse demonstrated high adaptability due to its modular design, scalability, and potential for secondary uses such as heat or hydrogen production. All three designs combine advanced safety, operational resilience, and strategic deployment advantages, making them strong candidates for future integration with digital infrastructure.

To test the robustness of the AHP results, a sensitivity analysis was performed by varying the weights of both top-level criteria and key subcriteria. This analysis demonstrated that the overall ranking of SMR designs remained relatively stable under different priority scenarios, such as increasing the emphasis on deployment feasibility, licensing progress, or operational flexibility. While minor changes in individual rankings occurred—such as position shifts between reactors with similar scores—the top three candidates consistently performed well across all scenarios. These findings confirm that the AHP framework provides a resilient and reliable decision-making structure for evaluating SMRs under diverse stakeholder preferences.

The economic analysis conducted in Chapter 7 highlighted both the challenges and the long-term opportunities associated with SMR deployment. While First-of-a-Kind (FOAK)

configurations exhibit high capital costs—frequently exceeding $14,000/kW installed—these are significantly reduced in BOAK (nth-of-a-kind) scenarios with multi-unit deployments. CAPEX reductions of up to 30% were observed, along with improvements in cost predictability. Operational expenditures remained competitive and stable across all scenarios, in the range of $18 to $42/MWh. LCOE analysis underscored the strong sensitivity of SMRs to financing conditions, with discount rates playing a critical role in cost viability. The Total Cost of Ownership (TCO) model further revealed how resilience, dispatchability, and emissions stability add economic value to SMRs, especially when powering mission-critical infrastructure like data centers. When resilience-related costs (storage, redundancy, downtime) were accounted for, the economic gap between SMRs and renewables narrowed significantly.

The profitability assessment showed that SMR co-location with data centers has the potential to generate substantial annual revenues, with positive EBITDA margins in most realistic deployment scenarios. However, the study also acknowledged the historical risk of cost overruns in nuclear projects—particularly in early-stage deployments. These risks reinforce the importance of conservative financial planning, streamlined regulatory pathways, and leveraging standardization and modularity to reduce uncertainty and accelerate cost reduction curves over time.

Finally, the environmental impact assessment confirmed that SMRs offer a highly favorable profile compared to fossil fuels and even some renewable sources. Life-cycle carbon emissions were estimated between 5 and 10 gCO_2-eq/kWh—orders of magnitude below coal, gas, and even solar PV in some cases. Additional advantages include minimal land use, manageable water consumption, and compact infrastructure requirements, making SMRs highly compatible with data center co-location. While radioactive waste remains a long-term consideration, current interim storage solutions and advancements in disposal technologies mitigate short-term risks. Overall, SMRs present a credible pathway for achieving deep decarbonization in digital infrastructure without compromising reliability or performance.

In conclusion, the integration of rigorous technical selection, detailed economic modeling, and life-cycle environmental analysis supports the main hypothesis of this thesis: SMRs are a technically, economically, and environmentally viable solution for powering data centers. Although barriers remain—particularly related to cost, regulation, and public perception—SMRs stand out for their resilience, low emissions, and ability to provide stable, dispatchable power. With appropriate support and continued technological progress, SMRs are well positioned to play a strategic role in the sustainable transformation of the digital economy.

12 REFERENCES

[1] International Energy Agency. (2024). *Energy demand from data AI*. Retrieved from https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai

[2] Kairos Power. (2024, October 14). *Google and Kairos Power partner to deploy 500 MW of clean electricity generation*. Retrieved from https://kairospower.com/external_updates/google-and-kairos-power-partner-to-deploy-500-mw-of-clean-electricity-generation/

[3] Amazon Web Services. (2024, October 16). *Amazon signs agreements for innovative nuclear energy projects to address growing energy demands*. Retrieved from https://www.aboutamazon.com/news/sustainability/amazon-nuclear-small-modular-reactor-net-carbon-zero

[4] Oklo Inc. (2024, November 14). *Oklo secures partnerships for up to 750 megawatts of power for U.S. data centers*. Retrieved from https://oklo.com/newsroom/news-details/2024/Oklo-Secures-Partnerships-for-Up-to-750-Megawatts-of-Power-for-U.S.-Data-Centers/default.aspx

[5] DatacenterDynamics. (2024, March 11). *Enel finalizing formation of SMR firm*. Retrieved from https://www.datacenterdynamics.com/en/news/enel-finalizing-formation-of-smr-firm/

[6] Dgtl Infra. (2024, March 25). *Data Center Power: A Comprehensive Overview of Energy*. Retrieved from https://dgtlinfra.com/data-center-power/

[7] United Nations. *Sustainable Development Goals (SDGs)*. Retrieved from https://sdgs.un.org/goals

[8] CloudOptimo. (2024, May 2). *Important role of data centers in cloud computing*. Retrieved from https://www.cloudoptimo.com/blog/important-role-of-data-centers-in-cloud-computing/

[9] McKinsey & Company. (2024, October 29). *AI power: Expanding data center capacity to meet growing demand*. Retrieved from https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/ai-power-expanding-data-center-capacity-to-meet-growing-demand

[10] Uptime Institute. *Tier classification system*. Retrieved from https://uptimeinstitute.com/tiers

[11] Datacenter Frontier. (2021, February 26). *Fit for the edge: Modular data centers*. Retrieved from https://www.datacenterfrontier.com/voices-of-the-industry/article/11428381/fit-for-the-edge-modular-data-centers

[12] International Energy Agency. (2025). *What the data centre and AI boom could mean for the energy sector*. Retrieved from https://www.iea.org/commentaries/what-the-data-centre-and-ai-boom-could-mean-for-the-energy-sector

[13] TechCrunch. (2025, July 1). *Google's data-center energy use doubled in four years*. Retrieved from https://techcrunch.com/2025/07/01/googles-data-center-energy-use-doubled-in-four-years/

[14] Levi, P. (2025, April 10). *Data centres will use twice as much energy by 2030 – driven by AI*. *Nature*. Retrieved from https://www.nature.com/articles/d41586-025-01113-z

[15] phoenixNAP. (2024, March 6). *Data Center Power Design Overview*. Retrieved from https://phoenixnap.com/blog/data-center-power

[16] International Atomic Energy Agency. (2023). *What are small modular reactors (SMRs)?* Retrieved from https://www.iaea.org/newscenter/news/what-are-small-modular-reactors-smrs

[17] GRS. (2023). *Small-scale nuclear power plants: What's behind SMR concepts?* Retrieved from https://www.grs.de/en/news/knowledge-dossier/small-scale-nuclear-power-plants-whats-behind-smr-concepts

[18] Innovation News Network. (2024, October 3). *Small modular reactors: Assessing the passive safety advantage*. Retrieved from https://www.innovationnewsnetwork.com/small-modular-reactors-assessing-the-passive-safety-advantage/51630/

[19] U.S. Department of Energy. *Benefits of small modular reactors (SMRs)*. Retrieved from https://www.energy.gov/ne/benefits-small-modular-reactors-smrs

[20] World Nuclear Association. (2023, September 29). *Economics of nuclear power*. Retrieved from https://www.world-nuclear.org/information-library/economic-aspects/economics-of-nuclear-power.aspx

[21] International Atomic Energy Agency. *Nuclear power in the clean energy transition*. Retrieved from https://www.iaea.org/newscenter/news/nuclear-energy-in-the-clean-energy-transition

[22] OECD Nuclear Energy Agency. *Small Modular Reactors. Nuclear Energy Market Potential for Near-term Deployment*. Retrieved from https://www.oecd.org/en/publications/small-modular-reactors_9789264266865-en.html

[23] International Atomic Energy Agency. (2022). *Lessons learned in regulating small modular reactors*. Retrieved from https://www.iaea.org/publications/15149/lessons-learned-in-regulating-small-modular-reactors

[24] World Nuclear Association. (2024). *Small nuclear power reactors*. Retrieved from https://world-nuclear.org/information-library/nuclear-fuel-cycle/nuclear-power-reactors/small-nuclear-power-reactors

[25] American Nuclear Society. (2023, August 3). *SMR survey suggests low awareness but high acceptability*. Retrieved from https://www.ans.org/news/article-5095/smr-survey-suggests-low-awareness-but-high-acceptability/

[26] U.S. Department of Energy. (2023). *Advanced small modular reactors (SMRs)*. Retrieved from https://www.energy.gov/ne/advanced-small-modular-reactors-smrs

[27] Oak Ridge National Laboratory. *Molten salt reactor*. Retrieved from https://www.ornl.gov/molten-salt-reactor

[28] OECD Nuclear Energy Agency. (2021). *High-temperature gas-cooled reactors*. Retrieved from https://www.oecd-nea.org/jcms/pl_20497/high-temperature-gas-cooled-reactors

[29] Wikipedia. *Liquid metal cooled reactor*. Wikipedia. Retrieved from https://en.wikipedia.org/wiki/Liquid_metal_cooled_reactor

[30] International Energy Agency. *District heating*. Retrieved from https://www.iea.org/energy-system/buildings/district-heating

[31] District Energy Association. (2024, June 11). *Finland to build its first SMR for district heating in 2025*. Retrieved from https://www.districtenergy.org/blogs/district-energy/2024/06/11/finland-to-build-its-first-smr-for-district-heatin

[32] U.S. Department of Energy. *Desalination basics*. Retrieved from https://www.energy.gov/eere/iedo/desalination-basics

[33] Vertiv. *Liquid cooling options for data centers*. Retrieved from https://www.vertiv.com/en-emea/solutions/learn-about/liquid-cooling-options-for-data-centers/

[34] ScienceDirect. *Absorption cooling*. Retrieved from https://www.sciencedirect.com/topics/chemical-engineering/absorption-cooling

[35] Phononic. *How thermoelectric cooling uses the Peltier effect*. Retrieved from https://phononic.com/resources/how-thermoelectric-cooling-uses-the-peltier-effect/

[36] International Atomic Energy Agency. *Advanced Reactors Information System (ARIS)*. Retrieved from https://aris.iaea.org/

[37] 1000minds. *Analytic Hierarchy Process (AHP): Multi-criteria decision-making*. Retrieved from https://www.1000minds.com/decision-making/analytic-hierarchy-process-ahp

[38] United Nations. (1997). *Treaty on the Non-Proliferation of Nuclear Weapons*. Retrieved from https://www.un.org/disarmament/wmd/nuclear/npt/text

[39] U.S. Congress. (2024). *H.R.8824 –The U.S-European Nuclear Energy Cooperation Act of 2024*. Retrieved from https://www.congress.gov/bill/118th-congress/house-bill/8824/text

[40] International Atomic Energy Agency. (2024). *Small Modular Reactors Catalogue 2024: A supplement to the IAEA advanced reactors information system (ARIS)*. Retrieved from https://aris.iaea.org/Publications/SMR_catalogue_2024.pdf

[41] Saaty, T. L. (1980). *The Analytic Hierarchy Process: Planning, priority setting, resource allocation*. New York: McGraw-Hill.

[42] U.S. Department of Energy. (2019, July 9). *TRISO particles: The most robust nuclear fuel on earth*. Retrieved from https://www.energy.gov/ne/articles/triso-particles-most-robust-nuclear-fuel-earth

[43] Stanford University. (2024). *Fission product removal in molten salt reactors*. Retrieved from http://large.stanford.edu/courses/2024/ph241/goodman1/

[44] Pacific Northwest National Laboratory. (2009). *Metallic fuels for advanced reactors*. Retrieved from https://www.pnnl.gov/sites/default/files/media/file/Metallic%20Fuels%20for%20Advanced%20Reactors.pdf

[45] OECD Halden Reactor Project. (2018). *Fission gas release from UO2 pellet fuel at high burn-up*. Retrieved from https://inis.iaea.org/records/a3bb0-29k15

[46] Gat, U., & Dodds, H. L. (2009). *Molten salt reactors – Safety options galore*. Oak Ridge National Laboratory & University of Tennessee at Knoxville. Retrieved from https://www.osti.gov/servlets/purl/469120

[47] International Atomic Energy Agency. (2024). *Sodium Coolant Handbook: Thermal Hydraulic Correlations*. Retrieved from https://www-pub.iaea.org/MTCD/publications/PDF/CRCP_SOD_003web.pdf

[48] International Atomic Energy Agency. (1992). *Coolant technology of water cooled reactors. Volume 3: Activity transport mechanisms in water cooled reactors* (IAEA-TECDOC-667). Retrieved from https://www-pub.iaea.org/MTCD/Publications/PDF/te_667v3_web.pdf

[49] Natesan, K., Purohit, A., & Tam, S. W. (2003). *Materials behavior in HTGR environments* (ANL-GenIV-029). Argonne National Laboratory. Retrieved from https://www.nrc.gov/docs/ML0323/ML032370015.pdf

[50] Compere, E. L., Kirslis, S. S., Bohlmann, E. G., Blankenship, F. F., & Grimes, W. R. (1973). *Fission product behavior in the Molten Salt Reactor Experiment* (ORNL-4865). Oak Ridge National Laboratory. Retrieved from https://moltensalt.org/references/static/downloads/pdf/ORNL-4865.pdf

[51] Matzie, R. A., Longo, J., Bradbury, R. B., Teare, K. R., & Hayns, M. R. (1992). *Design of the Safe Integral Reactor*. Nuclear Engineering and Design, 199(3), 255–267. Retrieeved from https://doi.org/10.1016/S0029-5493(00)00255-6

[52] U.S. Department of Energy. (2023). *Advanced Materials and Manufacturing Technologies 2022 Roadmap*. Retrieved from https://www.energy.gov/sites/default/files/2023-03/ne-ammt-roadmap-030823.pdf

[53] Zhang, S., Ji, X., Hou, G., Xu, D., Yang, C., & Cai, X. (2024). Seismic performance enhancement of small modular reactors using optimal design of inerter-based vibration

absorbers. *Nuclear Engineering and Technology, 56*(11), 4628–4641. Retrieved from https://www.sciencedirect.com/science/article/pii/S1738573324002857

[54] Royal HaskoningDHV. *Keeping data centres up to date*. Retrieved from https://www.haskoning.com/en/markets/data-centres/keeping-data-centres-up-to-date

[55] Avigni, P., & Petrovic, B. (2018). On-line refueling for the Advanced High Temperature Reactor (AHTR). *Nuclear Engineering and Design, 340*, 166–182. Retrieved from https://doi.org/10.1016/j.nucengdes.2018.10.006

[56] World Nuclear Association. (2023). *Uranium enrichment*. Retrieved from https://world-nuclear.org/information-library/nuclear-fuel-cycle/conversion-enrichment-and-fabrication/uranium-enrichment.aspx

[57] U.S. Nuclear Regulatory Commission. *Backgrounder on high burnup spent nuclear fuel*. Retrieved from https://www.nrc.gov/reading-rm/doc-collections/fact-sheets/bg-high-burnup-spent-fuel.html

[58] Mills, E., & Masanet, E. (2009). *A Simple Model for Determining True Total Cost of Ownership for Data Centers* (LBNL Report TUI3011B). Lawrence Berkeley National Laboratory. Retrieved from https://datacenters.lbl.gov/sites/default/files/%28TUI3011B%29SimpleModelDeterminingTrueTCO.pdf

[59] International Atomic Energy Agency. (2019). *Waste from innovative types of reactors and fuel cycles: A preliminary study* (IAEA Nuclear Energy Series No. NW-T-1.7). Vienna. Retrieved from https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1617_web.pdf

[60] Nuclear Innovation Alliance. (2024, December). *From reactors to repositories: Disposal pathways for advanced nuclear reactor waste*. Retrieved from https://nuclearinnovationalliance.org/reactors-repositories-disposal-pathways-advanced-nuclear-reactor-waste

[61] International Atomic Energy Agency. (2022, December 6). *Finland's spent fuel repository a game changer for the nuclear industry, Director General Grossi says*. Retrieved from https://www.iaea.org/newscenter/news/finlands-spent-fuel-repository-a-game-changer-for-the-nuclear-industry-director-general-grossi-says

[62] U.S. Department of Energy. (2022, October 5). *What is generation capacity?* Retrieved from https://www.energy.gov/ne/articles/what-generation-capacity

[63] U.S. Nuclear Regulatory Commission. (n.d.). *Emergency planning zones*. Retrieved from https://www.nrc.gov/about-nrc/emerg-preparedness/about-emerg-preparedness/planning-zones.html

[64] Subki, M. H. (2016, October). *Small Modular Reactors: Update on International Technology Development Activities*. Presentation at the 13th INPRO Dialogue Forum on Legal and Institutional Issues in the Global Deployment of SMRs, IAEA Headquarters, Vienna. Retrieved from

https://nucleus.iaea.org/sites/INPRO/df13/Presentations/007_Advances%20in%20Small%20Modular%20Reactor%20Technology%20developments.pdf

[65] Idaho National Laboratory. *Experimental Breeder Reactor I (EBR-I)*. Retrieved from https://inl.gov/ebr/

[66] International Atomic Energy Agency. (2002). *Operational and decommissioning experience with fast reactors* (IAEA-TECDOC-1405). Retrieved from https://www-pub.iaea.org/MTCD/Publications/PDF/te_1405_web.pdf

[67] Beck, J. M., & Pincock, L. F. (2011). *High Temperature Gas-Cooled Reactors: Lessons learned applicable to the Next Generation Nuclear Plant* (INL/EXT-10-19329, Rev. 1). Idaho National Laboratory. Retrieved from https://inldigitallibrary.inl.gov/sites/sti/sti/5021775.pdf

[68] Oak Ridge National Laboratory. *History of the Molten Salt Reactor Experiment (MSRE)*. U.S. Department of Energy. Retrieved from https://www.ornl.gov/molten-salt-reactor/history

[69] I-SMR. *Innovative Small Modular Reactor*. Retrieved from https://i-smrkr.com/

[70] NuScale Power. *NuScale Power Module™*. Retrieved from https://www.nuscalepower.com/products/nuscale-power-module

[71] Last Energy. *Technology*. Retrieved from https://www.lastenergy.com/technology

[72] Stratek Global. *HTMR-100*. Retrieved from https://stratekglobal.com/htmr-100/

[73] X-energy. *Xe-100 Reactor*. Retrieved from https://x-energy.com/reactors/xe-100

[74] ARC Clean Technology. *Technology*. Retrieved from https://www.arc-cleantech.com/technology

[75] MoltexFLEX. *FLEX Reactor*. Retrieved from https://www.moltexflex.com/flex-reactor/

[76] Terrestrial Energy. *Technology*. Retrieved from https://www.terrestrialenergy.com/technology

[77] Kairos Power. *Technology*. Retrieved from https://kairospower.com/technology/

[78] Stellaria. *Stellaria*. Retrieved from https://www.stellaria.fr/en/

[79] Oklo. *Energy*. Retrieved from https://oklo.com/energy/default.aspx

[80] Westinghouse Electric Company. *AP300 SMR*. Retrieved from https://westinghousenuclear.com/energy-systems/ap300-smr/

[81] Mignacca, B., & Locatelli, G. (2020). Economics and finance of Small Modular Reactors: A systematic review and research agenda. *Renewable and Sustainable Energy Reviews, 118*, 109519. Retrieved from https://doi.org/10.1016/j.rser.2019.109519

[82] Abou-Jaoude, A., Lin, L., Bolisetti, C., Worsham, E., Larsen, L. M., & Epiney, A. (2023). *Literature Review of Advanced Reactor Cost Estimates (Milestone M3CT-23IN1207065)*. Idaho National Laboratory. Retrieved from https://gain.inl.gov/content/uploads/4/2024/11/INL-RPT-23-72972-Literature-Review-of-Adv-Reactor-Cost-Estimates.pdf

[83] Official Data Foundation. (2023). *Inflation Calculator*. Alioth Finance. Retrieved from https://www.officialdata.org/us/inflation/2019?amount=843523&endYear=2023

[84] Organisation for Economic Co-operation and Development & Nuclear Energy Agency. (2020). *Unlocking Reductions in the Construction Costs of Nuclear: A Practical Guide for Stakeholders* (NEA No. 7530). Paris: OECD Publishing. Retrieved from https://www.oecd.org/content/dam/oecd/en/publications/reports/2020/08/unlocking-reductions-in-the-construction-costs-of-nuclear_2ca6777b/33ba86e1-en.pdf?utm_source=chatgpt.com

[85] Van Hee, N., Peremans, H., & Nimmegeers, P. (2024). *Economic potential and barriers of small modular reactors in Europe*. *Renewable and Sustainable Energy Reviews, 203*, 114743. Retrieved from https://doi.org/10.1016/j.rser.2024.114743

[86] Soto, G. J., Hanna, B., Toman, J., Guaita, N., Talbot, P., Epiney, A., & Lohse, C. S. (2024). *Powering data centers with clean energy: A techno-economic case study of nuclear and renewable energy dependability* (INL/RPT-24-79307, Rev. 0). Idaho National Laboratory. Retrieved from https://gain.inl.gov/content/uploads/4/2024/11/INL-RPT-24-79307-Powering-Data-Cneters-with-Clean-Energy.pdf

[87] Lazard. (2023, April). *LCOE+ April 2023*. Lazard Ltd. https://www.lazard.com/media/nltb551p/lazards-lcoeplus-april-2023.pdf

[88] Wang Esram, N., & Assadi, C. (2025, April). *Future-proof AI data centers, grid reliability, and affordable energy: Recommendations for states* (White Paper). American Council for an Energy-Efficient Economy (ACEEE). Retrieved from https://www.aceee.org/sites/default/files/pdfs/future-proof_ai_data_centers_grid_reliability_and_affordable_energy_-_recommendations_for_states.pdf

[89] Ponemon Institute. (2016). *Cost of Data Center Outages*. Uptime Institute / Emerson Network Power. Retrieved from https://www.nutanix.com/sg/blog/the-cost-of-downtime

[90] World Bank Group. (2024). *Carbon pricing in the power sector: Role and design for transitioning toward net-zero carbon development*. Retrieved from https://icapcarbonaction.com/system/files/document/carbon-pricing-in-the-power-sector_final.pdf

[91] Brightlio. (2024). *Colocation Pricing: What to Expect in 2024*. Retrieved from https://brightlio.com/colocation-pricing/#:~:text=According%20to%20CBRE%2C%20the%20average,18.6%25%20increase%20year%20over%20year.

[92] Lovering, J. R., Yip, A., & Nordhaus, T. (2016). Historical construction costs of global nuclear power reactors. *Energy Policy, 91*, 371–382. Retrieved from https://doi.org/10.1016/j.enpol.2016.01.011

[93] Yle. (2019, March 7). *Third Olkiluoto nuclear reactor granted operating license*. Retrieved from https://yle.fi/a/3-10680697

[94] AP News. (2023, August 31). *Georgia nuclear rebirth arrives 7 years late, $17B over cost*. Retrieved from https://apnews.com/article/georgia-nuclear-power-plant-vogtle-rates-costs-75c7a413cda3935dd551be9115e88a64

[95] Godsey, K. M. (2019). *Life cycle assessment of small modular reactors using U.S. nuclear fuel cycle* (Master's thesis, Clemson University). Clemson University. Retrieved from https://open.clemson.edu/cgi/viewcontent.cgi?article=4242&context=all_theses

[96] Intergovernmental Panel on Climate Change (IPCC). (2014). *Climate Change 2014: Mitigation of Climate Change*. Contribution of Working Group III to the Fifth Assessment Report. Retrieved from https://www.ipcc.ch/report/ar5/wg3/

[97] National Renewable Energy Laboratory (NREL). (2021). *Life Cycle Greenhouse Gas Emissions from Electricity Generation*. Retrieved from https://www.nrel.gov/docs/fy21osti/80580.pdf

[98] Sovacool, B. K. (2008). *Valuing the greenhouse gas emissions from nuclear power: A critical survey*. Energy Policy, 36(8), 2940–2953. Retrieved from https://doi.org/10.1016/j.enpol.2008.04.017

[99] International Energy Agency (IEA). (2023, March 22). *Global water consumption in the energy sector by fuel and power generation type in the Stated Policies Scenario, 2021 and 2030*. IEA. Retrieved from https://www.iea.org/data-and-statistics/charts/global-water-consumption-in-the-energy-sector-by-fuel-and-power-generation-type-in-the-stated-policies-scenario-2021-and-2030

[100] Sustainability Directory. (2025, April 25). *How does SMR waste compare to legacy nuclear?* Retrieved from https://energy.sustainability-directory.com/question/how-does-smr-waste-compare-to-legacy-nuclear/

[101] Ritchie, H. (2022, June 16). *How does the land use of different electricity sources compare?* Our World in Data. Retrieved from https://ourworldindata.org/land-use-per-energy-source

[102] International Atomic Energy Agency (IAEA). (2013). *INPRO Dialogue Forum on Global Nuclear Energy Sustainability: Licensing and safety issues for small and medium-sized reactors (SMRs) – Discussion Group 2: Siting considerations for SMRs*. IAEA Headquarters, Vienna, Austria. Retrieved from https://nucleus.iaea.org/sites/INPRO/df6/Working%20Documents/2_IAEA_INPRO_SMR_Discussion_Forum_Planning_Sheet_for_Group_2.pdf

APPENDIX 1: Global and Local Weights in the AHP Model

This appendix presents the detailed numerical results obtained from the AHP used in this study to evaluate the suitability of SMRs for data center applications. Two key tables are included:

- The **Global Weights Table** shows the overall contribution of each subcriterion to the final score of each reactor alternative. These values represent the product of each subcriterion's local weight and its corresponding criterion weight, providing a direct measure of influence on the final ranking.
- The **Local Weights Table** displays the normalized performance of each reactor across all subcriteria, independent of the global weight structure. These values indicate how well each alternative performs within individual subcriteria and were derived from expert judgment and standardized data inputs.

Both tables serve to ensure transparency and reproducibility of the multi-criteria evaluation process. They also enable further analysis, such as sensitivity assessments and benchmarking, by providing a disaggregated view of the decision framework.

Alternative	Module Independence	Unit Capacity Factor	Enrichment	Refueling Time	EPZ Size	Plant Size	Scalability	Design Status	Licensing Initiated	Non-Standard Design	Operational Experience	Component Qualification	Configuration	Seismic Resilience	Coolant Toxicity	Fuel Safety	Source Term	Secondary Applications	Discharge Burnup	Disposal Plan	Waste Streams
i-SMR	0.016	0.145	0.021	0.007	0.003	0.001	0.021	0.016	0.004	0.007	0.067	0.015	0.029	0.005	0.036	0.015	0.019	0.036	0.010	0.018	0.003
NuScale Power Module	0.016	0.145	0.021	0.007	0.003	0.008	0.021	0.016	0.036	0.007	0.067	0.015	0.029	0.005	0.036	0.015	0.019	0.036	0.010	0.018	0.003
PWR-20	0.048	0.145	0.021	0.007	0.003	0.003	0.050	0.016	0.004	0.019	0.067	0.015	0.003	0.005	0.036	0.015	0.019	0.036	0.007	0.018	0.003
HTMR100	0.016	0.048	0.011	0.079	0.008	0.001	0.021	0.007	0.004	0.007	0.015	0.002	0.003	0.002	0.078	0.156	0.039	0.012	0.028	0.009	0.003
Xe-100	0.016	0.145	0.006	0.079	0.008	0.003	0.021	0.007	0.011	0.007	0.015	0.002	0.003	0.005	0.078	0.156	0.039	0.012	0.045	0.009	0.003
ARC-100	0.048	0.048	0.006	0.035	0.020	0.001	0.005	0.003	0.004	0.007	0.031	0.005	0.003	0.002	0.016	0.029	0.007	0.036	0.017	0.009	0.028
FLEX Reactor	0.048	0.048	0.021	0.007	0.020	0.008	0.050	0.007	0.004	0.004	0.006	0.005	0.010	0.002	0.008	0.064	0.007	0.012	0.007	0.004	0.009
IMSR400	0.016	0.048	0.021	0.007	0.020	0.008	0.005	0.016	0.011	0.004	0.006	0.002	0.029	0.002	0.008	0.064	0.003	0.012	0.007	0.004	0.009
KP-FHR	0.016	0.145	0.006	0.079	0.020	0.001	0.021	0.036	0.036	0.004	0.006	0.002	0.003	0.002	0.008	0.156	0.007	0.012	0.017	0.004	0.009
Stellarium	0.016	0.048	0.006	0.035	0.020	0.003	0.005	0.003	0.004	0.004	0.006	0.002	0.003	0.005	0.008	0.064	0.007	0.012	0.045	0.004	0.009
Aurora	0.048	0.145	0.006	0.035	0.020	0.001	0.050	0.016	0.011	0.004	0.031	0.015	0.029	0.005	0.016	0.029	0.007	0.036	0.007	0.009	0.028
AP-300	0.048	0.145	0.021	0.007	0.003	0.008	0.005	0.007	0.004	0.019	0.067	0.015	0.003	0.002	0.036	0.015	0.019	0.036	0.010	0.018	0.003

Table 84. Global Weights

Alternative	Module Independence	Unit Capacity Factor	Enrichment	Refueling Time	EPZ Size	Plant Size	Scalability	Design Status	Licensing Initiated	Non-Standard Design	Operational Experience	Component Qualification	Configuration	Seismic Resilience	Coolant Toxicity	Fuel Safety	Source Term	Secondary Applications	Discharge Burnup	Disposal Plan	Waste Streams
i-SMR	0.333	1.000	0.536	0.084	0.146	0.146	0.411	0.451	0.099	0.354	1.000	1.000	1.000	1.000	0.464	0.093	0.483	1.000	0.232	1.000	0.108
NuScale Power Module	0.333	1.000	0.536	0.084	0.146	1.000	0.411	0.451	1.000	0.354	1.000	1.000	1.000	1.000	0.464	0.093	0.483	1.000	0.232	1.000	0.108
PWR-20	1.000	1.000	0.536	0.084	0.146	0.382	1.000	0.451	0.099	1.000	1.000	1.000	0.092	1.000	0.464	0.093	0.483	1.000	0.148	1.000	0.108
HTMR100	0.333	0.333	0.268	1.000	0.382	0.146	0.411	0.202	0.099	0.354	0.228	0.132	0.092	0.333	1.000	1.000	1.000	0.333	0.627	0.500	0.108
Xe-100	0.333	1.000	0.144	1.000	0.382	0.382	0.411	0.202	0.315	0.354	0.228	0.132	0.092	1.000	1.000	1.000	1.000	0.333	1.000	0.500	0.108
ARC-100	1.000	0.333	0.144	0.442	1.000	0.146	0.101	0.091	0.099	0.354	0.465	0.315	0.092	0.333	0.208	0.188	0.171	1.000	0.382	0.500	1.000
FLEX Reactor	1.000	0.333	0.536	0.084	1.000	1.000	1.000	0.202	0.099	0.188	0.092	0.315	0.339	0.333	0.098	0.409	0.171	0.333	0.148	0.250	0.329
IMSR400	0.333	0.333	0.536	0.084	1.000	1.000	0.101	0.451	0.315	0.188	0.092	0.132	1.000	0.333	0.098	0.409	0.084	0.333	0.148	0.250	0.329
KP-FHR	0.333	1.000	0.144	1.000	1.000	0.146	0.411	1.000	1.000	0.188	0.092	0.132	0.092	0.333	0.098	1.000	0.171	0.333	0.382	0.250	0.329
Stellarium	0.333	0.333	0.144	0.442	1.000	0.382	0.101	0.091	0.099	0.188	0.092	0.132	0.092	1.000	0.098	0.409	0.171	0.333	1.000	0.250	0.329
Aurora	1.000	1.000	0.144	0.442	1.000	0.146	1.000	0.451	0.315	0.188	0.465	1.000	1.000	1.000	0.208	0.188	0.171	1.000	0.148	0.500	1.000
AP-300	1.000	1.000	0.536	0.084	0.146	1.000	0.101	0.202	0.099	1.000	1.000	1.000	0.092	0.333	0.464	0.093	0.483	1.000	0.232	1.000	0.108

Table 85. Local Weights